

Supplementary Material for: Identifiability by Stationarity: Blind Gain–Object Separation in Self-Calibrating Ghost Imaging

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Abstract—This document supplements the main manuscript. Sections S1–S5 report the supplementary verification experiments promised in Sec. 9 of the main text: the permutation-whitening / factorized-randomization ablation (S1), the coherent-residual validation of the master identity (S2), the C_0 audit and the oracle-to-blind baselines (S3), and the extended flip-boundary detail with the low-photon drift-floor probe and replication tables (S4), and the shared simulation protocol (S5). Appendices A–F contain the complete proofs of every theorem, corollary, and proposition quoted in the main text. Citation numbers, section numbers (Sec. 1–10), figure numbers (Fig. 1–6), theorem labels, and equation tags (e.g. (F.10)) referring to the main manuscript retain their main-text meaning; figures and tables introduced here *continue* the main numbering (Figs. 7–12, Tables 2–3), and only the supplementary sections keep an S-prefix (S1–S5). All numerical values are descriptive results of the specified simulation grids unless a resampled interval is stated.

S1. PERMUTATION WHITENING AND THE FACTORIZED RANDOMIZATION ABLATION

This section carries the multi-permutation whitening-power experiment and the factorized randomization ablation summarized in Sec. 9.6 of the main text (data: results/perm_ablation_r1/; 32 draws per randomized arm $\times$ 10 objects under the Fig. 2 protocol — $K = 4096$, 64×64 objects, 8192 paired frames, noiseless carrier, Brown–Forsythe/Levene screen [29], [30] at 8 chunks, reject at $p < 10^{-3}$). Two randomization semantics must be kept separate (Appendix E.6): the *pixel-level* randomization of the SRHT theory — pixel permutation P_{col} and pixel signs D , whose Lemma E.1 covariance is invariant to acquisition order — and the *acquisition-order* reordering P_{row} , which controls temporal stationarity of the acquired series. Figure 7(a) reports the factorized attribution: either permutation alone substantially reduces the temporal-stationarity screen’s rejection rate — from 70% (ordered) to 5.6% (P_{col}) and 12.2% (P_{row}), all still far above the $\sim 0.1\%$ nominal rate of the $p < 10^{-3}$ test — the theory’s own randomization $P_{\text{col}}+D$ reduces it to 11.9%, the code’s SRHT arm ($P_{\text{row}}+D$) to 5.9%, and the historical combined arm ($P_{\text{col}}+P_{\text{row}}$) reproduces its $19/320 = 5.9\%$ headline — whereas pixel signs D alone barely move it (43.4%), because sign flips whiten carrier amplitudes but leave the coarse-to-fine sequency-energy ordering of natural-Hadamard rows intact. Figure 7(b) shows the honest per-draw certificate null quoted in the main text: on the historical

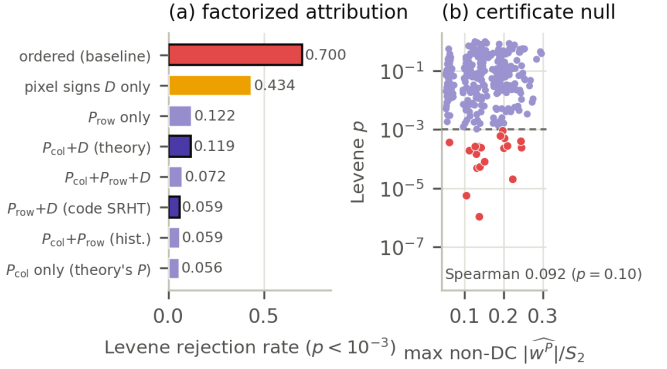


Fig. 7. Permutation whitening and factorized randomization ablation (32 draws per randomized arm $\times$ 10 objects; Levene screen at $p < 10^{-3}$). (a) Rejection rate per arm: either permutation alone substantially reduces the rejection rate (70% $\rightarrow$ 5.6–12.2%, all $\gg$ the $\sim 0.1\%$ nominal rate); pixel signs alone barely move it; outlined bars mark the theory’s randomization ($P_{\text{col}}+D$) and the code’s SRHT arm ($P_{\text{row}}+D$). (b) Per-draw certificate null on the historical $P_{\text{col}}+P_{\text{row}}$ arm (320 draws): max non-DC $|w^P|/S_2$ of the permuted squared object versus the Levene p -value (violet: accepted draws; red: rejected; dashed line: the $p = 10^{-3}$ gate). Walsh flatness of the realized draw does not predict rejection (Spearman 0.092, $p = 0.10$).

arm, the maximum non-DC Walsh coefficient of the permuted T^2 does not predict which draws reject (Spearman 0.092, $p = 0.10$), consistent with the probabilistic — not per-draw-certifiable — nature of permutation whitening (Sec. 7).

The acquisition-time ablation of the main text’s Sec. 9.4 (moved here from the main document for space) is reproduced as Fig. 8: at the single above-floor drift point $\rho = 10^{-3}$, acquisition-order permutation alone gains +5.39 dB over ordered Hadamard, pixel sign flips alone +5.47 dB, and the code’s full SRHT arm +5.45 dB — each ingredient by itself recovers essentially the full advantage, and combining them is not additive. All Fig. 8 arms randomize acquisition time only; the pixel-permutation (P_{col}) attribution is the subject of Fig. 7 above.

S2. VALIDATION OF THE MASTER IDENTITY: RECONSTRUCTION BRIDGE AND COHERENT RESIDUALS (E4)

This section carries two raw-MSE validations of Theorem 1 under injected, exogenous residuals. Figure 9 shows the balanced 8100-row fixed-design bridge; experiment E4 then

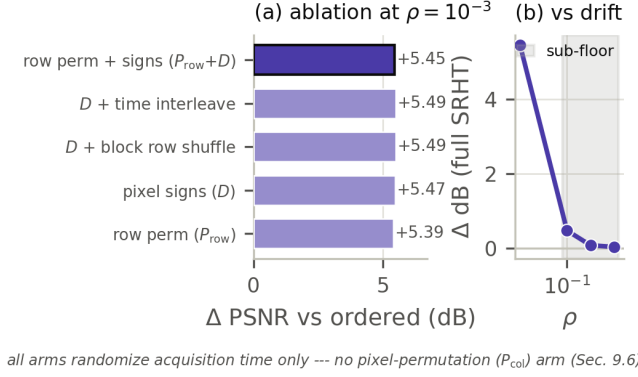


Fig. 8. Ablation isolating the acquisition-time randomization mechanism (moved from the main document). (a) PSNR gain over ordered Hadamard at the single above-floor drift point $\rho = 10^{-3}$ for five arms: acquisition-order (row) permutation P_{row} alone gains +5.39 dB, pixel sign flips D alone +5.47 dB, and the code’s full SRHT arm ($P_{\text{row}}+D$) +5.45 dB, with the block-shuffle and time-interleave sign variants in the same +5.5 dB band — each ingredient by itself recovers essentially the full advantage, and combining them is not additive. All five arms randomize acquisition time only: none applies the pixel/column permutation P_{col} of the Appendix E SRHT analysis, whose factorized attribution is Fig. 7. (b) The full-SRHT advantage versus drift rate ρ : the gain collapses toward 0 dB as every blind variant approaches the noise floor (shaded: sub-floor cells). What matters is that the chronology is randomized, not the particular form of the randomization. Descriptive values at single grid points (no resampled intervals).

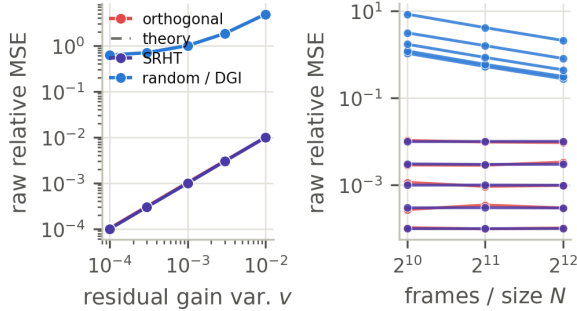


Fig. 9. Reconstruction bridge in total raw relative MSE (8100 rows: three reconstructors, $N \in \{1024, 2048, 4096\}$, six v levels including the $v = 0$ baseline, ten objects, fifteen seeds). Markers: measured R_{raw} ; dashed: fixed-design prediction $R_{\text{raw}}^{\text{raw}} + B_L v$. Deterministic inverses are flat in N ; random/DGI follows $1/N$. Exact leverage, ratios, and identity residual: main-text Sec. 9.3.

probes coherent residual covariance. Neither protocol contains detector photon noise.

Five residual-covariance structures — i.i.d., AR(1), sinusoidal, low-pass OU, blockwise — are injected at fixed $v = 10^{-3}$ across three reconstructors and ten objects (150 cells, $K = N = 1024$; data: results/coherent_residual_e4_r1/), using the same raw identity-verification metric. Figure 10(a) shows the full matrix law tracking measured relMSE within 4.2%. Panel (b) isolates the coherent clause: the scalar $v B_L$ shortcut is exact for complete orthogonal and SRHT inversions but fails by up to $3.65\times$ for random/DGI under low-pass-coherent residuals. E4 does not probe same-record δ - ξ dependence.

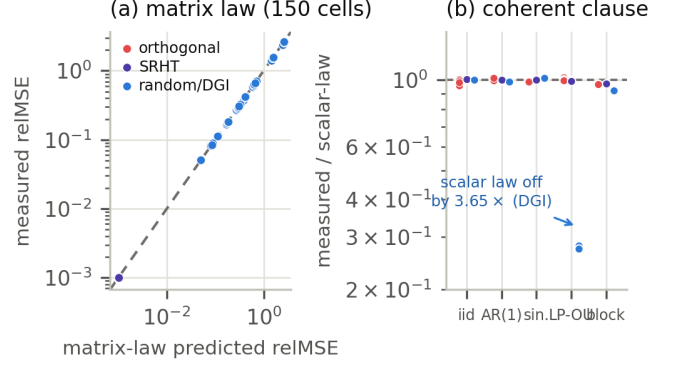


Fig. 10. Coherent-residual validation of Theorem 1 (E4; five injected residual structures $\times$ three reconstructors $\times$ ten objects at $v = 10^{-3}$, raw identity-verification metric). (a) Matrix-law prediction versus measured relMSE: agreement within 4.2% across all 150 cells. (b) Ratio of measured relMSE to the scalar $v B_L$ law by residual structure: exact for orthogonal and SRHT inversions, but up to $3.65\times$ off for random/DGI under low-pass OU residuals — a direct verification of Theorem 1’s coherent (full-matrix) clause.

S3. C_0 AUDIT AND ORACLE-TO-BLIND BASELINES

C_0 audit (E9). The one-sample DGI correlator constant $C_0 = \mathbb{E}\|Z\|^2/S_2 - 1$ is measured from 2×10^4 single-frame pattern draws per ensemble, for four pattern ensembles $\times$ five objects (data: results/c0_audit_e9_r1/). Figure 11(a) compares the measurement with the moment formula of Appendix F: mean relative error 0.35%, worst 2.01%, across values of C_0 spanning 10^3 to 5.7×10^6 . The strong ensemble dependence — $O(K)$ for signed Rademacher patterns versus $O(K_{\text{eff}}K)$ for the nonnegative offset ensembles — confirms the main text’s bookkeeping rule that C_0 is pipeline-specific, as Theorem 1’s DGI specialization insists: there is no universal correlator constant to tabulate.

Oracle, metered, and blind baselines (E11). On four held-out objects with OU drift ($\rho = 10^{-3}$, $s = 0.1$; five gain treatments $\times$ five seeds; data: results/oracle_baselines_e11_r1/), Fig. 11(b) shows the accountability ordering static $\geq$ oracle $\geq$ blind-AGC $\geq$ no-correction holding in CNR for 4 of 4 objects, with an oracle–blind CNR gap of only 0.035–0.147: at this drift point the stationarity-based ratio AGC concedes little to a gain oracle, while forgoing correction altogether costs the most. This is an estimator baseline, not a direct test of the Theorem B log estimator.

S4. EXTENDED QUANTITATIVE DETAIL: FLIP BOUNDARY, LOW-PHOTON FLOOR, AND REPLICATION TABLES

This section collects the extended flip-boundary detail, the low-photon floor probe, and the per-arm replication values behind main-text Fig. 3 and Table I.

Corrected- C_0 no-flip consistency. Section 9.4 of the main text reports that, once the drift-leverage constant of the mean-subtracted blind random+DGI pipeline is corrected to the nonnegative-background value $\Lambda_{\text{lev}} = 1 + C_0^{\text{lev,raw background}}$ of (F.10), the *conditional scalar proxy*

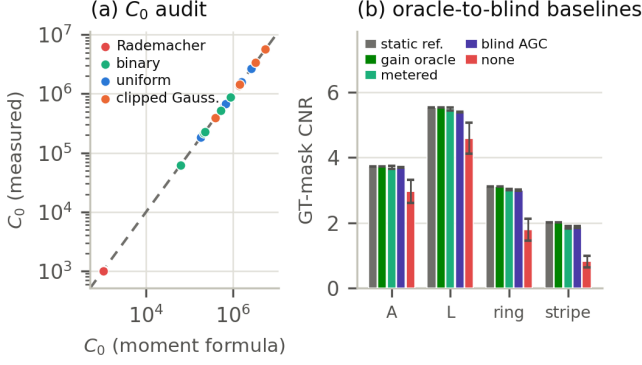


Fig. 11. C_0 audit and oracle-to-blind baselines. (a) Measured one-sample correlator constant C_0 versus the Appendix-F moment formula for four pattern ensembles $\times$ five objects (2×10^4 draws each): mean relative error 0.35% (worst 2.01%); the $O(K)$ Rademacher versus $O(K_{\text{eff}}K)$ offset-ensemble spread confirms that C_0 is pipeline-specific. (b) GT-mask CNR under five gain treatments on four held-out objects (OU drift, $\rho = 10^{-3}$, $s = 0.1$; error bars: s.d. over five seeds): the ordering static $\geq$ oracle $\geq$ blind-AGC $\geq$ none holds for 4 of 4 objects, with oracle-blind gaps of 0.035–0.147 CNR.

gives no flip anywhere on the simulated grid — consistent with the empirical 100/100 no-flip cells, but not a same-record scalar theorem validation. Figure 12(a) shows the historical per-cell margin behind that consistency check (data: `results/prop3_nofreeparam_r1/`): for both blind corrections (AGC and SCGI proxy), the minimum over the drift grid of $Q - r$ stays strictly positive in every one of the 2×50 (correction $\times$ cell) checks, with the margin narrowing as σ_a grows. The R15 projected mean/covariance decomposition, rather than this scalar proxy, is the rigorous empirical object for the same-record blind arm.

Low-photon drift-limited floor. Figure 12(b) carries the floor probe behind the closing claim of Sec. 9’s low-photon experiment (data: `results/paper_fig7_lowphoton_r5_final/`): at the high-photon end the log-domain gain-MSE floor is drift-limited rather than photon-limited — the floor of 2.434×10^{-4} at $\rho = 10^{-3}$ falls to 1.411×10^{-4} when the drift slows to $\rho = 10^{-4}$ for the proxy arms, and from 2.168×10^{-4} to 1.125×10^{-4} for the oracle-known-carrier estimator of Theorem C, whose floor sits *below* the proxies by 10.92% at $\rho = 10^{-3}$ and by 20.28% at $\rho = 10^{-4}$ (two distinct margins) because it deconvolves the known carrier exactly; the arms coincide through the photon-limited part of the probed $\bar{\lambda} \geq 4$ range and separate only at the floor.

Table II collects the secondary measured constants and grid conventions behind main-text Fig. 4 and Table I (moved here from the main document’s Table I for space; drift leverage, D_H , same-record residual diagnostics, and the Prop. 3 no-free-parameter agreement remain in the main text).

Per-arm replication values (main-text Fig. 3 and Sec. 9.2). Table III tabulates the best-window blind gain errors and the slow-drift variance-segment slopes per arm from the 20-seed uncertainty replication (fixed $W \leq 16$ variance-segment convention; 31,600 rows; medians with two-way seeds-and-objects crossed cluster bootstrap 95% CIs), together with the

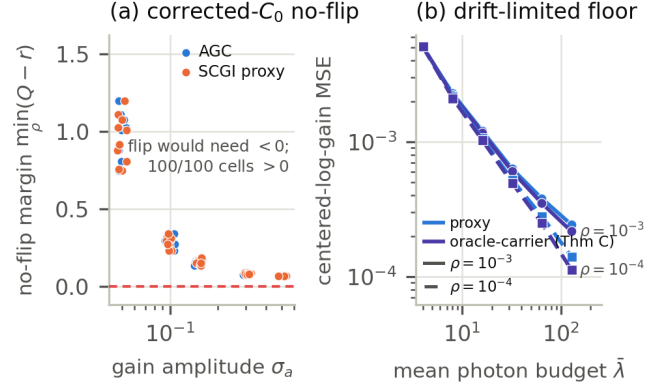


Fig. 12. Extended flip-boundary detail and the low-photon floor probe. (a) Corrected- C_0 no-flip margins for the blind random+DGI arm: per-cell minimum over the drift grid of $Q - r$ for the conditional scalar proxy with drift-leverage constant $1 + C_0^{\text{lev,raw}}$ background, for the AGC and SCGI-proxy corrections; a flip would require a negative value, and all 100/100 cells are positive, matching the empirical no-flip outcome of Sec. 9.4. This is a consistency check, not a same-record scalar theorem validation. (b) High-photon floor probe: log-domain gain MSE versus mean photon budget $\bar{\lambda}$ for the uncalibrated proxy and the oracle-known-carrier calibrated soft-log (Theorem C) at $\rho = 10^{-3}$ (solid) and $\rho = 10^{-4}$ (dashed); the two drift rates share the photon-limited regime and separate only at the floor (2.434×10^{-4} versus 1.411×10^{-4} for the proxy, 2.168×10^{-4} versus 1.125×10^{-4} for the oracle-carrier arm — 10.92% lower at $\rho = 10^{-3}$, 20.28% lower at $\rho = 10^{-4}$), which is therefore drift-limited, not photon-limited.

raw-arm single-pass diagnostics and the exploratory 5-seed adaptive-segment slopes.

S5. SHARED SIMULATION PROTOCOL: PER-FIGURE ENUMERABLES

Bucket-signal conventions (full statement; main-text Sec. 5): (i) *signed coefficient model* — B_n is a signed transform coefficient (e.g. a bare Walsh–Hadamard coefficient); it can vanish or change sign, so $\log B_n$ is undefined and no log-domain gain estimator is available without further structure; (ii) *physical nonnegative offset model* — the optical bucket is a nonnegative intensity, with signed patterns realized with a bias/offset so that $B_n \geq B_{\min} > 0$; Theorem B and the main-text Sec. 6 estimators are stated in this model; (iii) $\pm$ *pairwise model* — each signed pattern is acquired as a complementary pair of nonnegative patterns whose buckets are differenced; the difference is signed again, but the drift enters through adjacent-pair increments controlled by ρ_{pair} (main-text Sec. 8). The raw signed arms of main-text Sec. 9 follow convention (i) and are diagnostic only.

Notation (consolidated from main-text Sec. 2): K pixels, N frames, drift class $S \ni 1$ with $p = \dim S$; gauge $(a, T) \mapsto (ca, T/c)$; bandwidth ρ_{bw} ($p \approx \rho_{\text{bw}}N$) versus adjacent-pair decorrelation ρ_{pair} ; object functionals $K_{\text{eff}} = S_1^2/S_2$, $K_{\infty} = S_2/\|T\|_{\infty}^2$, $K_4 = S_2^2/\sum_j T_j^4$; pipeline constants $C_0 = \mathbb{E}\|Z\|^2/S_2 - 1$, D_H , and the reconstructor leverage $B_L = (1/S_2) \sum_n b_n^2 \|Le_n\|^2$.

All supplementary and main-text experiments share the protocol of Sec. 9 of the main text (matched acquisition budgets, shared OU drift traces within each seed, a single

TABLE II

SECONDARY MEASURED CONSTANTS AND CONVENTIONS BEHIND MAIN-TEXT FIG. 4 (COMPANION TO MAIN-TEXT TABLE I). EVERY CONSTANT IS PIPELINE-SPECIFIC; CONFIDENCE CONVENTIONS AS IN THE MAIN TEXT (TWO-WAY SEEDS-AND-OBJECTS CROSSED CLUSTER BOOTSTRAP WHERE STATED).

Quantity	Value in this pipeline	Uncertainty
K_{eff} (object-panel range)	59.8–864.3	—
Drift / noise / flux conventions	OU log-gain, $\rho \in [10^{-3}, 10]$, drift s.d. 0.030–0.104; noise $\sigma \in [0.05, 0.5]$; equal frames and equal total flux per arm	—
Above-floor gate	relMSE < 0.5; 29 of 45 cells above floor (16 at noise floor)	cell count, not resampled; no bootstrap CI computed
Empirical SRHT-paired vs. ordered-Hadamard crossover, two-parameter power-law fit	$R^2 = 0.989$ – 0.9995 , fitted exponents -0.94 to -1.93	descriptive fits of a comparison Prop 3 does not rank; survival-style CIs left to future work

TABLE III

PER-ARM BEST-WINDOW BLIND GAIN ERRORS AND $\log(\text{err})$ -VS- $\log W$ SLOPES BEHIND MAIN-TEXT FIG. 3 ($\rho_{\text{pair}} = 10^{-3}$, $s = 0.1$, 2048-FRAME BUDGET; MEDIAN OVER 20 SEEDS $\times$ 10 OBJECTS WITH TWO-WAY CLUSTERED 95% CIs; EXPLORATORY SLOPES FROM THE EARLIER 5-SEED ADAPTIVE-SEGMENT CONVENTION). THE RAW SIGNED-WALSH ARMS ARE DIAGNOSTIC-ONLY (BUCKET CONVENTION (I)); THEIR SINGLE-PASS 1024-FRAME MEDIAN ARE GIVEN FOR THE BUDGET CONTROL.

Arm	Best-window error (ratio AGC)	Best-window error (log-window implementation)	Slope, ratio [95% CI]	AGC	Explor. slope
random uniform	— (bundle 0.007–0.017)	agrees to $\sim 2\%$	$-0.39 [-0.42, -0.31]$	—	—
random binary	0.0150 [0.0130, 0.0165]	agrees to $\sim 2\%$	$-0.41 [-0.44, -0.37]$	—	-0.39
SRHT-paired	0.0077 [0.0066, 0.0083]	agrees to $\sim 2\%$	$-0.58 [-0.66, -0.41]$	—	-0.60
randomly permuted paired Hadamard	0.0073 [0.0063, 0.0079]	0.0119 [0.0090, 0.0133]	$-0.62 [-0.67, -0.44]$	—	-0.61
naturally ordered paired Hadamard	—	shallow decay $-0.222 [-0.306, -0.054]$ (slope)	$+0.03 [-0.06, +0.17]$ (no decay)	—	-0.24
raw ordered (2048; single-pass 0.896)	0.899 [0.892, 0.903], 0.950 at $W=512$	0.708 [0.563, 0.840], ≥ 0.71 , non-decaying	—	—	—
raw shuffled (2048; single-pass 0.788)	0.794 [0.775, 0.816]	0.136 [0.083, 0.196]	—	—	—

sliding-window-mean AGC estimator for all arms with no per-arm tuning, the ten-object panel spanning $K_{\text{eff}} = 59.8$ – 864.3 , base seed 20240708). Per-figure enumerables (main-text figure numbers):

- **Fig. 2 (carrier stationarity):** 8192 frames per arm at 64×64 , split into 8 chunks with a 128-frame transient cutoff, AGC window $W = 64$, five arms.
- **Fig. 3 (blind gain identifiability):** 32×32 objects ($K = 1024$), 2048 frames per arm, nine arms — the raw signed-Walsh arms at both the frame-matched 2048 budget (sequence acquired twice) and the single-pass 1024 diagnostic — $\rho_{\text{pair}} \in \{10^{-3}, 10^{-2}\}$, drift scale $s = 0.1$, W swept over a log-spaced grid from 4 to 1024; every row is scored under both the ratio AGC and the log-domain estimator of Theorem B; the final uncertainty replication (Table III) uses 20 drift seeds per (object, ρ) pair — 31,600 rows — while the exploratory R^2 analysis used 5 seeds.
- **Reconstruction bridge (Fig. 9):** injected $v \in \{0, 10^{-4}, 3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}, 10^{-2}\}$, $N \in \{1024, 2048, 4096\}$, 15 seeds, three reconstruction arms.
- **Fig. 5 (low-photon robustness):** mean photon budgets $\bar{\lambda} \in \{0.25, \dots, 128\}$ on a 2×2 grid, soft-log $\alpha = 0.5$, requested $W = 64$ (effective interior width 65), $\rho_{\text{pair}} = 10^{-3}$, $s = 0.1$, 5 seeds (error bars: SD over the 50 object-by-seed cells), with five estimator arms — naive clipped

log, Anscombe, the uncalibrated soft-log proxy, the mean-Poisson calibrated proxy $m_{\alpha}^{-1}(\text{mean}_W \psi_{\alpha})$ (retained for continuity), and the oracle-known-carrier calibrated soft-log of Theorem C (per-window carrier-calibration root, carriers supplied from the simulation truth as a flat-field-calibrated benchmark; m_{α} tabulated exactly by truncated Poisson summation, tail bound $\leq 3.5 \times 10^{-34}$) — compared on the same paired Poisson draw at each budget; all windowed arms use truncated, renormalized windows at the record boundaries (no replicate padding).

- **Fig. 6 (threshold scan):** rank/solver scan at $K = 64$ (20 seeds, 1300 cells) and $K = 128$ (30 seeds, 1950 cells), $p \in \{3, 5, 9, 17, 33\}$, margins $N - K - p \in [-8, +16]$ (lower end truncated at $N > K$ for small p).
- **Pattern conventions:** i.i.d. uniform $[0, 1]$; i.i.d. binary Bernoulli(0.5); paired Hadamard as complementary $(1 \pm H)/2$ frames (2 frames per coefficient); SRHT as $H \cdot \text{diag}(\pm 1)$ with an optional row permutation, paired the same way; raw signed Walsh arms are diagnostic-only (main-text Sec. 5, bucket convention (i)) and not a valid physical estimator without offset or pairing.
- **Metrics:** blind gain error is scale-aligned after unit-mean normalization; mask contrast uses GT-mask CNR at threshold 0.5. Theorem 1/Prop. 3 validation uses raw relative MSE; scale-aligned reconstruction MSE is retained only as a

separately labeled image-quality metric. The OU drift trace uses `normalize_mean=1`.

Appendix A supports Sec. 4 (Theorem A and Corollaries 1–2); Appendix B supports Sec. 4 (Theorem A', parametric-transversality proof); Appendix C supports Sec. 5 (Propositions 1–2, condition (★), Theorem B); Appendix D supports Sec. 6 (Theorems C–D, Fisher geometry, low-photon); Appendix E supports Sec. 7 (SRHT Walsh condition, permutation bounds); Appendix F supports Sec. 8 (Theorem 1, specializations, Prop 3). Heuristic or genericity steps are labeled as such inside each proof; every appendix ends its theorems with a “What this does not claim” paragraph.

APPENDIX A

SQUARE-DESIGN NON-IDENTIFIABILITY

This appendix proves Theorem A of Sec. 4 together with its two corollaries, in the notation of the main text: noiseless bucket record $R_n = a_n B_n$, carrier $B_n = (MT)_n$, log-gain $\ell = \log a$ (entrywise), drift class S a linear subspace of $\mathbb{R}^N$ with $\text{span}\{1\} \subseteq S$ and $\dim S = p$. This appendix is the archival form of these results under the manuscript labels Theorem A, Corollary 1, Corollary 2.

A. A.0 Standing assumptions

- **(A1) Model.** Noiseless record $R_n = a_n(MT)_n$, $n = 1, \dots, N$, with $N = K$ and $M \in \mathbb{R}^{K \times K}$ **invertible** — Hadamard, Fourier, random square, or otherwise. Write $c = MT$ for the carrier coefficient vector, so $R = e^\ell \odot c$ ($\odot$ = entrywise product).
- **(A2) Gain class.** $a_n > 0$ for all n ; the admissible log-gains form (or contain a neighborhood inside) the linear subspace $S \ni 1$, $\dim S = p \geq 2$; write $p - 1 = \dim(S/\text{span}\{1\})$ for the nonconstant drift dimension.
- **(A3) Object class.** $T \in \mathbb{R}^K$ unconstrained (Theorem A, Corollary 1); constrained classes are treated in Corollary 2.
- **(A4) Carrier support.** For the clean dimension statements we assume full carrier support, $c_n = (MT)_n \neq 0$ for all n ; this holds for generic T (each $\{c_n = 0\}$ is a hyperplane). Zero buckets are discussed in Remark A.2.
- **(A5) Gauge.** The global-scale gauge orbit of a pair (ℓ, T) is $\mathcal{G}(\ell, T) = \{(\ell + t1, e^{-t}T) : t \in \mathbb{R}\}$; identifiability is always meant modulo $\mathcal{G}$.

B. A.1 Theorem A (square non-identifiability, any invertible design)

Theorem A. Assume (A1)–(A5). For every $s \in S$ the pair

$$\ell' = \ell + s, \quad c' = e^{-s} \odot c, \quad T'(s) = M^{-1}(e^{-s} \odot MT)$$

reproduces the bucket record exactly: $e^{\ell'_n} c'_n = R_n$ for all n . If s is nonconstant on $\text{supp}(MT)$ — under (A4), simply nonconstant — then $(\ell', T'(s)) \notin \mathcal{G}(\ell, T)$ and $T'(s) \neq \gamma T$ for every $\gamma \in \mathbb{R}$. The set of admissible pairs consistent with the record therefore contains a real-analytic family of dimension $\dim S = p$, of which the gauge orbit accounts for only $\dim(S \cap \text{span}\{1\}) = 1$; the non-gauge ambiguity dimension is at least $p - 1 \geq 1$. In particular, gain–object separation from

a square design is not identifiable modulo gauge whenever S contains a nonconstant element.

Proof. (Data equality.) $e^{\ell'_n} c'_n = e^{\ell_n + s_n} e^{-s_n} c_n = e^{\ell_n} c_n = R_n$, and $T'(s) = M^{-1}c'$ is well defined because M is invertible. Both steps are exact; no smallness of s is used.

(Distinctness of the object.) $T'(s) = \gamma T$ for some γ iff $MT'(s) = \gamma MT$, i.e. $e^{-s_n} c_n = \gamma c_n$ for every n , i.e. $e^{-s_n} = \gamma$ for every $n \in \text{supp}(c)$. Since $x \mapsto e^{-x}$ is injective, this forces s constant on $\text{supp}(c)$. Contrapositively, s nonconstant on $\text{supp}(MT)$ implies $T'(s)$ is not proportional to T .

(Distinctness modulo gauge.) $(\ell + s, T'(s)) \in \mathcal{G}(\ell, T)$ requires $\ell + s = \ell + t1$ in the gain coordinate, i.e. $s = t1$ (log is injective on positive gains). A nonconstant s therefore leaves the orbit; note that when $s = t1$ the construction reduces exactly to the gauge, $T'(t1) = e^{-t}T$, so the family contains the orbit and nothing of the orbit is double-counted.

(Dimension.) The map $s \mapsto (\ell + s, T'(s))$ is injective (read off the first coordinate) and real-analytic in s , hence its image is a p -dimensional analytically parametrized family of exact solutions. Intersecting with the gauge orbit uses up $\dim(S \cap \text{span}\{1\}) = 1$ dimension, leaving non-gauge ambiguity dimension $\geq p - 1$. ■

The display in the main text (Sec. 4) is the instance $s = \bar{\ell}1 - \ell$: the drift is absorbed in *one step* into the relabelled object $T' = M^{-1} \text{diag}(a)MT$ (up to the gauge constant), leaving a constant gain. The hypothesis “ a nonconstant on the support of MT ” in the main-text statement is exactly the distinctness condition established above.

a) *What this does not claim.*: Theorem A is a statement about the square case $N = K$ and an unconstrained object; for $N > K$ the construction fails unless $e^{-s} \odot MT$ remains in $\text{range}(M)$, and the correct analysis is the tall-design identifiability theory of Theorem A' (see Appendix B). It does not say the record is uninformative about the gain in a statistical model: under random illumination, the *relative* gain is consistently estimable (Theorem B; see Appendix C), a different — statistical, asymptotic — notion of identifiability which Theorem A neither contradicts nor implies. Finally, Theorem A is about uniqueness, not conditioning: it makes no claim about how well- or ill-posed any estimator is.

C. A.2 Corollary 1 (slowness does not help)

Corollary 1. Assume (A1)–(A5).

(i) If the admissible log-gain class is the linear space S itself, the entire ambiguity family of Theorem A lies inside the class: $\ell \in S$, $s \in S \Rightarrow \ell + s \in S$. In particular, choosing $s = \bar{\ell}1 - \ell \in S$ produces a **constant-gain** exact explanation of the record with the static object $T' = M^{-1}(e^{\bar{\ell}-\ell} \odot MT)$: a slowly drifting gain over the true object is indistinguishable from no drift at all over a relabelled object.

(ii) If the admissible class is a slowness ball $\mathcal{B}_L = \{\ell \in S : \|\ell\|_* \leq L\}$ for any seminorm $\|\cdot\|_*$ vanishing on constants (e.g. a Hölder or low-pass-energy seminorm), and ℓ is interior ($\|\ell\|_* < L$), then every nonconstant $s \in S$ with $\|s\|_*$ small enough keeps $\ell + s \in \mathcal{B}_L$. The local non-gauge ambiguity dimension remains $p - 1$: slowness bounds the *size* of the ambiguity, never its *dimension*.

Proof. (i) is linearity of S plus Theorem A applied with the stated s ; nonconstancy of ℓ on $\text{supp}(c)$ makes the constant-gain alternative distinct modulo gauge. (ii) Subadditivity of the seminorm gives $\|\ell + s\|_* \leq \|\ell\|_* + \|s\|_* \leq L$ once $\|s\|_* \leq L - \|\ell\|_*$; the set of such s is a neighborhood of 0 in S and Theorem A applies to each nonconstant member. ■

a) *What this does not claim.*: Slowness is not useless in general — in the tall-design regime the whole point is that slow drift means small p , making the threshold $N \geq K + p$ cheap to satisfy (Sec. 4). The corollary only asserts that at $N = K$ a slowness prior *by itself* removes nothing from the ambiguity, because the ambiguity directions live in the slow class itself.

D. A.3 Corollary 2 (nonnegativity is insufficient; known support zeros can restore local identifiability)

Throughout this corollary write H for the square invertible design ($H = M$; the letter matches the fixed-basis usage, e.g. Hadamard), and define the gain-absorption operator

$$M_s := H^{-1} \text{diag}(e^{-s})H, \quad T'(s) = M_s T,$$

so that the ambiguity set at T under an object prior class $\mathcal{T}$ is $V_{\mathcal{T}}(T) = \{s \in S : M_s T \in \mathcal{T}\}$, and identifiability modulo gauge is the assertion $V_{\mathcal{T}}(T) = S \cap \text{span}\{\mathbf{1}\}$.

Corollary 2. Assume (A1)–(A5).

(i) *(Positivity insufficient.)* Let $\mathcal{T}_+ = \{T : T_j \geq 0 \forall j\}$ and let T have full support, $T_j > 0$ for all j . Then there is $r > 0$ such that every $s \in S$ with $\|s\|_{\infty} < r$ satisfies $M_s T > 0$; explicitly, it suffices that

$$\|H^{-1}\|_2 \|HT\|_2 (e^r - 1) < \min_j T_j.$$

Hence $V_{\mathcal{T}_+}(T)$ contains a full neighborhood of 0 in S and the local non-gauge ambiguity dimension is still $p - 1$.

(ii) *(Known support zeros; differential criterion and ordinary-local sufficiency.)* Let $\Omega \subset \{1, \dots, K\}$, $|\Omega| = q$, and $\mathcal{T}_{\Omega} = \{T : T_{\Omega^c} = 0, T \geq 0\}$, with the true $T \in \mathcal{T}_{\Omega}$, $T_{\Omega} > 0$. The constrained model is locally differentially identifiable modulo gauge at T if and only if

$$\ker(P_{\Omega^c} H^{-1} \text{diag}(HT)|_S) = S \cap \text{span}\{\mathbf{1}\},$$

where P_{Ω^c} is coordinate projection onto Ω^c . The gauge direction always lies in the kernel. Equality of the kernel with the gauge is also a sufficient condition for ordinary local identifiability: with γ the smallest singular value of the linearized map on a complement $S_{\perp}$ of $\text{span}\{\mathbf{1}\}$ in S and C_H the second-order remainder constant below, every s with

$$0 < \|s_{\perp}\|_2 < \min\{1, \gamma/C_H\}$$

violates the support constraint. The exact necessary-and-sufficient condition for ordinary local identifiability is instead that 0 be an isolated zero of the gauge-fixed nonlinear map $F|_{S_{\perp}}$ defined below; derivative injectivity is sufficient for this isolation but is not necessary.

(iii) *(Rank counts — dimension-counting arguments, labeled as such.)* The linearized map in (ii) sends the $(p - 1)$ -dimensional quotient $S/\text{span}\{\mathbf{1}\}$ into $\mathbb{R}^{K-q}$. Thus $K_0 :=$

$K - q \geq p - 1$ is necessary for the full-rank differential criterion. It is not a necessary condition for ordinary local identifiability, because a rank-deficient analytic zero may still be isolated, and the count alone is not sufficient without an explicit full-rank witness. Conditional on such a witness, full column rank is generic in (H, T_{Ω}) . For unknown support of size $\leq q$, $K \geq 2q + (p - 1)$ remains only a dimension-counting prediction, up to endpoint conventions.

Proof. (i) $M_0 T = T$ and $\|M_s T - T\|_2 = \|H^{-1}((e^{-s} - \mathbf{1}) \odot HT)\|_2 \leq \|H^{-1}\|_2 \|HT\|_2 \max_n |e^{-s_n} - 1| \leq \|H^{-1}\|_2 \|HT\|_2 (e^{\|s\|_{\infty}} - 1)$. If this is below $\min_j T_j$ then $M_s T > 0$ entrywise. Since the positive orthant is open and $s \mapsto M_s T$ is continuous, positivity excludes nothing locally; Theorem A supplies the distinctness. ■ (i)

(ii) Define the support-constraint map $F : S \rightarrow \mathbb{R}^{K-q}$, $F(s) = P_{\Omega^c} M_s T$. It is real-analytic, $F(0) = P_{\Omega^c} T = 0$, and its differential at 0 is

$$DF_0 u = -P_{\Omega^c} H^{-1} \text{diag}(HT) u,$$

by differentiating $e^{-tu} = \mathbf{1} - tu + O(t^2)$ inside M_{tu} . The gauge is always in the kernel: $H^{-1} \text{diag}(HT)\mathbf{1} = H^{-1}(HT) = T$ and $P_{\Omega^c} T = 0$; consistently, $F(t\mathbf{1}) = e^{-t} F(0) = 0$ exactly, for all t .

Differential criterion. If the kernel strictly exceeds the gauge, some nonconstant $u \in S$ satisfies the linearized support constraint, so differential identifiability on the gauge quotient fails. Conversely, if the kernel is exactly the gauge, $DF_0|_{S_{\perp}}$ is injective, hence the gauge-fixed constrained map is an immersion at 0. A null derivative need not integrate to an exact ambiguity curve: an analytic zero can be isolated even when its derivative is singular. This is why the displayed kernel equality is an iff only for differential identifiability.

Sufficiency in a ball. Suppose the kernel equals the gauge and let $\gamma = \sigma_{\min}(DF_0|_{S_{\perp}}) > 0$. Entrywise Taylor with remainder, $|e^{-x} - 1 + x| \leq \frac{x^2}{2} e^{|x|}$, gives for $\|s\|_{\infty} \leq 1$

$$\|F(s) - DF_0 s\|_2 \leq C_H \|s\|_2^2,$$

$$C_H := \frac{e}{2} \|H^{-1}\|_2 \|HT\|_2.$$

For gauge-fixed $s = s_{\perp} \in S_{\perp}$ (any s may be reduced to this form using $F(s + t\mathbf{1}) = e^{-t} F(s)$, which rescales but never annihilates F),

$$\|F(s_{\perp})\|_2 \geq \gamma \|s_{\perp}\|_2 - C_H \|s_{\perp}\|_2^2 > 0$$

$$\text{for } 0 < \|s_{\perp}\|_2 < \min\{1, \gamma/C_H\}.$$

A nonzero $F(s)$ means $M_s T \notin \mathcal{T}_{\Omega}$ (support constraint violated), so no non-gauge ambiguity exists in that ball. More generally, after gauge fixing, ordinary local identifiability holds exactly when $F^{-1}(0) \cap S_{\perp}$ contains no nonzero point in some neighborhood of 0, i.e. exactly when 0 is an isolated zero of $F|_{S_{\perp}}$. This proves both the quantitative sufficient condition and the exact nonlinear criterion. ■ (ii)

(iii) *These are rank/dimension arguments, not an ordinary-identifiability theorem.* A linear map from a $(p - 1)$ -dimensional space cannot be injective into $\mathbb{R}^{K-q}$ if $K - q < p - 1$, proving the stated necessity for differential full rank. When $K - q \geq p - 1$, clear denominators in the maximal minors of $P_{\Omega^c} H^{-1} \text{diag}(HT)|_{S_{\perp}}$: if one admissible pair (H, T_{Ω})

has a nonzero maximal minor, that minor is not identically zero, so full column rank holds off a proper algebraic set. The dimension count does not supply that witness, and a fixed named basis must be checked directly. Neither a failed rank count nor a singular derivative rules out higher-order zero isolation. The unknown-support count $K \geq 2q + (p - 1)$ arises from demanding that the q -sparse model avoid all of its gain-transformed copies $M_s \Sigma_q$, in analogy with the generic sparsity thresholds of Kech–Krahmer [13]; it is a heuristic prediction, unverified for structured designs, and the uniform no-support-switching condition it summarizes is strictly stronger than (ii). ■ (heuristic scope)

a) *What this does not claim.*: Part (i) says nonnegativity fails to restore *identifiability*; it may still improve estimator conditioning and is routinely useful as regularization — no claim is made either way. Part (ii) is local and fixed-support: derivative full rank is a checkable sufficient condition for ordinary local identifiability, whereas zero isolation is the exact nonlinear condition. Neither excludes isolated global aliases. Part (iii) concerns differential rank only; it supplies neither a necessary zero count for ordinary local identifiability nor a witness for any fixed basis. For unknown support only its heuristic count is offered. These qualifications apply only to Corollary 2; they do not alter the generic ordinary-local iff or the exact below-wall alias manifold of Theorem A'(i) (Appendix B).

E. A.4 Remarks

a) *Remark A.1 (Fisher-theoretic reading).*: Because the ambiguity of Theorem A is an explicit real-analytic curve $t \mapsto (\ell + ts, T'(ts))$ along which the noiseless data are constant, every nonconstant $s \in S / \text{span}\{\mathbf{1}\}$ yields an exact null direction of the (gauge-quotient) Fisher information in any regular dominated noise model built on this record: the Cramér–Rao bound is infinite for any functional that varies along the curve. Here the implication “explicit ambiguity curve $\Rightarrow$ Fisher null direction” is exact, not merely first-order.

b) *Remark A.2 (zero buckets).*: If some $c_n = (MT)_n = 0$, two separate pathologies arise, which the statements above deliberately quarantine via (A4): the gain values a_n at zero-carrier frames are entirely unobservable (a trivial extra ambiguity, of dimension $\dim(S \cap \{s : s = 0 \text{ on } \text{supp}(c)\})$ beyond the count of Theorem A), and log-domain processing of $R_n = 0$ is undefined, an estimation-side issue treated by the calibrated soft-log machinery of the low-photon analysis (Sec. 6; see Appendix D), not an identifiability issue.

APPENDIX B

TALL-DESIGN IDENTIFIABILITY: THE PARAMETRIC-TRANSVERSALITY PROOF

This appendix proves the tall-design identifiability results quoted as **Theorem A'** in Sec. 4, restated in full as Theorem A'(i)–(iv) in Sec. B.8: the generic local threshold with a genuine iff at $N = K + p - 1$ for $K \geq 2$, genuine (exact, positive-dimensional) failure below that wall, generic exact identifiability at $N \geq K + p$, uniform identifiability over the nonvanishing-carrier set at $N \geq 2K + p - 1$, and the

degenerate $K = 1$ stratum. The proof route is **parametric transversality** (a rowwise design-perturbation witness plus Fubini), verified adversarially in review round R11; it holds for **every fixed drift subspace** $S \ni \mathbf{1}$ — no genericity of S is assumed anywhere, so the physical Fourier low-pass space S_{LP} is covered directly, with no transfer lemmas. Sharpness (necessity) of the two global thresholds is **not** claimed as a theorem; its status is stated precisely after Theorem A'. This appendix **supersedes** the earlier incidence-variety/stratum-counting route (see Sec. B.9); the superseded “sharp iff” statements at $N = K + p$ and $N = 2K + p - 1$ are withdrawn.

A. B.0 Setting, notation, and the reduction

The bucket record is $R_n = a_n B_n$ with $B = MT$: in vector form the data map is

$$\Phi_M(T, s) = D_{e^s} M T, \quad M \in \mathbb{R}^{N \times K}, \quad N \geq K, \\ s \in S, \quad T \in \mathbb{R}^K,$$

where $D_x = \text{diag}(x)$, e^s is componentwise, and the log-gain trajectory is confined to the fixed, known drift class $S \subset \mathbb{R}^N$ with $\dim S = p \geq 2$ and $\mathbf{1} \in S$. Write $y = MT$ for the carrier and define

$$\mathcal{U} = \{(M, T) : (MT)_n \neq 0 \ \forall n\}, \\ \mathcal{U}_M = \{T \in \mathbb{R}^K : (MT)_n \neq 0 \ \forall n\}, \\ \mathcal{R} = \{M \in \mathbb{R}^{N \times K} : \text{rank } M < K\}.$$

$\mathcal{U}$ is open with null complement (each $\{m_n^\top T = 0\}$ is a proper algebraic subset); $\mathcal{R}$ is a proper algebraic subset (vanishing of all $K \times K$ minors), hence Lebesgue-null. **Every exceptional set below explicitly includes $\mathcal{R}$ (or $\mathcal{R} \times \mathbb{R}^K$).** The gauge is $(T, s) \sim (cT, s - \log c \mathbf{1})$, $c > 0$; identifiability is always modulo this gauge. “For a.e.” always refers to Lebesgue measure on the stated parameter space.

Lemma B.0 (reduction). *Assume $\text{rank } M = K$ and $(M, T) \in \mathcal{U}$, and fix any $s \in S$. Then $(T', s') \in \mathbb{R}^K \times S$ is an alias of (T, s) — i.e. $\Phi_M(T', s') = \Phi_M(T, s)$ — iff, with $d := s - s' \in S$,*

$$MT' = D_{e^d} MT. \quad (\text{B.1})$$

If $d = a\mathbf{1}$ is constant, then (B.1) reads $MT' = e^a MT$ and full column rank gives $T' = e^a T$ — exactly the gauge orbit with $c = e^a$; conversely every gauge point solves (B.1) with constant d . Hence (M, T) is identifiable iff (B.1) has no solution (T', d) with $d \in S \setminus \mathbb{R}\mathbf{1}$. Moreover, fixing a complement S_0 of $\mathbb{R}\mathbf{1}$ in S ($\dim S_0 = p - 1$) and writing $d = d_0 + a\mathbf{1}$, the normalization $(T', d) \mapsto (e^{-a}T', d_0)$ maps solutions to solutions; so (M, T) is identifiable iff (B.1) has no solution with $d \in S_0 \setminus \{0\}$.

Proof. D_{e^s} is invertible, so record equality is equivalent to (B.1) with $d = s - s'$. Constant $d = a\mathbf{1}$: row n reads $m_n^\top T' = e^a y_n$, i.e. $MT' = e^a MT$, and $\text{rank } M = K$ forces $T' = e^a T$, which is the gauge point $(cT, s - \log c \mathbf{1})$ with $c = e^a > 0$. The equivariance $(T', d) \mapsto (e^a T', d + a\mathbf{1})$ is a direct substitution in (B.1). ■

Two structural remarks. (a) *The reduction eliminates s* : identifiability is a property of (M, T, S) alone, uniform over the true drift — this is the source of the “for every $s \in S$ simultaneously” quantifier in every theorem below. (b) *Full column rank is necessary*: for rank-deficient M the equivalence “constant $d \Leftrightarrow$ gauge-trivial” is **false** even on $\mathcal{U}$. Counterexample (rank $M = 1$): $N = K = 2$, both rows of M equal to $(0, 1)$, $T = (0, 1)^\top$, $T' = (1, 1)^\top$: $MT = MT' = (1, 1)^\top$, so $(T', d = 0)$ solves (B.1) with constant d , yet $T' \notin \mathbb{R}T$ — an object-kernel ambiguity invisible to the normalized $d \neq 0$ incidence set. This is why $\mathcal{R}$ is carried explicitly in every exceptional set.

a) *What this does not claim.*: The reduction is exact but presupposes rank $M = K$ and a nonvanishing carrier; neither is a restriction for the a.e. statements (both failure sets are null), but both must be — and are — carried explicitly.

B. B.1 Lemma B.1 (row dichotomy)

Lemma B.1. *Let $(M, T) \in \mathcal{U}$ and let (T', d) solve (B.1) with d nonconstant ($d \notin \mathbb{R}1$). Then T' is not a scalar multiple of T ; consequently $T' - e^{d_n}T \neq 0$ for every $n = 1, \dots, N$.*

Proof. Suppose $T' = \beta T$. Row n of (B.1): $\beta y_n = e^{d_n} y_n$; since $y_n \neq 0$ on $\mathcal{U}$, $e^{d_n} = \beta$ for all n , so d is constant — contradiction. For the display: if $T' = e^{d_n}T$ for a single n , then $T' \parallel T$ ($T \neq 0$ because $y = MT \neq 0$). ■

Only *global* nonconstancy of d is used — partial constancy (some equal components) is harmless — and the only division is by y_n , legitimate on $\mathcal{U}$.

C. B.2 Lemma B.2 (joint submersivity)

Define $F : \mathbb{R}^{N \times K} \times \mathbb{R}^K \times \mathbb{R}^K \times S_0 \rightarrow \mathbb{R}^N$,

$$\begin{aligned} F(M, T, T', d) &= MT' - D_{e^d}MT, \\ F_n &= m_n^\top (T' - e^{d_n}T). \end{aligned} \quad (\text{B.2})$$

Lemma B.2. *At every zero of F with $(M, T) \in \mathcal{U}$ and $d \in S_0 \setminus \{0\}$, the differential of F with respect to M alone is surjective onto $\mathbb{R}^N$.*

Proof. Perturb row n only: $\delta M = e_n \delta m_n^\top$ gives $\delta F = e_n \cdot \delta m_n^\top (T' - e^{d_n}T)$, which spans $\mathbb{R}e_n$ as δm_n ranges over $\mathbb{R}^K$, provided $T' - e^{d_n}T \neq 0$ — which holds at every such zero for every n by Lemma B.1. Rows are decoupled, so the image contains every e_n . ■

Only δM is used: no genericity of S , no structure of d beyond nonconstancy, and the object is untouched. This rowwise witness is uniform over the whole zero set — the ingredient the superseded route lacked.

D. B.3 A measure lemma, and generic exact identifiability at $N \geq K + p$

Lemma B.3 (C^1 image of a lower-dimensional manifold is null). *Let X be a second-countable C^1 manifold with $\dim X = m < q$, and $f : X \rightarrow \mathbb{R}^q$ a C^1 map. Then $f(X)$ has q -dimensional Lebesgue measure zero.*

Proof. Second countability gives a countable chart cover $\varphi_i : U_i \rightarrow \mathbb{R}^m$, and each chart domain is exhausted by countably

many compact cubes; by countable subadditivity it suffices to treat one compact cube Q of side ℓ . On Q , $g = f \circ \varphi_i^{-1}$ is C^1 , hence Lipschitz with some constant L . Subdividing Q into k^m subcubes of side ℓ/k , each image lies in a ball of radius $L\sqrt{m}\ell/k$ in $\mathbb{R}^q$, so the outer measure of $g(Q)$ is at most $k^m c_q (2L\sqrt{m}\ell/k)^q = C k^{m-q} \rightarrow 0$ since $m < q$. ■

No properness or compactness of f is needed; the zero manifolds below are embedded in Euclidean space, hence second countable and σ -compact automatically. The statement is **false** for merely continuous maps (space-filling curves) — the C^1 regularity supplied by Lemma B.2 is load-bearing.

Theorem B.4 (generic exact identifiability; Theorem A'(ii)).

Fix $S \ni 1$, $\dim S = p \geq 2$. If $N \geq K + p$, there is a Lebesgue-null set $E_{\text{gen}} \subset \mathbb{R}^{N \times K} \times \mathbb{R}^K$ such that every $(M, T) \notin E_{\text{gen}}$ is identifiable: for every $s \in S$, the only $(T', s') \in \mathbb{R}^K \times S$ with $\Phi_M(T', s') = \Phi_M(T, s)$ are the gauge orbit of (T, s) .

Proof. Work on the open set $\Omega = \{(M, T, T', d) : (M, T) \in \mathcal{U}, T' \in \mathbb{R}^K, d \in S_0 \setminus \{0\}\}$. By Lemma B.2, 0 is a regular value of F on Ω , so $Z = F^{-1}(0) \cap \Omega$ is an embedded real-analytic submanifold of codimension N :

$$\dim Z = (NK + K) + K + (p - 1) - N.$$

The projection $\pi : Z \rightarrow \mathbb{R}^{N \times K} \times \mathbb{R}^K$ onto (M, T) is C^1 and

$$\dim Z - \dim(\mathbb{R}^{N \times K} \times \mathbb{R}^K) = K + p - 1 - N \leq -1$$

when $N \geq K + p$, so $\pi(Z)$ is Lebesgue-null by Lemma B.3. By the reduction (Lemma B.0, which requires rank $M = K$),

$$\{\text{non-identifiable pairs in } \mathcal{U}\} \subset (\mathcal{R} \times \mathbb{R}^K) \cup \pi(Z),$$

and the full non-identifiable set is contained in $(\mathcal{R} \times \mathbb{R}^K) \cup \pi(Z) \cup \mathcal{U}^c$ — three null sets. Set E_{gen} to their union. ■

a) *Remark B.4.1 (quantifier order, proved).*: For a.e. (M, T) : for **every** $s \in S$, no non-gauge alias — the null set E_{gen} is independent of s , because the reduction eliminated s before any measure-theoretic step. This does **not** imply that one design works for every object; that is Theorem B.5 at its own larger threshold. ($T' = 0$ needs no special handling: $T' = 0$ forces $D_{e^d}y = 0$, impossible on $\mathcal{U}$.)

b) *Remark B.4.2 (fixed-object slice).*: For each **fixed** $T \neq 0$: a.e. M identifies that particular T when $N \geq K + p$ (the exceptional design set may depend on T). Proof: freeze T , run the same argument with $F_T(M, T', d) = MT' - D_{e^d}MT$ on $\{M : (M, T) \in \mathcal{U}\} \times \mathbb{R}^K \times (S_0 \setminus \{0\})$; Lemma B.2's witness uses only δM , the zero set has dimension $NK + K + (p - 1) - N \leq NK - 1$ when $N \geq K + p$, and Lemma B.3 applies; add $\mathcal{R}$ and the proper algebraic set $\{M : (M, T) \notin \mathcal{U}\}$. This slice form is what covers the fixed structured objects of the Fig. 6 protocol.

c) *What this does not claim.*: No claim of uniformity over objects for a fixed design (see Theorem B.5 and its counterexample); no claim at the boundary $N = K + p - 1$ (necessity is open — Sec. B.8, sharpness paragraph); nothing about conditioning, noise, or estimators; deterministic structured designs are not covered by any a.e. statement.

E. B.4 Uniform identifiability on the nonvanishing-carrier set at $N \geq 2K + p - 1$

Theorem B.5 (uniform sufficiency on $\mathcal{U}_M$; Theorem A'(iii)). Fix $S \ni \mathbf{1}$, $\dim S = p \geq 2$. If $N \geq 2K + p - 1$, there is a Lebesgue-null set $E_{\text{unif}} \subset \mathbb{R}^{N \times K}$ such that for every $M \notin E_{\text{unif}}$: every $T \in \mathcal{U}_M$ is identifiable, for every $s \in S$ simultaneously.

Proof. The solution set of (B.1) is scaling-equivariant — (T, T', d) solves iff (cT, cT', d) does, $c \neq 0$ — and nonvanishing carrier and nonconstancy are scale-invariant, so it suffices to rule out solutions with $|T| = 1$. Let

$$\Omega' = \{(M, T, T', d) : |T| = 1, (M, T) \in \mathcal{U}, T' \in \mathbb{R}^K, d \in S_0 \setminus \{0\}\},$$

an open subset of $\mathbb{R}^{N \times K} \times S^{K-1} \times \mathbb{R}^K \times S_0$. Lemma B.2 holds verbatim on Ω' (the sphere constrains T , not M), so $Z' = F^{-1}(0) \cap \Omega'$ is an embedded real-analytic manifold of codimension N with

$$\begin{aligned} \dim Z' &= NK + (K - 1) + K + (p - 1) - N \\ &= NK + (2K + p - 2 - N), \end{aligned}$$

and the projection $\pi' : Z' \rightarrow \mathbb{R}^{N \times K}$ onto M has deficit $2K + p - 2 - N \leq -1$ when $N \geq 2K + p - 1$; $\pi'(Z')$ is null by Lemma B.3. Set $E_{\text{unif}} = \mathcal{R} \cup \pi'(Z')$ (rank-deficient designs added explicitly: the reduction needs $\text{rank } M = K$). If $M \notin E_{\text{unif}}$ and (T', d) solved (B.1) for some $T \in \mathcal{U}_M, d \in S_0 \setminus \{0\}$, then rescaling by $c = 1/|T|$ would put $(M, cT, cT', d) \in Z'$, i.e. $M \in \pi'(Z')$ — contradiction. Lemma B.0 concludes. ■

a) Remark B.5.1 (why $\mathcal{U}_M$ cannot be dropped: a zero-carrier counterexample at the threshold): “Every nonzero object” is **false** even at $N = 2K + p - 1$. Take $K = 2, p = 2, N = 5 = 2K + p - 1, S = \text{span}\{\mathbf{1}, e_1\}, T = (1, 1)^\top$, and M with rows $(1, -1), (1, 0), (0, 1), (1, 1), (1, 2)$, so $MT = (0, 1, 1, 2, 3)^\top$ — full column rank, first carrier coordinate zero. For every $t \neq 0, d = te_1 \in S$ is nonconstant and $(T', d) = (T, te_1)$ solves (B.1): row 1 reads $m_1^\top T' = e^t \cdot 0 = 0$, rows 2–5 are unchanged. So (T, s) and $(T, s - te_1)$ produce identical records for every t — a one-parameter family of non-gauge aliases. This mechanism is available for **any** design: every object on the hyperplane $\{m_1^\top T = 0\}$ acquires such aliases whenever S contains a vector supported on the zero-carrier row. The correct conclusion is the stated one: for a.e. M , every $T \in \mathcal{U}_M$, every $s \in S$.

b) What this does not claim: Uniformity is over nonvanishing-carrier objects and drifts, not over designs: an adversarially chosen M can defeat the bound. Necessity of $2K + p - 1$ is not claimed (Sec. B.8). No claim about recovery algorithms or noise.

F. B.5 Lemma W: position of the twisted column space

For $(M, T) \in \mathcal{U}$ with $\text{rank } M = K$ define

$$W(M, T) := D_y^{-1} \text{col}(M) \subset \mathbb{R}^N, \quad y = MT,$$

a K -dimensional subspace with $\mathbf{1} \in W$ (since $D_y \mathbf{1} = y \in \text{col}(M)$). Aliases relate to W by: $d \in S$ solves (B.1) for some T' iff $e^d \in W(M, T)$ — indeed $D_{e^d} y = D_y e^d$, so (B.1)

reads $MT' = D_y e^d$, i.e. $e^d = D_y^{-1} MT' \in W$; conversely $e^d = D_y^{-1} M u$ gives the (unique, by full column rank) solution $T' = u$.

Lemma W. Fix any subspace $S \ni \mathbf{1}$ with $\dim S = p \leq N$, and $K \geq 1$. For a.e. $(M, T) \in \mathcal{U}$, with $W = W(M, T)$:

$$\begin{aligned} \dim(S \cap W) &= 1 + \max\{0, (p - 1) + (K - 1) - (N - 1)\} \\ &= 1 + \max\{0, K + p - N - 1\}. \end{aligned} \tag{B.3}$$

No genericity of S is assumed or used — only $\mathbf{1} \in S$.

Proof. (Step 1: basis completion; fixed T .) Fix $T \neq 0$ and complete it to a basis $B_T = [T, b_2, \dots, b_K] \in GL_K$. The map $M \mapsto MB_T = (y, v_2, \dots, v_K)$ is a linear isomorphism of $\mathbb{R}^{N \times K}$, hence maps null sets to null sets both ways: a.e. $M \Leftrightarrow$ a.e. $(y, v_2, \dots, v_K)$.

(Step 2: fixed- y isomorphism.) For fixed $y \in (\mathbb{R}^*)^N$ (a co-null condition), set $w_j := D_y^{-1} v_j$. For fixed y this is a linear isomorphism of $\mathbb{R}^{N(K-1)}$ in $(v_2, \dots, v_K)$ — it preserves “a.e.” and introduces no correlation between the w_j and the fixed S . Then $W = D_y^{-1} \text{span}(y, v_2, \dots, v_K) = \text{span}(\mathbf{1}, w_2, \dots, w_K)$.

(Step 3: quotient.) Let $q : \mathbb{R}^N \rightarrow \mathbb{R}^N / \mathbb{R}\mathbf{1}$ (dimension $N - 1$). $A := q(S)$ has dimension $p - 1$ (the kernel $\mathbb{R}\mathbf{1}$ lies in S); $q(W) = \text{span}(qw_2, \dots, qw_K)$.

(Step 4: general position against the fixed A .) For a.e. $(w_2, \dots, w_K)$: the $q(w_j)$ are independent and $\dim(A \cap q(W)) = \max\{0, (p - 1) + (K - 1) - (N - 1)\}$. Failure is the vanishing of suitable maximal minors of the matrix $[A\text{-basis} \mid qw_2 \dots qw_K]$ — a polynomial condition in the w_j , not identically zero (a witness exists: against one fixed subspace, vectors can be chosen stepwise outside a finite union of proper subspaces), hence a proper algebraic — null — subset. This is general position with respect to the fixed A only; no genericity of S .

(Step 5: $q(S \cap W) = q(S) \cap q(W)$, via $\mathbf{1} \in W$.) “ $\subseteq$ ” is trivial. “ $\supseteq$ ”: if $x \in S, x' \in W$ and $q(x) = q(x')$, then $x - x' \in \mathbb{R}\mathbf{1} \subseteq W$, so $x \in W$, i.e. $x \in S \cap W$. The kernel of q on $S \cap W$ is exactly $\mathbb{R}\mathbf{1}$ (as $\mathbf{1} \in S \cap W$), whence $\dim(S \cap W) = 1 + \dim(q(S) \cap q(W))$.

(Step 6: Fubini twice, with Borel measurability.) The quantifier order so far: for every fixed $T \neq 0$, for a.e. y , for a.e. (v_j) — hence, by Steps 1–2 and Fubini on $\mathbb{R}^N \times \mathbb{R}^{N(K-1)}$, for every fixed $T \neq 0$, for a.e. M , formula (B.3) holds. The failure set $B = \{(M, T) \in \mathcal{U} : (B.3) \text{ fails}\}$ is Borel: the dimension condition is a Boolean combination of rank conditions (minor vanishing/nonvanishing) on matrices whose entries are rational in (M, T) on $\mathcal{U}$ (the w_j are entries of M divided by coordinates of MT). Measurability licenses a second Fubini, over T : every T -slice of B is M -null, so B is null in $\mathbb{R}^{N \times K} \times \mathbb{R}^K$. ■

a) Remark W.1 ($K = 1$): For $K = 1$ the lemma **holds**: $W = \mathbb{R}\mathbf{1}$ identically, $\dim(S \cap W) = 1$, and (B.3) also gives $1 + \max\{0, p - N\} = 1$ since $p \leq N$. What degenerates at $K = 1$ is the necessity of the global thresholds (Sec. B.7), not Lemma W.

b) What this does not claim: (B.3) is an a.e. statement over (M, T) ; it says nothing about any *particular* design (e.g.

structured M), and nothing about the conditioning of the intersection, only its dimension.

G. B.6 The generic local threshold ($K \geq 2$) and genuine failure below the wall

a) Kernel derivation (independent; no import from the superseded route).: Fix $(M, T) \in \mathcal{U}$ with $\text{rank } M = K$ and any $s \in S$. The differential of Φ_M at (T, s) is

$$D\Phi_M(T, s)[\delta T, \delta s] = D_{e^s}(M \delta T + D_y \delta s), \quad y = MT. \quad (\text{B.4})$$

The diagonal factor is invertible, so $(\delta T, \delta s) \in \ker \Leftrightarrow M \delta T = -D_y \delta s$; the right side lies in $\text{col}(M)$ iff $\delta s \in D_y^{-1} \text{col}(M) = W(M, T)$, and then $\delta T = -(M^\top M)^{-1} M^\top D_y \delta s$ is unique. Hence $\ker D\Phi_M(T, s) \cong S \cap W$ via $(\delta T, \delta s) \mapsto \delta s$, and

$$\begin{aligned} \dim \ker D\Phi_M(T, s) &= \dim(S \cap W), \\ \text{rank } D\Phi_M(T, s) &= K + p - \dim(S \cap W), \end{aligned}$$

independently of s . The gauge tangent is the line spanned by $(-T, 1)$ (from $c \mapsto (cT, s - \log c \mathbf{1})$), and it lies in the kernel since $MT = D_y \mathbf{1}$; as $\mathbf{1} \in S \cap W$ always, the maximal gauge-compatible rank is $K + p - 1$ and the defect from it is $\dim(S \cap W) - 1$. This derivation is self-contained: nothing is imported from the superseded route, and no genericity of S is used. By Lemma W, for a.e. $(M, T) \in \mathcal{U}$ and every s :

$$\text{rank } D\Phi_M(T, s) = \min\{N, K + p - 1\}.$$

Theorem B.6 (generic local iff; Theorem A'(i), rank part). Fix $S \ni \mathbf{1}$, $\dim S = p \geq 2$, and $K \geq 2$. There is a Lebesgue-null set $E_{\text{loc}} \subset \mathbb{R}^{N \times K} \times \mathbb{R}^K$ — take $E_{\text{loc}} = (\mathcal{R} \times \mathbb{R}^K) \cup \mathcal{U}^c \cup B$ with B the failure set of Lemma W — such that for every $(M, T) \notin E_{\text{loc}}$ with $T \in \mathcal{U}_M$ and every $s \in S$: $\text{rank } D\Phi_M(T, s) = \min\{N, K + p - 1\}$, and (T, s) is locally identifiable modulo gauge **iff** $N \geq K + p - 1$.

Proof. The rank formula is the kernel derivation plus Lemma W. *Sufficiency at $N \geq K + p - 1$ (ordinary local injectivity modulo gauge):* write $S = \mathbb{R}\mathbf{1} \oplus S_0$; the bookkeeping

$$S \cap W = \mathbb{R}\mathbf{1} \oplus (S_0 \cap W)$$

holds because for $x \in S \cap W$, $x = a\mathbf{1} + x_0$ with $x_0 \in S_0$ gives $x_0 = x - a\mathbf{1} \in W$. At maximal rank $\dim(S \cap W) = 1$, so $S_0 \cap W = 0$. Restrict Φ_M to the gauge slice $\mathbb{R}^K \times (s + S_0)$, of dimension $K + p - 1$: by (B.4) its differential at (T, s) has kernel $\cong S_0 \cap W = 0$, so the restricted map is an immersion there, hence (constant-rank/immersion theorem) a local embedding — locally injective on the slice. Any nearby alias is gauge-normalized into the slice by a factor c near 1, so ordinary local identifiability modulo gauge holds. *Necessity at $N \leq K + p - 2$:* Theorem B.7 below supplies a positive-dimensional manifold of exact aliases through (T, s) — genuine failure, not merely a rank diagnostic. ■

Theorem B.7 (genuine failure below the wall; Theorem A'(i), below-wall part). Fix $S \ni \mathbf{1}$, $K \geq 2$, $N \leq K + p - 2$. For every $(M, T) \notin E_{\text{loc}}$ with $T \in \mathcal{U}_M$ and every $s \in S$: through (T, s) passes a real-analytic manifold of exact aliases of dimension $K + p - N - 1 \geq 1$; every point with $d \neq 0$

is gauge-inequivalent to (T, s) , and the aliased objects satisfy $T'(d) \rightarrow T$ as $d \rightarrow 0$.

Proof. Let $P = \Pi_{W^\perp}$ be the orthogonal projection onto $W^\perp$ ($\dim W^\perp = N - K$ by full column rank) and define the real-analytic map $\varphi : S_0 \rightarrow W^\perp$, $\varphi(d) = P(e^d)$. Then $e^d \in W \Leftrightarrow \varphi(d) = 0$ and $\varphi(0) = P\mathbf{1} = 0$. The differential is $D\varphi_0 = P|_{S_0} (\frac{d}{dt}|_0 e^{td} = d$ componentwise; P linear). By the bookkeeping $S \cap W = \mathbb{R}\mathbf{1} \oplus (S_0 \cap W)$ and Lemma W with $N \leq K + p - 2$ (the max in (B.3) is attained), $\dim(S_0 \cap W) = K + p - N - 1$, so

$$\text{rank } D\varphi_0 = (p - 1) - (K + p - N - 1) = N - K = \dim W^\perp :$$

$D\varphi_0$ is surjective. By the (real-analytic) submersion/implicit-function theorem, $\varphi^{-1}(0)$ is, near 0, a real-analytic manifold of dimension $(p - 1) - (N - K) = K + p - N - 1 \geq 1$. For each d in it, $e^d \in W$; with the left inverse $L = (M^\top M)^{-1} M^\top$ set

$$T'(d) := L D_y e^d,$$

so that $MT'(d) = D_y e^d$ ($D_y e^d = D_{e^d} y \in \text{col}(M)$ precisely when $e^d \in W$): $(T'(d), d)$ solves (B.1), $T'(0) = Ly = T$, and $T'(d) \rightarrow T$ by continuity. Every $d \neq 0$ lies in $S_0 \setminus \{0\}$, hence $d \notin \mathbb{R}\mathbf{1}$: a genuine non-gauge alias — (T, s) and $(T'(d), s - d)$ produce identical records but are gauge-inequivalent. ■

b) Consistency check at $N = K$: At $N = K$ (square, $\text{rank } M = K$): $W = \mathbb{R}^N$, $W^\perp = 0$, $\varphi \equiv 0$, and the alias manifold is a full neighborhood of 0 in S_0 — ambiguity dimension $p - 1$, exactly Theorem A. There is no hidden $N \geq K + 1$ assumption, and no contradiction ($p \geq 2$ gives $K < K + p \leq 2K + p - 1$, so the square case lies strictly below both global thresholds).

c) What this does not claim: The local iff is generic in (M, T) — a particular structured design must be checked separately. Local identifiability at $N \geq K + p - 1$ does not imply global uniqueness (far aliases at the boundary are exactly the open sharpness question, Sec. B.8). Nothing about the singular values of the Jacobian, i.e. conditioning.

H. B.7 The $K = 1$ stratum

Theorem B.8 ($K = 1$: global identifiability; Theorem A'(iv)). Let $K = 1$. For every M , every $T \in \mathcal{U}_M$, and every $s \in S$, the pair (T, s) is globally identifiable modulo gauge, for every admissible N .

Proof. $T \in \mathcal{U}_M$ means $T \neq 0$ and $m_n \neq 0$ for all n (scalars $m_n T \neq 0$). Any solution of (B.1) satisfies $e^{d_n} = m_n T' / (m_n T) = T' / T$ for every n , so d is constant and $c := T' / T = e^{d_1} > 0$ — the gauge. ■

The local threshold at $K = 1$ reads $N \geq K + p - 1 = p$, which is **automatically satisfied** ($S \subseteq \mathbb{R}^N$ forces $p \leq N$) — the threshold is vacuous, not “inapplicable”. Consequently neither $N \geq K + p$ nor $N \geq 2K + p - 1$ is necessary at $K = 1$: corner strata break naive universal-necessity claims, which is why all necessity statements in this appendix are restricted to $K \geq 2$.

I. B.8 Theorem A' — the manuscript statement, and sharpness status

Theorem A' (proved tall-design identifiability results). *Let $K \geq 1$, $N \geq K$, and let $S \leq \mathbb{R}^N$ be a fixed, known linear subspace of dimension $p \geq 2$ with $\mathbf{1} \in S$. For $M \in \mathbb{R}^{N \times K}$ define $\Phi_M(T, s) = D_{e^s}MT$ and $\mathcal{U}_M = \{T \in \mathbb{R}^K : (MT)_n \neq 0 \text{ for every } n\}$. Parameters are identified modulo the positive global-scale gauge $(T, s) \sim (cT, s - \log c \mathbf{1})$, $c > 0$.*

(i) (Generic local threshold and genuine failure below it, $K \geq 2$.) There is a Lebesgue-null set $E_{\text{loc}} \subset \mathbb{R}^{N \times K} \times \mathbb{R}^K$ such that for every $(M, T) \notin E_{\text{loc}}$ with $T \in \mathcal{U}_M$ and every $s \in S$: $\text{rank } D\Phi_M(T, s) = \min\{N, K + p - 1\}$. Consequently (T, s) is locally identifiable modulo gauge iff $N \geq K + p - 1$. If $N \leq K + p - 2$ then a gauge-normalized neighborhood of (T, s) contains a real-analytic manifold of exact aliases of dimension $K + p - N - 1 \geq 1$, and every other sufficiently nearby point of it is gauge-inequivalent.

*(ii) (Generic exact-identifiability sufficiency.) If $N \geq K + p$, there is a Lebesgue-null set $E_{\text{gen}} \subset \mathbb{R}^{N \times K} \times \mathbb{R}^K$ such that for every $(M, T) \notin E_{\text{gen}}$ and every $s \in S$: $D_{e^{s'}}MT' = D_{e^s}MT$ with $(T', s') \in \mathbb{R}^K \times S$ implies $T' = cT$, $s' = s - \log c \mathbf{1}$ for a unique $c > 0$. Exact identifiability holds simultaneously for every possible true drift $s \in S$. This statement is **not** uniform over all objects for one fixed M .*

(iii) (Uniform exact-identifiability sufficiency on the nonvanishing-carrier set.) If $N \geq 2K + p - 1$, there is a Lebesgue-null set $E_{\text{unif}} \subset \mathbb{R}^{N \times K}$ such that for every $M \notin E_{\text{unif}}$, every $T \in \mathcal{U}_M$, every $s \in S$, and every $(T', s') \in \mathbb{R}^K \times S$: $D_{e^{s'}}MT' = D_{e^s}MT$ implies $T' = cT$, $s' = s - \log c \mathbf{1}$ for a unique $c > 0$. The same generic design works simultaneously for all nonvanishing-carrier objects and all true drifts in S .

(iv) (Degenerate one-dimensional object stratum.) If $K = 1$ then for every M and every $T \in \mathcal{U}_M$ and every $s \in S$, the pair (T, s) is globally identifiable modulo gauge, for every admissible N . Neither $N \geq K + p$ nor $N \geq 2K + p - 1$ is necessary when $K = 1$.

Proof. (i) = Theorems B.6–B.7 (E_{loc} from Lemma W); (ii) = Theorem B.4; (iii) = Theorem B.5; (iv) = Theorem B.8 (which holds for every M , not just a.e.). ■

Sharpness status. Parts (ii) and (iii) are sufficient conditions, not proved if-and-only-if thresholds. For $K \geq 2$, the below-wall part of (i) proves generic exact failure whenever $N \leq K + p - 2$. The remaining necessity assertion for the $K + p$ bound is the boundary case $N = K + p - 1$: dimension counting predicts nonlocal aliases there, but dominance of the bad-pair projection has not been proved. Likewise, dimension counting predicts that a generic design fails uniform identifiability for some object when $N \leq 2K + p - 2$, but this necessity direction has not been proved in the intermediate regime. Thus $K + p$ and $2K + p - 1$ are conjecturally sharp for $K \geq 2$ under generic/nondegenerate drift geometry, not universal sharp thresholds for every fixed S . The $K = 1$ stratum explicitly rules out any universal necessity claim.

a) What this does not claim.: Necessity of the two global thresholds (see the sharpness paragraph); anything about deterministic structured designs (the ordered-Hadamard obstruction of Theorem A is unaffected, and no a.e. statement covers a fixed named design); anything about conditioning, noise robustness, or estimators — uniqueness at a threshold

is compatible with arbitrarily bad stability; and no coverage of objects constrained to lower-dimensional sets (e.g. exactly sparse) — open constraint sets (e.g. positivity with nonvanishing carrier) inherit all a.e. statements by restriction. The Fig. 6 experiment (Sec. 9) is evidence for the *local rank wall* of part (i) — with the fixed-object slice covered by Remark B.4.2 — and probes algorithmic recovery; it neither tests nor proves the sharpness predictions.

J. B.9 Relation to the superseded incidence-variety route

Earlier versions of this appendix proved Theorem A' through a collision incidence variety: a stratum-counting lemma over the gain-ratio manifold, a dominant-stratum rank hypothesis $\text{rank}[M, -D_b M] = \min(N, 2K)$, a determinantal-transversality argument for the rank-drop locus, and two low-pass lemmas (a level-multiplicity bound for S_{LP} and a generic-rank lemma for $[M, -DM]$) feeding a three-level transfer proposition. **That route is superseded (R11), and its claims are withdrawn or replaced as follows.**

- The former “sharp iff” statements at $N \geq K + p$ (generic) and $N \geq 2K + p - 1$ (uniform) are **withdrawn**: their necessity directions rested on unproved dominant-projection and determinantal-transversality assertions. The proved statements are the sufficiency directions, now established by Theorems B.4–B.5 above, together with the genuinely sharp *local* iff at $N = K + p - 1$ (Theorem B.6) and genuine below-wall failure (Theorem B.7). Sharpness of the two global thresholds is a labeled prediction (Sec. B.8).
- The former uniform claim quantified over “every nonzero object”; that is **false** at the stated threshold (Remark B.5.1) and is replaced by uniformity over the nonvanishing-carrier set $\mathcal{U}_M$.
- The former route needed genericity of the drift family (or the low-pass transfer lemmas to emulate it). The parametric-transversality proof holds for **every fixed** $S \ni \mathbf{1}$, so the low-pass transfer machinery (level-multiplicity bound, generic-rank lemma, three-level transfer hierarchy, and the L_S multiplicity check) is no longer needed for Theorem A' and is not restated here; the withdrawn material remains available in the archived drafts for provenance.
- The $K = 1$, $N = p = 2$ counterexample that broke the old claimed generic threshold is absorbed as the degenerate stratum, Theorem A'(iv).

a) What this section does not claim.: Nothing in the superseded route is relied upon anywhere in Secs. B.0–B.8; conversely, no assertion of the superseded route beyond the ones re-proved above should be quoted from earlier drafts.

APPENDIX C

CARRIER STATISTICS AND THE STATIONARITY ANCHOR

This appendix proves Proposition 1, formalizes condition (★) quantitatively, proves Proposition 2 (with counterexamples in both directions), and proves Theorem B. Throughout, $R_n = a_n B_n$ is the bucket record, $B_n = \langle I_n, T \rangle = \sum_{j=1}^K I_{n,j} T_j$ the carrier, $\ell_n = \log a_n$ the log-gain, S the drift class with $p = \dim S$, $S_1 = \sum_j T_j$, $S_2 = \|T\|_2^2$,

$K_{\text{eff}} = S_1^2/S_2$, $K_\infty = S_2/\|T\|_\infty^2$, $K_4 = S_2^2/\sum_j T_j^4$, and W denotes the window length of the blind estimator $\hat{g}_n = W^{-1} \sum_{k \in W_n} Y_k$ applied to $Y_n = \log R_n$. Asymptotic statements are for a triangular array indexed by (K, N) : the object $T = T^{(K)}$ and pattern law may vary with K , and limits are taken along $K, N \rightarrow \infty$.

A. C.1 Proposition 1 — random-basis carrier statistics

Proposition 1 (carrier moments, stationarity, CLT, Berry–Esseen). Assume:

- (P1) $T_j \geq 0$ for all j , $S_1 > 0$, $S_2 > 0$.
- (P2) The pattern entries $\{I_{n,j}\}$ are i.i.d. across n and j with mean $\mu_I > 0$ and variance $\sigma_I^2 > 0$; the entries are nonnegative (so $B_n \geq 0$ a.s., and $B_n > 0$ a.s. whenever $P(I = 0)^{|\text{supp } T|} = 0$).
- (P3) (for the CLT) Spikiness $K_\infty \rightarrow \infty$ along the array, with the family $\{(I - \mu_I)^2\}$ uniformly integrable (automatic for a fixed pattern law).
- (P4) (for Berry–Esseen) $\nu_3 := \mathbb{E}|I - \mu_I|^3/\sigma_I^3 < \infty$.

Then:

(i) **Exact moments and stationarity.** $\mathbb{E}B_n = \mu_I S_1$, $\text{Var } B_n = \sigma_I^2 S_2$, hence $\text{CV}_B = (\sigma_I/\mu_I)/\sqrt{K_{\text{eff}}}$. The sequence $\{B_n\}_{n \geq 1}$ is i.i.d., hence strictly stationary and white.

(ii) **CLT.** Under (P1)–(P3), $(B_n - \mu_I S_1)/(\sigma_I \sqrt{S_2}) \Rightarrow \mathcal{N}(0, 1)$.

(iii) **Berry–Esseen.** Under (P4), for a universal constant C ,

$$\sup_x \left| P\left(\frac{B_n - \mu_I S_1}{\sigma_I \sqrt{S_2}} \leq x\right) - \Phi(x) \right| \leq C \nu_3 \frac{\sum_j |T_j|^3}{S_2^{3/2}} \leq C \nu_3 K_\infty^{-1/2}.$$

The correct CLT small parameter is $K_\infty^{-1/2}$, **not** K_{eff}^{-1} .

(iv) **Higher cumulants and a conditional log-carrier expansion.** For every $r \geq 2$ for which $\kappa_r(I)$ exists, $\kappa_r(B_n) = \kappa_r(I) \sum_j T_j^r$; thus the standardized law can depend on all existing ratios $\sum_j T_j^r/S_2^{r/2}$, not on K_{eff} alone. Write $B_n = \mu_I S_1(1 + u_n)$. The following log expansion needs additional assumptions not implied by (P1)–(P4): along the array, assume $B_n > 0$ a.s., $\mathbb{E}|\log B_n|^2 < \infty$, $\mathbb{E}|u_n|^3 \rightarrow 0$, and

$$\mathbb{E}[\log(1 + u_n)]^2 \mathbf{1}\{|u_n| > \frac{1}{2}\} = O(\mathbb{E}|u_n|^3).$$

Then $\mathbb{E}u_n = 0$, $\text{Var } u_n = (\sigma_I/\mu_I)^2/K_{\text{eff}}$, and

$$\begin{aligned} \mathbb{E} \log B_n &= \log(\mu_I S_1) - \frac{1}{2} \text{Var}(u_n) + O(\mathbb{E}|u_n|^3), \\ \text{Var}(\log B_n) &= \frac{(\sigma_I/\mu_I)^2}{K_{\text{eff}}} + O(\mathbb{E}|u_n|^3). \end{aligned}$$

Proof. (i) Linearity of expectation and independence across j give the mean and (Bienaymé) the variance; independence across n gives the i.i.d. claim. (ii) Apply the Lindeberg–Feller CLT [46, Thm. 27.2] to the triangular array $X_j = T_j(I_{n,j} - \mu_I)$ with $s^2 = \sigma_I^2 S_2$. Since $|X_j| \leq \|T\|_\infty |I - \mu_I|$, the event $\{|X_j| > \varepsilon s\}$ is contained in $\{|I - \mu_I| > \varepsilon \sigma_I \sqrt{K_\infty}\}$, so the Lindeberg fraction is

$$\begin{aligned} & \frac{1}{\sigma_I^2 S_2} \sum_j T_j^2 \mathbb{E}[(I - \mu_I)^2 \mathbf{1}\{|X_j| > \varepsilon s\}] \\ & \leq \frac{1}{\sigma_I^2} \mathbb{E}[(I - \mu_I)^2 \mathbf{1}\{|I - \mu_I| > \varepsilon \sigma_I \sqrt{K_\infty}\}] \rightarrow 0 \end{aligned}$$

by uniform integrability as $K_\infty \rightarrow \infty$. (iii) Esseen’s inequality for sums of independent, non-identically distributed variables [47, Ch. V] gives the bound with third-moment ratio $\nu_3 \sum_j |T_j|^3/S_2^{3/2}$, and $\sum_j |T_j|^3 \leq \|T\|_\infty S_2$ yields the $K_\infty^{-1/2}$ form. (iv) Cumulant homogeneity ($\kappa_r(cX) = c^r \kappa_r(X)$) and additivity over independent summands give the cumulant formula whenever the stated cumulants exist. On $\{|u_n| \leq \frac{1}{2}\}$, Taylor’s theorem gives $\log(1 + u_n) = u_n - u_n^2/2 + O(|u_n|^3)$ and $\log^2(1 + u_n) = u_n^2 + O(|u_n|^3)$. On the complement, the displayed log-tail assumption, Cauchy–Schwarz, and $|u|^j \mathbf{1}\{|u| > 1/2\} \leq 2^{3-j}|u|^3$ for $j = 1, 2$ control the mean and second-moment remainders by $O(\mathbb{E}|u_n|^3)$. Subtracting the squared mean gives the variance expansion. ■

Status of the log expansion (conditional). Step (iv) is a controlled delta-method expansion under its explicit positivity, integrability, and lower-tail assumptions; it is **not** a consequence of (P1)–(P4). In particular, an atom at $B_n = 0$ makes $\mathbb{E} \log B_n = -\infty$ even if its probability decays with K . A strictly positive pattern floor together with suitable concentration is one sufficient route to the displayed log-tail control. The expansion enters finite-sample variance and rate constants, never algebraic identifiability. Photon records with zeros instead use the separate oracle-known-carrier soft-log theorem of Appendix D, not a blind extension of Theorem B.

a) *What this does not claim.*: The proposition does not say the carrier law is determined by (μ_B, σ_B) — that reduction holds only at the Gaussian-approximation level; the exact law feels every weighted cumulant of T . For gain recovery, the shape-dependent contribution to $m_T = \mathbb{E} \log B_n$ is a *time-constant* scalar absorbed by the global-scale gauge, but the shape-dependent log variance and tails remain in finite-sample concentration and rate constants. Because the iid law is still time-homogeneous, this extra dependence does not create temporal bias or defeat consistency once (★) holds. The proposition also does not assert cross-row decorrelation or window-energy whitening for structured (SRHT) carriers — those require the Walsh-spectrum/ K_4 conditions of Appendix E, where K_{eff} alone is again insufficient.

B. C.2 The quantitative stationarity condition (★)

Definition (★) (W, ϵ, δ) . Let $\{B_n\}_{n \leq N}$ be the carrier and W_n length- W windows covering $\{1, \dots, N\}$. The carrier satisfies the *quantitative stationarity anchor* with tolerance ϵ and confidence $1 - \delta$ if there exists a scalar $m_T \in \mathbb{R}$ (allowed to depend on T , not on n) such that

$$P\left(\max_{1 \leq n \leq N} \left| W^{-1} \sum_{k \in W_n} \log B_k - m_T \right| \leq \epsilon\right) \geq 1 - \delta.$$

We say (★) **holds asymptotically** if for every $\delta > 0$ there are windows $W = W_N \rightarrow \infty$, $W/N \rightarrow 0$, with $\epsilon = \epsilon_{W,N,\delta} \rightarrow 0$. The scalar m_T is exactly the quantity absorbed by the global-scale gauge $a \mapsto ca$, $T \mapsto T/c$ (which shifts $m_T \mapsto m_T - \log c$, $\ell \mapsto \ell + \log c$, leaving Y invariant).

C. C.3 Proposition 2 — (★) is a window-estimator criterion, not a universal one

Proposition 2. Assume the drift is slow in the sense of hypothesis (B3) below, with $C_\beta L (W/N)^\beta \rightarrow 0$. Then:

(a) **Equivalence for the windowed corrector.** The blind windowed estimator $\hat{g}_n = W^{-1} \sum_{k \in W_n} Y_k$ is uniformly consistent for $\ell_n + m_T$ (in probability, uniformly over n) if and only if $(\star)$ holds asymptotically.

(b) $(\star)$ is not necessary for identifiability. There are designs violating $(\star)$ that are nonetheless identifiable: a tall design with $N \geq K + p$ is exactly algebraically identifiable modulo scale for a.e. design-object pair (M, T) , simultaneously for every drift $s \in S$ (Theorem A'(ii); see Appendix B.4 and Remark B.4.1), with no stationarity whatsoever — the a.e. (M, T) quantifier does not cover deterministic structured designs. Likewise, at fixed known support, ordinary local identifiability may follow from the explicit full-rank criterion of Corollary 2; the count $K_0 \geq p - 1$ alone is only necessary for that differential full-rank test.

(c) $(\star)$ is not sufficient for algebraic identifiability. There are settings where $(\star)$ holds yet exact separation of (a, T) is impossible: (c1) any square iid random design $N = K$ whose positive carrier satisfies the stated log-tail and window-scaling conditions — those conditions give $(\star)$, yet Theorem A applies to every invertible realization, so a nonconstant $s \in S$ still maps (ℓ, T) to an exact alias; (c2) a uniform row re-ordering π of a positive-offset full square Hadamard record with finite log range and variance obeys $(\star)$ under the scaling below, while the row-permuted matrix remains invertible and therefore still falls under Theorem A. Quantitatively, Serfling's inequality for sampling without replacement [31] gives, for the finite population $g_k = \log B_k$ with variance τ_g^2 and range R_g ,

$$P\left(\left|W^{-1} \sum_{i=n}^{n+W-1} g_{\pi(i)} - \bar{g}\right| \geq t\right) \leq 2 \exp\left[-\frac{Wt^2}{2\tau_g^2 + \frac{2}{3}R_g t}\right],$$

so a union bound over the $\leq N$ windows yields $(\star)(W, \epsilon, \delta)$ whenever $\log N \ll W$ (and W remains below the drift coherence time). This device fails when some g_k are undefined (nonpositive buckets) or the population log-variance/range is uncontrolled, and it does **not** deliver the small carrier variance $\sigma_{LR}^2 \asymp 1/K_{\text{eff}}$ of genuinely random nonnegative patterns.

Proof. (a) Decompose exactly

$$\begin{aligned} \hat{g}_n - (\ell_n + m_T) &= \underbrace{\left(W^{-1} \sum_{k \in W_n} \ell_k - \ell_n\right)}_{\text{drift bias}} \\ &\quad + \underbrace{\left(W^{-1} \sum_{k \in W_n} \log B_k - m_T\right)}_{\text{carrier deviation}}. \end{aligned}$$

The drift bias is $\leq C_\beta L(W/N)^\beta \rightarrow 0$ deterministically by (B3). Hence $\max_n |\hat{g}_n - (\ell_n + m_T)| \rightarrow 0$ in probability iff the maximal carrier deviation does — which is verbatim the asymptotic form of $(\star)$. Both implications are immediate from this identity. (b) Theorem A' (see Appendix B) is proved without any distributional assumption on the carrier order; the support-zero criterion is Corollary 2 (see Appendix A). (c1) The iid log-tail/window hypotheses give $(\star)$, while Theorem A (see Appendix A) holds for every invertible realized square matrix. (c2) Serfling's bound plus the union over windows gives $(\star)$ under the displayed positive-log scaling;

row permutation preserves invertibility, so Theorem A still supplies the exact alias. ■

a) *What this does not claim.*: Part (a) is an equivalence for one estimator family (windowed local averages of $\log R_n$; the same proof covers kernel weights of matching order). It is not a characterization of identifiability by all mechanisms — (b) and (c) are precisely the two failure modes of that over-reading. Nor does $(\star)$ by itself bound the *rate*: rates require the variance model of Theorem B, and reconstruction quality after gain correction additionally requires the corrected linear inverse problem to be well posed (see Appendix F).

D. C.4 Theorem B — statistical relative-gain identifiability

Theorem B. For each record length N , assume:

- **(B1) Triangular-array signal model.** $Y_{N,n} = \log R_{N,n} = \ell_{N,n} + m_{T,N} + z_{N,n}$, $n = 1, \dots, N$, where $m_{T,N} = \mathbb{E} \log B_{N,n}$ is constant within row, $B_{N,n} > 0$ a.s., and $z_{N,n} = \log B_{N,n} - m_{T,N}$ is centered.
- **(B2) Carrier noise.** Each row $\{z_{N,n}\}_{n=1}^N$ is strictly stationary. Under (B2a) it is α -mixing with a finite $(2+\eta)$ moment and summable Davydov bound, so its rowwise autocovariances $\gamma_N(h)$ are absolutely summable. Write

$$\sigma_{\text{abs},N}^2 := \sum_{h \in \mathbb{Z}} |\gamma_N(h)|,$$

$$V_{N,n}(W) := \text{Var} \left(W^{-1} \sum_{k \in W_{N,n}} z_{N,k} \right).$$

Thus $V_{N,n}(W) \leq \sigma_{\text{abs},N}^2/W$ for an unweighted window of W distinct frames. Under (B2b), used for high-probability bounds, the rowwise constants are uniform in N : $\beta_N(k) \leq \beta_0 \exp[-(k/b)^\kappa]$ and $\|z_{N,n}\|_{\psi_1} \leq M$.

- **(B3) Slow drift and window order.** Define the actual deterministic window bias

$$b_{N,n}(W) := W^{-1} \sum_{k \in W_{N,n}} \ell_{N,k} - \ell_{N,n}.$$

For a normalized-time Hölder radius L_N and a matched window, $|b_{N,n}(W)| \leq C_\beta L_N(W/N)^\beta$. This bound is valid for arbitrary truncated or one-sided windows when $\beta \in (0, 1]$; for $\beta \in (1, 2]$ it is asserted only at indices whose weights annihilate the first moment (e.g. centered symmetric interior windows), or for a separately defined boundary local-polynomial rule.

- **(B4) Vanishing bias and variance.** $W = W_N \rightarrow \infty$, $W_N/N \rightarrow 0$, and, at every index covered by the conclusion, $b_{N,n}(W_N) \rightarrow 0$ and $V_{N,n}(W_N) \rightarrow 0$. Sufficient conditions are $L_N(W_N/N)^\beta \rightarrow 0$ and $\sigma_{\text{abs},N}^2/W_N \rightarrow 0$. A uniform-in- n conclusion requires the corresponding suprema over the stated index set to vanish.

Then:

- (i) **Pointwise consistency with rate.** For $\hat{g}_{N,n} = W_N^{-1} \sum_{k \in W_{N,n}} Y_{N,k}$, (B2a) gives the exact bias-variance control

$$\begin{aligned} \mathbb{E}[\hat{g}_{N,n} - (\ell_{N,n} + m_{T,N})]^2 &= b_{N,n}(W_N)^2 + V_{N,n}(W_N) \\ &\leq CL_N^2(W_N/N)^{2\beta} + C \frac{\sigma_{\text{abs},N}^2}{W_N}, \end{aligned}$$

and (B4), not rowwise finiteness alone, makes it tend to zero. For a fixed stationary law, $V_{N,n}(W) = \sigma_{\text{LR},n}^2/W + o(W^{-1})$. For a triangular array the same additive expansion holds uniformly only under the additional uniform covariance-tail condition $\lim_{H \rightarrow \infty} \sup_N \sum_{|h| \geq H} |\gamma_N(h)| = 0$:

$$V_{N,n}(W_N) = \frac{\sigma_{\text{LR},n}^2}{W_N} + o(W_N^{-1}), \quad \sigma_{\text{LR},n}^2 = \sum_{h \in \mathbb{Z}} \gamma_N(h).$$

Under (B2b), set $q = \lceil b \log \{\max(e, 4\beta_0 W_N / \delta)\} \rceil^{1/\kappa}$. With probability at least $1 - \delta$,

$$\begin{aligned} & |\hat{g}_{N,n} - (\ell_{N,n} + m_{T,N})| \\ & \leq |b_{N,n}(W_N)| + C \sqrt{\frac{\nu_{N,q}^2 q \log(4/\delta)}{W_N}} \\ & \quad + C \frac{M q \log(4/\delta)}{W_N}, \end{aligned}$$

where $\nu_{N,q}^2$ is the rowwise block-variance proxy; uniformity over the covered indices follows with $\delta \mapsto \delta/N$ and a uniform bias bound.

(ii) **Centered-gain recovery.** If the bounds in (B4) hold uniformly over all indices used in the centering, then $\hat{\ell}_{N,n}^c := \hat{g}_{N,n} - N^{-1} \sum_m \hat{g}_{N,m}$ satisfies

$$\begin{aligned} & \sup_n \mathbb{E} [\hat{\ell}_{N,n}^c - (\ell_{N,n} - \bar{\ell}_N)]^2 \\ & \leq 4 \sup_m \{b_{N,m}(W_N)^2 + V_{N,m}(W_N)\} \longrightarrow 0. \end{aligned}$$

Thus the relative log-gain is consistently estimable on this triangular array.

(iii) **Gauge.** The scalar $m_{T,N} + \bar{\ell}_N$ is not identifiable: for every $c > 0$, $a \mapsto ca$, $T \mapsto T/c$ leaves the bucket law invariant while shifting $\ell \mapsto \ell + \log c$ and $m_T \mapsto m_T - \log c$. Within the additive bucket-only experiment (B1) with its carrier-noise law held fixed, part (ii) recovers the centered-gain estimand; no claim is made that scale is the only ambiguity in a richer joint pattern-bucket model.

Proof. Put $e_{N,n} = \hat{g}_{N,n} - (\ell_{N,n} + m_{T,N}) = b_{N,n}(W_N) + W_N^{-1} \sum_{k \in W_{N,n}} z_{N,k}$. The second term is centered, so its mean-square adds to the squared deterministic bias exactly. Stationarity gives

$$V_{N,n}(W) = \frac{1}{W} \sum_{|h| < W} \left(1 - \frac{|h|}{W}\right) \gamma_N(h) \leq \frac{\sigma_{\text{abs},N}^2}{W}.$$

For a fixed law, dominated convergence yields $WV(W) \rightarrow \sum_h \gamma(h)$. For a triangular array, split the difference $WV_N(W) - \sum_h \gamma_N(h)$ at a fixed lag H : the finite-lag taper error vanishes as $W_N \rightarrow \infty$, uniformly when the finite-lag covariances are uniformly bounded, while the two tails are bounded by the stated uniform covariance-tail condition. This proves the additive $o(W_N^{-1})$ expansion; a multiplicative $1+o(1)$ form would be false when the signed long-run variance vanishes.

For (B2b), partition the window into alternating blocks of length q . Berbee coupling costs at most $\beta_0 W_N e^{-(a/b)^\kappa} / q \leq \delta/4$ with the displayed choice (enlarging the universal constant

for the ceiling); Bernstein's inequality for the coupled sub-exponential block sums and a union bound over the two families give the probability display. For (ii), $\hat{\ell}_{N,n}^c - (\ell_{N,n} - \bar{\ell}_N) = e_{N,n} - N^{-1} \sum_m e_{N,m}$; Jensen and $(x-y)^2 \leq 2x^2 + 2y^2$ give the factor 4. Part (iii) is the direct scale transformation; part (ii) identifies only the centered estimand in the stated additive experiment. ■

a) *What this does not claim.*: Theorem B is a **statistical, asymptotic** statement about relative-gain recovery under explicit triangular-array bias and variance conditions; rowwise stationarity and a finite rowwise covariance sum do not by themselves imply consistency. It is not exact finite-sample algebraic identifiability of (a, T) — Theorem A still holds verbatim for a square random design. The all-index conclusion for $\beta > 1$ requires an actually implemented boundary rule with the stated moment cancellation; ordinary truncated averages are covered only for $\beta \leq 1$. The theorem presupposes positive carriers and integrable logs. Photon-counting records with empty bins instead require the oracle-known-carrier calibrated soft-log result of Appendix D; that result must not be read as a blind extension of this theorem. The constants $\sigma_{\text{abs},N}^2$ and $\sigma_{\text{LR},N}^2$ are model inputs; identifying them with $(\sigma_I / \mu_I)^2 / K_{\text{eff}}$ invokes the flagged expansion of Proposition 1(iv). Finally, consistent gain correction does not itself guarantee good reconstruction; Appendix F treats the propagation step.

APPENDIX D

ESTIMATION RATES, MINIMAX LOWER BOUNDS, AND FISHER GEOMETRY

This appendix proves the rate statements of Sec. 6: a high-probability windowed-estimator upper bound under mixing, a matching interior minimax rate for a separately specified fixed iid-Gaussian quotient experiment (Theorem D), the quotient-Fisher geometry underlying the identifiability dichotomy, and the direct oracle-known-carrier Poisson soft-log rate. Notation is locked to the main text: $R_n = a_n B_n$, $\ell = \log a \in S$ with $\dim S = p$ and $\text{span}\{1\} \subset S$, $Y_n = \log R_n = \ell_n + m_T + z_n$ with $m_T = \mathbb{E} \log B_n$ a time-independent scalar, windows W , and the spread functionals K_{eff} , K_∞ as in Appendix C. The Hölder smoothness order is written $\beta \in (0, 2]$ with seminorm L_a ; the symbol α is reserved for the soft-log offset $\psi_\alpha(c) = \log(c + \alpha)$.

Throughout, the *discrete Hölder ball* $H_\beta(L_a)$ is defined operationally: ℓ belongs to $H_\beta(L_a)$ relative to a window family $\{W_n\}$ of order β if the deterministic smoothing bias obeys

$$\left| W^{-1} \sum_{k \in W_n} \ell_k - \ell_n \right| \leq C_\beta L_a W^\beta \quad \text{for all } n. \quad (\text{D.1})$$

For $\beta \in (0, 1]$ a one-sided moving average of ℓ with modulus $|\ell_k - \ell_n| \leq L_a |k - n|^\beta$ satisfies (D.1). For $\beta \in (1, 2]$, (D.1) requires a *centered symmetric* window (or a kernel of order $\lfloor \beta \rfloor$), so that the first-order bias cancels and only curvature L_2 survives; a one-sided window does not attain $\beta > 1$.

A. D.1 *Lemma D.1 (windowed gain-estimation rate under mixing)*

a) *Statement.*: Assume:

- 1) (*Model*) $Y_n = \ell_n + m_T + z_n$, with $\{z_n\}$ strictly stationary and $\mathbb{E}z_n = 0$.
- 2) (*Mixing*) $\{z_n\}$ is β -mixing with $\beta(k) \leq \beta_0 \exp[-(k/b)^\kappa]$ for some $\beta_0, b > 0, \kappa > 0$.
- 3) (*Tails*) $\|z_n\|_{\psi_1} \leq M$ (sub-exponential).
- 4) (*Smoothness*) $\ell \in H_\beta(L_a)$ in the sense (D.1), with the window order matched to β as above.

Let $\hat{g}_n = W^{-1} \sum_{k \in W_n} Y_k$ and set $q = \lceil b \log\{\max(e, 4\beta_0 W/\delta)\} \rceil^{1/\kappa}$. Then for each fixed n and $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$|\hat{g}_n - (\ell_n + m_T)| \leq C_\beta L_a W^\beta + C \left[\sqrt{\frac{\nu_q^2 q \log(4/\delta)}{W}} + \frac{M q \log(4/\delta)}{W} \right], \quad (\text{D.2})$$

where $\nu_q^2 = \text{Var}(q^{-1/2} \sum_{i=1}^q z_i)$ is the block-variance proxy; when the autocovariances are absolutely summable, $\nu_q^2 \leq C' \sigma_{\text{abs}}^2$ with the summable-autocovariance constant $\sigma_{\text{abs}}^2 = \sum_h |\gamma(h)|$ of Appendix C (hypothesis (B2a)) — the signed long-run variance $\sigma_{\text{LR}}^2 \leq \sigma_{\text{abs}}^2$ of Appendix C is the asymptotic variance of window means and plays no role in these non-asymptotic bounds. Uniformity over $n = 1, \dots, N$ follows by replacing δ with δ/N (a $\log N$ inflation inside the brackets).

(*Expectation version, weaker hypotheses.*) If instead $\{z_n\}$ is only α -mixing with $\sum_{h \geq 1} \alpha(h)^{\eta/(2+\eta)} < \infty$ and $\|z_0\|_{2+\eta} < \infty$ for some $\eta > 0$, then

$$\mathbb{E}[\hat{g}_n - (\ell_n + m_T)]^2 \leq C L_a^2 W^{2\beta} + C \sigma_{\text{abs}}^2 / W. \quad (\text{D.3})$$

b) *Proof.*: Bias. $\hat{g}_n - (\ell_n + m_T) = [W^{-1} \sum_k \ell_k - \ell_n] + W^{-1} \sum_k z_k$; the first bracket is bounded by $C_\beta L_a W^\beta$ by (D.1).

Fluctuation, high probability. Partition the window W_n into $m = \lceil W/q \rceil$ consecutive blocks of length q and separate them into odd- and even-indexed families. By Berbee's coupling lemma [34] (see also Rio [35]), on an enlarged probability space each family can be replaced by a sequence of *independent* blocks with the same marginals, at total-variation cost at most $m\beta(q) \leq \beta_0 W \exp[-(q/b)^\kappa]$. The displayed choice of q , including β_0 , makes this cost at most $\delta/4$ (the case $q \geq W$ is a one-block, possibly vacuous boundary bound). On the coupling event, each family is a sum of independent block sums $S_j = \sum_{i \in \text{block } j} z_i$ with $\|S_j\|_{\psi_1} \leq CqM$ (triangle inequality for the ψ_1 norm) and $\text{Var}(S_j) = q\nu_q^2$. Bernstein's inequality for sums of independent sub-exponential variables (e.g. Vershynin [36], Thm. 2.8.1) applied to each family, with a union bound over the two families, yields the bracketed term in (D.2). When $\sum_h |\text{Cov}(z_0, z_h)| < \infty$, $\nu_q^2 = \text{Var}(z_0) + 2 \sum_{h=1}^{q-1} (1 - h/q) \text{Cov}(z_0, z_h) \leq C' \sigma_{\text{abs}}^2$.

Fluctuation, expectation. Davydov's covariance inequality [32] gives $|\text{Cov}(z_0, z_h)| \leq 8\alpha(h)^{\eta/(2+\eta)} \|z_0\|_{2+\eta}^2$; the summability hypothesis makes the autocovariance series absolutely summable, so $\text{Var}(W^{-1} \sum_{k \in W_n} z_k) \leq \bar{\sigma}^2 / W$ with $\bar{\sigma}^2 \leq C\sigma_{\text{abs}}^2$. Combining with the squared bias via $(x+y)^2 \leq 2x^2 + 2y^2$ gives (D.3). ■

Corollary D.2 (optimized blind high-count upper rate). Minimizing the right side of (D.3) over W gives

$$W^* \asymp (\sigma_{\text{abs}}^2 / L_a^2)^{1/(2\beta+1)}, \quad \text{MSE}^* \asymp L_a^{2/(2\beta+1)} \sigma_{\text{abs}}^{4\beta/(2\beta+1)}, \quad (\text{D.4})$$

valid when $W^* \in [1, N]$ and W^* is shorter than the gain coherence time; otherwise the boundary window $W = 1$ or $W = N$ applies. For iid random carriers, $\sigma_{\text{abs}}^2 = \sigma_{\text{LR}}^2 \approx (\sigma_I / \mu_I)^2 / K_{\text{eff}}$ to leading Gaussian/log-variance order under the spikiness condition $K_\infty \rightarrow \infty$ of Prop. 1. No identification of the deterministic Hölder constant L_a with the OU decorrelation parameter is made; the OU protocol has the separate $\rho^{1/2}$ analysis in Appendix F.

c) *What this does not claim.*: The constants C, C_β are not tracked and depend on the mixing constants (β_0, b, κ) and M . The bound is for the *relative* gain $\ell_n + m_T$; the scalar $m_T + \text{gauge}(\ell)$ is not estimable (Theorem B). For $\beta > 1$, the asserted bias order covers only windows or kernels whose weights cancel the first moment; a truncated one-sided edge average is not covered without a defined boundary local-polynomial rule. Nothing is claimed for $\beta > 2$, for non-stationary carriers, or about adaptive window selection when (β, L_a) are unknown. The step $\nu_q^2 \lesssim \sigma_{\text{abs}}^2$ requires summable autocovariances, which the stated mixing plus moment hypotheses supply.

B. D.2 Theorem D (fixed-Gaussian interior minimax equivalence)

a) *Experiment, gauge, class, and loss.*: Observe the fixed regression experiment

$$Y_n = m + f(t_n) + \varepsilon_n, \quad t_n = n/N, \quad \varepsilon_n \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2),$$

with unknown $m \in \mathbb{R}$. Fix the gauge by $N^{-1} \sum_n f(t_n) = 0$. For $0 < \beta \leq 1$, let $\mathcal{H}_\beta^0(L, A)$ contain the gauge-fixed functions with $\|f\|_\infty \leq A$ and $|f(x) - f(y)| \leq L|x - y|^\beta$. For $1 < \beta \leq 2$, require $f \in C^1$, $\|f\|_\infty \leq A$, and $|f'(x) - f'(y)| \leq L|x - y|^{\beta-1}$; centered weights remove the affine term. The loss at a fixed interior grid point t_{n_0} is $\mathbb{E}_f(\hat{f}_{n_0} - f(t_{n_0}))^2$.

Choose a fixed $h_0 < \frac{1}{4} \text{dist}(t_{n_0}, \{0, 1\})$ and the feasible centered bandwidths $\mathcal{H}_N = \{h : W = 2\lfloor Nh/2 \rfloor + 1, 3 \leq W \leq Nh_0\}$. Put

$$h_* := \left(\frac{\sigma^2}{NL^2} \right)^{1/(2\beta+1)}, \quad r_N := L^{2/(2\beta+1)} \left(\frac{\sigma^2}{N} \right)^{2\beta/(2\beta+1)}.$$

Theorem D (fixed-Gaussian interior rate). Assume $0 < \beta \leq 2$, $h_* \in \mathcal{H}_N$ up to a fixed multiplicative rounding constant, and $Lh_*^\beta \leq A/4$. Then constants $0 < c_\beta < C_\beta < \infty$, depending only on β , the interior margin, and the fixed rounding constants, satisfy

$$c_\beta r_N \leq \inf_f \sup_{f \in \mathcal{H}_\beta^0(L, A)} \mathbb{E}_f(\hat{f}_{n_0} - f(t_{n_0}))^2 \leq C_\beta r_N. \quad (\text{D.5})$$

The upper bound is attained by the centered local average at bandwidth h_* (or the corresponding order-one local polynomial), after subtracting the record mean. Thus the equivalence is for this fixed iid-Gaussian quotient experiment and this interior pointwise loss.

b) *Proof.*: For the upper bound, let $W \asymp Nh_*$ and average a centered window around n_0 , then subtract the record mean. The gauge constraint removes m without bias. Symmetry cancels the linear term for $\beta > 1$, so squared smoothing bias is at most $CL^2h_*^{2\beta}$; the Gaussian variance is at most $C\sigma^2/(Nh_*) + C\sigma^2/N$. The feasibility conditions make both leading terms of order r_N and the record-mean term no larger.

For the lower bound, take a smooth bump $g_h(t) = cLh^\beta\phi((t-t_{n_0})/h)$ supported inside the interior window, and subtract its discrete mean to remain on the gauge slice. With c small, 0 and $g_h - \bar{g}_h$ lie in $\mathcal{H}_\beta^0(L, A)$. At $h \asymp h_*$ their pointwise separation is at least $c'Lh_*^\beta$, while the Gaussian Kullback–Leibler divergence is

$$\text{KL}(P_0, P_1) = \frac{\|g_h - \bar{g}_h\|_2^2}{2\sigma^2} \leq C \frac{NL^2h_*^{2\beta+1}}{\sigma^2} \leq C_0.$$

Le Cam's two-point lemma [25] yields the lower bound in (D.5). ■

c) *Integrated Sobolev counterpart.*: In the same iid-Gaussian experiment, expand in a discrete Fourier basis orthonormal for the empirical inner product $N^{-1} \sum_n u_n v_n$, omit the constant coefficient (the gauge), and let $\Theta_\beta(L) = \{\theta : \sum_{j \neq 0} |j|^{2\beta} \theta_j^2 \leq L^2\}$. Integrated loss is $N^{-1} \sum_n \mathbb{E}(\hat{f}_n - f_n)^2 = \sum_{j \neq 0} \mathbb{E}(\hat{\theta}_j - \theta_j)^2$, and each empirical Fourier coefficient has noise variance σ^2/N . When the optimal cutoff $J_* \asymp (NL^2/\sigma^2)^{1/(2\beta+1)}$ satisfies $1 \ll J_* \ll N$,

$$\inf_{\hat{f}} \sup_{\theta \in \Theta_\beta(L)} \frac{1}{N} \sum_n \mathbb{E}(\hat{f}_n - f_n)^2 \asymp_\beta L^{2/(2\beta+1)} (\sigma^2/N)^{2\beta/(2\beta+1)}. \quad (\text{D.6})$$

Fourier truncation proves the upper bound; the standard Assouad/Pinsker Gaussian-sequence lower bound [27], [25] proves the reverse inequality. No claim to the sharp Pinsker constant is needed here.

d) *What this does not claim.*: Theorem D is not a minimax theorem for every fixed mixing or non-Gaussian carrier law; Lemma D.1 supplies upper bounds there. It makes no boundary-point claim and no adaptive claim. Outside the feasible interior regime, the one-frame noise floor, the finite class diameter, and full-record averaging give separate lower or upper caps, but we do *not* combine them into a three-segment finite- N minimax equivalence. Such an equivalence would require matching bounds in every saturation regime, which are not proved here.

C. D.3 Proposition D.4 (Theorem-A ambiguity implies quotient-Fisher singularity)

a) *Statement.*: (a) *Linear-Gaussian window model.* Let $\ell = U\theta$ with $U \in \mathbb{R}^{N \times p'}$, and observe $Y = U\theta + m\mathbf{1} + z$, $z \sim \mathcal{N}(0, \Sigma)$, $\Sigma \succ 0$ known, $m \in \mathbb{R}$ the unknown carrier log-mean

(confounded with global gain scale). The Fisher information for θ after eliminating the nuisance m is

$$I_\theta = U^\top \Sigma^{-1/2} P_\perp \Sigma^{-1/2} U, \quad P_\perp = I - \frac{\Sigma^{-1/2} \mathbf{1} \mathbf{1}^\top \Sigma^{-1/2}}{\mathbf{1}^\top \Sigma^{-1} \mathbf{1}}. \quad (\text{D.7})$$

For a linear functional $L\theta$: if $\text{row}(L) \subseteq \text{range}(I_\theta)$, every unbiased estimator obeys $\text{Cov}(L\hat{\theta}) \succeq L I_\theta^+ L^\top$ (Moore–Penrose CRB for singular information [38], [39]); if L has a nonzero component along $\ker I_\theta$, **no unbiased estimator of $L\theta$ exists and the CRB is infinite** — the functional is not locally estimable.

(b) *Quotient statement.* Let Θ be the parameter manifold, G the global-scale group $(a, T) \mapsto (ca, T/c)$, and P_θ a dominated model differentiable in quadratic mean. Define the quotient score operator $S_\theta : T_{[\theta]}(\Theta/G) \rightarrow L_0^2(P_\theta)$ and $I_\theta = S_\theta^* S_\theta$. Then: $I_\theta \succ 0$ on $T_{[\theta]}(\Theta/G)$ **iff** $[\theta] \mapsto P_\theta$ is an immersion, i.e. iff the model is *locally differentially identifiable* modulo scale. Any C^1 ambiguity curve $[\theta(t)]$ with $P_{\theta(t)} = P_{\theta(0)}$ satisfies $S_\theta \theta'(0) = 0$, hence produces a Fisher null direction. For the Theorem-A ambiguity this implication is *exact*: the explicit data-preserving path $\ell' = \ell + ts$, $T' = M^{-1}(c \odot e^{-ts})$, $s \in S \setminus \text{span}\{\mathbf{1}\}$ — the *exact collision curve* $\ell_u = \ell + u$, $T_u = M^{-1} D_{e^{-u}} M T$ in the notation of Appendix A — is a genuine ambiguity curve for a square invertible design, so the Fisher matrix in the full (ℓ, T) parameterization is singular along its tangent, and the CRB is $+\infty$ for every functional that varies along it. For a *tall* full-column-rank design the exact collision exists for a given $u \in S$ iff the range condition $(I - P_M) D_{e^{-u}} M T = 0$ holds (P_M the column-space projector); under the nonvanishing-carrier assumption $T_u \propto T$ only when u is constant, and the quotient tangent null space is $\mathcal{N}_{\ell, T} = \{(u, h) : D_{MT} u + Mh = 0\} \supseteq \text{span}\{(\mathbf{1}, -T)\}$, with local differential identifiability modulo scale exactly when equality holds.

(c) *Local-window corollary.* For a window over which the target is a common shift t and the Gaussian covariance is the known positive-definite matrix Σ_W , the exact information and CRB are

$$I_W(t) = \mathbf{1}^\top \Sigma_W^{-1} \mathbf{1}, \quad \text{Var}(\hat{t}) \geq (\mathbf{1}^\top \Sigma_W^{-1} \mathbf{1})^{-1}.$$

Only under standard Toeplitz/short-range conditions does this become $(\mathbf{1}^\top \Sigma_W^{-1} \mathbf{1})^{-1} = \sigma_{\text{LR}}^2/W + o(W^{-1})$, or heuristically $\sigma_z^2/W_{\text{eff}}$ with $W_{\text{eff}} = W/(1 + 2 \sum_{h \geq 1} \rho_z(h))$. The exact inverse-covariance expression, not the heuristic effective sample size, is the finite-window Fisher object.

b) *Proof.*: (a) The joint Fisher matrix of (θ, m) in the Gaussian model is the block matrix $\begin{pmatrix} U^\top \Sigma^{-1} U & U^\top \Sigma^{-1} \mathbf{1} \\ \mathbf{1}^\top \Sigma^{-1} U & \mathbf{1}^\top \Sigma^{-1} \mathbf{1} \end{pmatrix}$. Nuisance elimination is the Schur complement with respect to the m -block [40], which is exactly (D.7). The pseudoinverse CRB for singular FIM is Stoica–Marzetta [39]. The infinite-CRB claim is cleanest without CRB machinery: a direction $u \in \ker I_\theta$ that is tangent to an exact ambiguity curve leaves the data law invariant, so any estimator has *identical* distribution under θ and $\theta + tu$; unbiasedness for a functional with $Lu \neq 0$ would force $L\theta = L(\theta + tu)$, a contradiction — hence no unbiased finite-variance estimator exists.

(b) Differentiability in quadratic mean makes S_θ well defined and $I_\theta = S_\theta^* S_\theta$ the pullback metric [48, Ch. 7]. $I_\theta \succ 0$ iff S_θ is injective iff the differential of $[\theta] \mapsto \sqrt{dP_\theta}$ is injective, i.e. an immersion. The ambiguity-curve implication is the chain rule: $t \mapsto P_{\theta(t)}$ constant $\Rightarrow$ its L^2 derivative $S_\theta \theta'(0)$ vanishes. For Theorem A the curve is explicit (see Appendix A), so no genericity is needed. ■

c) *What this does not claim.*: The converse direction — Fisher singularity implying an actual *ambiguity manifold* — requires constant-rank (or analytic) regularity so that the rank defect integrates (rank theorem); Fisher singularity by itself only defeats *local differential* identifiability. Nonsingular Fisher does not exclude isolated *global* aliases. Part (a) assumes Gaussian noise with known Σ ; part (c) is a Gaussian-model computation used as an order-of-magnitude floor, not a bound proven for the actual log-carrier distribution. No efficiency claim is made for the windowed estimator beyond rate (constants open, Theorem D).

D. D.4 Theorem C (low-photon calibrated soft-log rate and the conditional $1/(W\bar{\lambda})$ law)

a) *Statement (exact finite-window identity with known carriers, weighted form).*: Let counts follow the calibrated model $C_n \mid \ell_n \sim \text{Pois}(\lambda_n)$, $\lambda_n = \Lambda_0 e^{\ell_n} b_n + d$, and assume that counts at distinct frames are conditionally independent given the gain path and carriers. Here $d \geq 0$ is the dark count and $\{b_n\}$ are the *known* per-frame carrier intensities ($b_n > 0$); in the Fig. 5 protocol these are *oracle* values supplied from the simulation truth — a flat-field-calibrated benchmark, not programmed design intensities — and $d = 0$ — Remark D.4.1; the unknown-random-carrier and known-law variants are Remarks D.4.3 and D.4.5). Fix $\alpha > 0$, let $\psi_\alpha(c) = \log(c + \alpha)$, and let $m_\alpha(\lambda) = \mathbb{E}[\psi_\alpha(C)]$ for $C \sim \text{Pois}(\lambda)$ — the homogeneous calibration curve, strictly increasing. For a fixed window n , let $\mathcal{I}_n$ be the set of *distinct* frame indices used by the estimator and let $w_{nk} \geq 0$, $\sum_{k \in \mathcal{I}_n} w_{nk} = 1$, be its weights (for the implementation's windows, $\mathcal{I}_n = W_n$ and $w_{nk} = 1/W$). If a padding convention repeats a frame, all repeated occurrences are first aggregated into its single weight w_{nk} . Put

$$\begin{aligned} q_k(t) &= m_\alpha(\Lambda_0 e^t b_k + d), \\ v_k(t) &= \text{Var}[\psi_\alpha(\text{Pois}(\Lambda_0 e^t b_k + d))], \\ M_n(t) &= \sum_{k \in \mathcal{I}_n} w_{nk} q_k(t), \quad y_n = \sum_{k \in \mathcal{I}_n} w_{nk} \psi_\alpha(C_k), \end{aligned}$$

and define

$$B_n = \sum_{k \in \mathcal{I}_n} w_{nk} \{q_k(\ell_k) - q_k(\ell_n)\}, \quad V_n = \sum_{k \in \mathcal{I}_n} w_{nk}^2 v_k(\ell_k).$$

Fix a compact operating range $\Theta = [\theta_{\min}, \theta_{\max}]$ and let

$$\underline{\kappa}_n = \inf_{t \in \Theta} M'_n(t) > 0, \quad D_\Theta = \theta_{\max} - \theta_{\min}.$$

In the implementation Θ is the *calibration-safe interval*: if the calibration curve is certified only for $\lambda \in [\lambda_-, \lambda_+]$ and $d < \lambda_-$, the largest safe interval before intersection with the physical parameter range is

$$\left[\max_{k \in \mathcal{I}_n} \log \frac{\lambda_- - d}{\Lambda_0 b_k}, \quad \min_{k \in \mathcal{I}_n} \log \frac{\lambda_+ - d}{\Lambda_0 b_k} \right],$$

and **nonemptiness of this interval is an assumption of the theorem** — a clamp cannot repair an empty calibration-safe interval. The estimator is the *endpoint-clamped generalized inverse* applied to the window statistic:

$$\hat{\theta}_n = G_n(y_n), \quad G_n(y) = M_n^{-1}[\Pi_{M_n(\Theta)} y], \quad (\text{D.8a})$$

where $\Pi_{M_n(\Theta)}$ projects onto the compact interval $M_n(\Theta)$ before inversion, so G_n returns the corresponding endpoint of Θ when y_n falls outside $M_n(\Theta)$ (an all-zero window has $y_n = \log \alpha$ and returns $\theta_{\min}$; without the clamp the inverse would diverge to $-\infty$ there — the clamp is part of the estimator, not an implementation afterthought). Assume:

- 1) (*Operating range; per-frame sensitivities and noise*) for every k and every $\theta \in \Theta$, $\lambda_k(\theta) = \Lambda_0 e^\theta b_k + d \in [\lambda_-, \lambda_+] \subset (0, \infty)$; consequently the per-frame sensitivity bound $\bar{\kappa}_k = \sup_{t \in \Theta} q'_k(t)$ and the per-frame noise $v_k(t)$ are finite, with $v_k(t) \leq \bar{\sigma}_\alpha^2 := \sup_{\lambda \in [\lambda_-, \lambda_+]} \text{Var}[\psi_\alpha(\text{Pois}(\lambda))] < \infty$, $\kappa_{\max} := \sup_k \bar{\kappa}_k < \infty$, and $\underline{\kappa}_n > 0$;
- 2) (*Smoothness*) $\ell \in H_\beta(L_a)$ with the pointwise modulus $|\ell_k - \ell_n| \leq L_a |k - n|^\beta$ and $\beta \in (0, 1]$ (the paper's operative case is $\beta = 1$; the precise obstruction above one and its partial repair are Remark D.4.2);
- 3) (*Membership*) $\ell_n \in \Theta$ for all n . (A positive margin $\Delta := \min_n \text{dist}(\ell_n, \partial\Theta) > 0$ enters only the clamp-probability diagnostic of Remark D.4.4; the risk bound below does not use it.)

Then, conditionally on the gain path and known carriers,

$$\text{Var} \left[\sum_{k \in \mathcal{I}_n} w_{nk} \{\psi_\alpha(C_k) - q_k(\ell_k)\} \right] = V_n$$

exactly, and no mixing or stationarity assumption is required. Then, conditionally on the gain path and the known carriers,

$$\mathbb{E}[(\hat{\theta}_n - \ell_n)^2 \mid \ell, b] \leq \frac{B_n^2 + V_n}{\underline{\kappa}_n^2}. \quad (\text{D.8})$$

There is **no separate clamp-event risk term**: the endpoint-clamped generalized inverse G_n is globally $\underline{\kappa}_n^{-1}$ -Lipschitz on all of $\mathbb{R}$ (proof below), so all-zero and out-of-range windows are already covered by (D.8); the clamp probability survives only as a diagnostic (Remark D.4.4). This is a finite-window statement: it requires neither stationarity nor mixing of the known carrier sequence. Moreover, the pointwise modulus gives

$$|B_n| \leq L_a \sum_{k \in \mathcal{I}_n} w_{nk} \bar{\kappa}_k |k - n|^\beta.$$

For an unpadded average of W distinct frames, $w_{nk} = 1/W$ and hence

$$V_n = \frac{\bar{v}_{n,\alpha}}{W}, \quad \bar{v}_{n,\alpha} = \frac{1}{W} \sum_{k \in W_n} v_k(\ell_k) \leq \bar{\sigma}_\alpha^2,$$

while $|B_n| \leq C \kappa_{\max} L_a W^\beta$, so (D.8) specializes to

$$\mathbb{E}(\hat{\theta}_n - \ell_n)^2 \leq C \left(\frac{\kappa_{\max}}{\underline{\kappa}_n} \right)^2 L_a^2 W^{2\beta} + \frac{\bar{\sigma}_\alpha^2}{\underline{\kappa}_n^2 W}$$

(no clamp term: the global Lipschitz property of G_n already covers clamped windows). Let $\mathcal{W}_n$ be the set of integer

widths for which the frames are distinct after aggregation, the calibration-safe interval is nonempty, and the displayed smoothness bound applies. Then the finite-window theorem gives the feasible infimum

$$\inf_{W \in \mathcal{W}_n} \left\{ C \left(\frac{\kappa_{\max}}{\kappa_n(W)} \right)^2 L_a^2 W^{2\beta} + \frac{\bar{\sigma}_\alpha^2}{\kappa_n(W)^2 W} \right\}.$$

Only when the sensitivity ratios are uniformly controlled over the candidate widths and the unconstrained optimizer lies in $\mathcal{W}_n$ does the power-law simplification read

$$\text{MSE}_\alpha^* \leq C \kappa_n^{-2} [\kappa_{\max} L_a]^{2/(2\beta+1)} \bar{\sigma}_\alpha^{4\beta/(2\beta+1)} \quad (\text{D.9})$$

with κ_n a common lower sensitivity bound on those feasible widths. This is Theorem C of the main text. **The pointwise loss is log-domain squared error:** $\hat{\theta}_n - \ell_n$ is an increment of log-gain (equivalently log-intensity). Because Λ_0 and b_k are fixed and known, this oracle experiment identifies the absolute log-gain scale. Record-mean centering is an evaluation choice made to match the blind task and Fig. 5, not a gauge forced by the oracle model. In the photon-starved regime the variance-to-sensitivity ratio obeys the *general window law*

$$\frac{V_n}{M_n'(\ell_n)^2} = \frac{\sum_k w_{nk}^2 \lambda_k(\ell_k)}{[\sum_k w_{nk} \lambda_k(\ell_n)]^2} \{1 + O_\alpha(\varepsilon)\}$$

(low-photon corollary below), with the *actual* intensities $\lambda_k(\ell_k)$ in the numerator and the *anchor* intensities $\lambda_k(\ell_n)$ in the sensitivity denominator; it simplifies to $\{1 + O_\alpha(\varepsilon)\}/(W\bar{\lambda}_n)$ only for distinct equal-weight frames under *local gain flatness* (see the counterexample below), and in that flat regime it is minimax-order sharp in this *centered log-domain loss* at $d = 0$ by the centered-loss lower bound following the proof. As the expected intensity tends to zero, the Poisson information for log intensity vanishes; with the clamp (D.8a), empty observed bins never make the estimator diverge.

b) Proof.: Sensitivity and variance of the soft-log. The Poisson derivative identity $\frac{d}{d\lambda} \mathbb{E}f(C) = \mathbb{E}[f(C+1) - f(C)]$ (differentiate $\sum_c e^{-\lambda} \lambda^c f(c)/c!$ term by term) gives

$$\kappa_\alpha(\lambda) := \frac{d}{d \log \lambda} \mathbb{E} \log(C + \alpha) = \lambda \mathbb{E} \log \frac{C + 1 + \alpha}{C + \alpha},$$

with the expansions $\kappa_\alpha(\lambda) = 1 + O(1/\lambda)$ for $\lambda \gg 1$ and $\kappa_\alpha(\lambda) = c_\alpha \lambda + O(\lambda^2)$ for $\lambda \ll 1$, where $c_\alpha = \log(1 + 1/\alpha)$. Since $q_k'(t) = \kappa_\alpha(\lambda_k(t)) \cdot \Lambda_0 e^t b_k / \lambda_k(t)$ ($= \kappa_\alpha(\lambda_k(t))$ at $d = 0$), hypothesis 1 holds on any compact operating range. No uniform identification $\kappa_n \asymp c_\alpha \bar{\lambda}_n$ is claimed: under unrestricted heteroscedastic carriers the infimum of M_n' over Θ can sit far below $c_\alpha \bar{\lambda}_n$ (two-frame example $\lambda_1 = \varepsilon^2$, $\lambda_2 = 2\varepsilon - \varepsilon^2$), so κ_n -based displays are conservative bounds only; the correct low-photon law is the window-average corollary below. The Poisson Poincaré inequality [41], [42], $\text{Var } f(C) \leq \lambda \mathbb{E}[(f(C+1) - f(C))^2]$, yields $v_k = O(1/\lambda)$ at high λ and $v_k = c_\alpha^2 \lambda + O(\lambda^2)$ at low λ , whence $\bar{\sigma}_\alpha^2 < \infty$ on $[\lambda_-, \lambda_+]$.

Risk bound. Write

$$y_n - M_n(\ell_n) = B_n + Z_n, \quad Z_n = \sum_{k \in \mathcal{I}_n} w_{nk} \{\psi_\alpha(C_k) - q_k(\ell_k)\}.$$

The variables in Z_n are independent and centered conditionally on the gain path and known carriers (Poisson counts are independent across frames given their intensities — the carrier is known, so no carrier randomness remains; this *conditional count independence* is an assumption of the theorem), so $\mathbb{E}Z_n = 0$ and $\text{Var}(Z_n) = V_n$ exactly — a finite-window identity requiring no mixing and no long-run variance.

Global Lipschitz property of the clamped inverse. M_n is continuous and strictly increasing on the compact Θ with $M_n' \geq \kappa_n > 0$, so its ordinary inverse is κ_n^{-1} -Lipschitz on $M_n(\Theta)$. Projection onto the interval $M_n(\Theta)$ is nonexpansive, hence the composition $G_n = M_n^{-1} \circ \Pi_{M_n(\Theta)}$ satisfies

$$|G_n(y_2) - G_n(y_1)| \leq \frac{|y_2 - y_1|}{\kappa_n} \quad \text{for all } y_1, y_2 \in \mathbb{R}.$$

Since $\ell_n \in \Theta$ gives $G_n\{M_n(\ell_n)\} = \ell_n$,

$$|\hat{\theta}_n - \ell_n| = |G_n(y_n) - G_n\{M_n(\ell_n)\}| \leq \frac{|B_n + Z_n|}{\kappa_n}$$

with no case split on whether y_n lands inside $M_n(\Theta)$: taking conditional second moments and using $\mathbb{E}[(B_n + Z_n)^2 | \ell, b] = B_n^2 + V_n$ gives (D.8) directly. The bound on B_n follows by applying the mean value theorem to each q_k and then the pointwise modulus; for equal weights the triangle inequality gives $|B_n| \leq C \kappa_{\max} L_a W^\beta$ (valid for the pointwise modulus with $\beta \leq 1$; no cancellation across the window is invoked — cf. Remark D.4.2), and the standard bias–variance optimization gives (D.9). ■

c) Approximate calibration and numerical inversion.:

Let $\tilde{M}_n$ be a continuous strictly increasing approximation to M_n on the same interval Θ with $\sup_{t \in \Theta} |\tilde{M}_n(t) - M_n(t)| \leq \varepsilon_{\text{cal},n}$, let $\tilde{G}_n$ be its clamped generalized inverse (same endpoint convention), and suppose the numerical solver returns $\hat{\theta}_n^{\text{num}}$ with $|\hat{\theta}_n^{\text{num}} - \tilde{G}_n(y_n)| \leq \varepsilon_{\text{bis},n}$, where $\varepsilon_{\text{bis},n}$ is measured in the *log-parameter* domain. Inverse bracketing under the shared endpoint convention gives $G_n(y - \varepsilon_{\text{cal},n}) \leq \tilde{G}_n(y) \leq G_n(y + \varepsilon_{\text{cal},n})$, hence $\sup_y |\tilde{G}_n(y) - G_n(y)| \leq \varepsilon_{\text{cal},n}/\kappa_n$, and the conservative implementation bound is the RMSE (Minkowski) form

$$\{\mathbb{E}[(\hat{\theta}_n^{\text{num}} - \ell_n)^2 | \ell, b]\}^{1/2} \leq \frac{\sqrt{B_n^2 + V_n}}{\kappa_n} + \frac{\varepsilon_{\text{cal},n}}{\kappa_n} + \varepsilon_{\text{bis},n}. \quad (\text{D.8b})$$

The quantities with the same (log-parameter) units are $\varepsilon_{\text{cal},n}/\kappa_n$ and $\varepsilon_{\text{bis},n}$; a response-domain calibration error and a log-parameter bisection error must *not* be added before division by the sensitivity. If tabulation and interpolation have separately certified forward-error budgets, replace $\varepsilon_{\text{cal},n}$ by their sum. For J midpoint bisections on a log-parameter interval of diameter D_n , the numerical error is certified by $\varepsilon_{\text{bis},n} \leq D_n/2^{J+1}$.

d) A theorem-level interpolation certificate.: A midpoint probe of the interpolation error is an empirical diagnostic, not by itself a uniform bound over the calibration interval. Put $x = \log \lambda$ and $f_\alpha(x) = m_\alpha(e^x)$. For a log-intensity grid of maximum spacing Δ_x , piecewise-linear interpolation from exact node values obeys

$$\|f_\alpha - I_{\Delta_x} f_\alpha\|_\infty \leq \frac{\Delta_x^2}{8} \|f_\alpha''\|_\infty, \quad (\text{D.8c})$$

and with $\Delta g(c) = g(c+1) - g(c)$ the Poisson difference identities give

$$f''_\alpha(x) = \lambda \mathbb{E} \Delta \psi_\alpha(C) + \lambda^2 \mathbb{E} \Delta^2 \psi_\alpha(C), \quad \lambda = e^x.$$

A certified bound $K_{\alpha,2} \geq \sup_{\lambda \in [\lambda_-, \lambda_+]} |f''_\alpha|$ together with a certified node-evaluation error ε_{tab} yields $\varepsilon_{\text{cal}} \leq \varepsilon_{\text{tab}} + K_{\alpha,2} \Delta_x^2/8$; the expectations and Poisson tails may be enclosed analytically or by interval arithmetic. **Until such an enclosure is generated, the measured midpoint maximum of Remark D.4.1 must be labeled empirical**, and the manuscript so labels it.

e) Low-photon corollary (the general leading term).: Let $d = 0$ and define the *anchor* and *actual* per-frame intensities

$$\lambda_{k,n}^0 = \Lambda_0 e^{\ell_n} b_k = \lambda_k(\ell_n), \quad \lambda_k^* = \Lambda_0 e^{\ell_k} b_k = \lambda_k(\ell_k),$$

$$A_n = \sum_k w_{nk} \lambda_{k,n}^0, \quad D_n = \sum_k w_{nk}^2 \lambda_k^*.$$

Fix $\alpha > 0$, put $c_\alpha = \log(1 + 1/\alpha)$, and suppose $\max_{k \in \mathcal{I}_n} \{\lambda_{k,n}^0, \lambda_k^*\} \leq \varepsilon$. The fixed- α Poisson expansions of the proof give, uniformly over arbitrary positive carrier heterogeneity,

$$M'_n(\ell_n) = c_\alpha A_n \{1 + O_\alpha(\varepsilon)\}, \quad V_n = c_\alpha^2 D_n \{1 + O_\alpha(\varepsilon)\},$$

and therefore

$$\begin{aligned} \frac{V_n}{M'_n(\ell_n)^2} &= \frac{D_n}{A_n^2} \{1 + O_\alpha(\varepsilon)\} \\ &= \frac{\sum_k w_{nk}^2 \lambda_k(\ell_k)}{[\sum_k w_{nk} \lambda_k(\ell_n)]^2} \{1 + O_\alpha(\varepsilon)\}. \end{aligned} \quad (\text{D.10})$$

For $m = |\mathcal{I}_n|$ equal-weight *distinct* frames, writing $I_{n,\text{ref}} = \sum_k \lambda_{k,n}^0$ and $I_{n,\text{act}} = \sum_k \lambda_k^*$, (D.10) reads $I_{n,\text{act}}/I_{n,\text{ref}}^2 \cdot \{1 + O_\alpha(\varepsilon)\}$, and with $h_{\ell,n} = \max_k |\ell_k - \ell_n|$,

$$e^{-h_{\ell,n}} \leq I_{n,\text{act}}/I_{n,\text{ref}} \leq e^{h_{\ell,n}}.$$

Only under *local gain flatness* $h_{\ell,n} = o(1)$ may the leading term be simplified to $\{1 + O_\alpha(\varepsilon + h_{\ell,n})\}/I_{n,\text{ref}}$; when the gain is constant in the window, $I_{n,\text{act}} = I_{n,\text{ref}} = m\bar{\lambda}_n$ and only then does (D.10) become $\{1 + O_\alpha(\varepsilon)\}/(m\bar{\lambda}_n)$ — the $1/(W\bar{\lambda})$ statement quoted in the main text. Carrier comparability is *not* needed; local gain flatness *is*. These expansions are stated for fixed α ; $c_\alpha \rightarrow \infty$ as $\alpha \rightarrow 0$, so the fixed- α small- λ statement must not be read as a triangular $\alpha \rightarrow 0$ asymptotic.

f) Counterexample to the unqualified $1/(W\bar{\lambda})$ claim.: Take two equal-weight frames with equal carriers, target gain $\ell_n = 0$, the other frame's gain $\ell = \log 2$, and reference intensity ε . Then $I_{n,\text{ref}} = 2\varepsilon$ but $I_{n,\text{act}} = 3\varepsilon$, so (D.10) gives $3/(4\varepsilon)$ whereas the unqualified formula gives $1/(2\varepsilon)$: the ratio is $3/2$ and does *not* converge to one as $\varepsilon \downarrow 0$. The unconditional window-average law asserted in earlier drafts is therefore false as stated and is withdrawn; (D.10) is the correct general statement.

g) From a variance factor to estimator asymptotics.:

Display (D.10) is an algebraic expansion of the local variance-to-sensitivity factor. To identify it with the estimator's leading MSE one additionally needs a local variance regime — $I_{n,\text{act}} \rightarrow \infty$, $|B_n| = o(\sqrt{V_n})$, a Lindeberg condition, negligible clamp probability, and inverse-map linearization such as $\sup_\Theta |M''_n| \sqrt{V_n}/M'_n(\ell_n)^2 \rightarrow 0$. For bounded total information the compact clamp makes the risk saturate; one cannot globally identify the MSE with $1/I_n$.

h) Centered-loss low-photon lower bound.: To compare with the blind task, evaluate the oracle experiment under the scale-invariant centered loss

$$\begin{aligned} d_{\text{quot}}^2(\hat{g}, g) &= \inf_{a \in \mathbb{R}} \frac{1}{N} \|\log \hat{g} - \log g - a \mathbf{1}\|_2^2 \\ &= \frac{1}{N} \|P_N(\log \hat{g} - \log g)\|_2^2, \quad P_N = I - \frac{1}{N} \mathbf{1} \mathbf{1}^\top. \end{aligned}$$

This is the centered-log-gain loss reported in Fig. 5. Multiplying either gain profile by an arbitrary positive scalar leaves the loss unchanged, even though known Λ_0 and carriers make absolute scale identifiable in this oracle experiment.

Pointwise lower bound (Le Cam two-point). Let $0 < \beta \leq 1$ and, conditionally on known positive carriers $q_1, \dots, q_N$, observe independent counts $C_i \sim \text{Pois}(q_i e^{\ell_i})$. Estimate the quotient coordinate $P_N \ell$. Fix an interior index n_0 , a baseline $\ell^{(0)}$ lying strictly inside both the amplitude and Hölder constraints (e.g. $\|\ell^{(0)}\|_\infty \leq a_0/2$ and seminorm at most $L_a/2$), and a window A of order W containing n_0 . Choose a contrast $\psi^{(W)}$ with

$$\begin{aligned} \text{supp } \psi^{(W)} &\subseteq A, \quad \mathbf{1}^\top \psi^{(W)} = 0, \quad \|\psi^{(W)}\|_\infty \leq 1, \\ \psi_{n_0}^{(W)} &= 1, \quad \|\psi^{(W)}\|_2^2 \geq cW, \\ |\psi_i^{(W)} - \psi_j^{(W)}| &\leq CW^{-\beta} |i - j|^\beta. \end{aligned}$$

Put $\lambda_i^{(0)} = q_i e^{\ell_i^{(0)}}$ and $I_A(\psi) = \sum_{i \in A} \lambda_i^{(0)} \psi_i^2 \leq W \bar{\lambda}_A$ with $\bar{\lambda}_A = W^{-1} \sum_{i \in A} \lambda_i^{(0)}$. For amplitude $a \leq c_0 \min\{a_0, L_a W^\beta, I_A(\psi)^{-1/2}\}$, both $\ell^{(0)}$ and $\ell^{(1)} = \ell^{(0)} + a\psi^{(W)}$ lie in a rescaled $H_\beta(L_a)$ ball; the Poisson KL obeys

$$\text{KL}(P_0, P_1) = \sum_i \lambda_i^{(0)} \{e^{a\psi_i} - 1 - a\psi_i\} \leq \frac{e^{a_0}}{2} a^2 I_A(\psi),$$

kept below a numerical constant by the amplitude choice. Because $\mathbf{1}^\top \psi = 0$, quotient projection does not reduce the separation at n_0 , and Pinsker followed by Le Cam's two-point lemma [25] gives

$$R_{n_0}^{\text{quot}} \geq c \min \left\{ a_0^2, L_a^2 W^{2\beta}, \frac{1}{W \bar{\lambda}_A} \right\}. \quad (\text{D.11})$$

When local photon rates have common order $\bar{\lambda}$, optimizing W yields

$$R_{n_0}^{\text{quot}} \gtrsim L_a^{2/(2\beta+1)} \bar{\lambda}^{-2\beta/(2\beta+1)},$$

the direct Poisson counterpart of the upper rate (D.9); these optimized displays assume $1 \lesssim W_* \lesssim N$, and otherwise (D.11) stays in the one-frame or finite-record saturation regime. Thus the $1/(W\bar{\lambda})$ law is a *local-minimax* statement in the regime $W\bar{\lambda} \rightarrow \infty$; for bounded total information the risk saturates at the squared diameter of the local parameter neighborhood.

It does *not* route through Theorem D's Gaussian bound (a Gaussian model is not a submodel of the mixed-Poisson experiment).

Integrated lower bound (guarded Assouad hypercube). Le Cam's two-point argument controls the pointwise loss only; the integrated quotient loss needs a hypercube. Partition an interior fraction of $\{1, \dots, N\}$ into $m \asymp N/W$ cells of length CW ; inside each cell reserve a guard band of order W and place a translated, rescaled copy ψ_j of one fixed compactly supported, mean-zero H_β template that vanishes on the cell boundary, with active supports separated by at least cW and $\|\psi_j\|_2^2 \geq cW$. Set $\ell^\omega = \ell^{(0)} + a \sum_j \omega_j \psi_j$, $\omega \in \{0, 1\}^m$, with

$$\bar{\lambda}_* = \max_j |A_j|^{-1} \sum_{i \in A_j} \lambda_i^{(0)},$$

$$a \leq c_0 \min\{a_0, L_a W^\beta, (W \bar{\lambda}_*)^{-1/2}\}.$$

The guard bands and the vanishing boundary keep the Hölder constant of the sum bounded independently of m , and the interior baseline keeps every vertex in the ball. Adjacent vertices differ on one block with $\text{KL} \leq Ca^2 W \bar{\lambda}_* \leq \kappa < 1$; all pairwise differences are mean-zero, so quotient projection loses no distance, and Assouad's lemma [25] gives

$$\begin{aligned} R_{\text{int}}^{\text{quot}} &= \inf_{\hat{\ell}} \sup_{\omega} \frac{1}{N} \mathbb{E}_\omega \|P_N(\hat{\ell} - \ell^\omega)\|_2^2 \\ &\geq c \min \left\{ a_0^2, L_a^2 W^{2\beta}, \frac{1}{W \bar{\lambda}_*} \right\}. \end{aligned} \quad (\text{D.12})$$

Random carriers: choose alternatives before observing B. If $q_i = \Lambda_0 B_i$ is random with a parameter-independent law and $\mathbb{E} B_i < \infty$, the alternatives and their amplitude must be chosen *before* observing B , using the *expected* block information $\bar{\mathcal{I}}_A = \mathbb{E}_B \sum_{i \in A} \lambda_i^{(0)} \psi_i^2$ — not the realized one. The calculation then applies verbatim to the carrier-observed *joint oracle* experiment (C, B) , whose KL is $\text{KL}(P_0^{C,B}, P_1^{C,B}) = \mathbb{E}_B \text{KL}(P_0^{C|B}, P_1^{C|B})$; marginalization gives $\text{KL}(P_0^C, P_1^C) \leq \text{KL}(P_0^{C,B}, P_1^{C,B})$, so the oracle lower bound transfers to the less-informative count-only experiment. This transfers *hardness*; it does not construct a blind estimator. For dark count $d > 0$, replace the total count information by

$$\bar{\mathcal{I}}_A(\psi) = \mathbb{E}_B \sum_{i \in A} \frac{s_i^2}{s_i + d} \psi_i^2, \quad s_i = \Lambda_0 e^{\ell_i} b_i,$$

the local quadratic-KL information in the contrast direction (the bounded-perturbation remainder must be controlled uniformly); the identity $I(\theta) = \lambda$ does not hold for log gain in the presence of additive dark counts.

i) Oracle local-shift information (the Fig. 5 reference line). For a fixed window A , suppose independently $C_k \sim \text{Pois}\{\lambda_k(t)\}$ with $\lambda_k(t) = s_k e^t + d_k$, where $s_k > 0$ and $d_k \geq 0$ are *known*. The score and Fisher information for the one-dimensional common shift t are

$$\begin{aligned} S_t(C) &= \sum_{k \in A} \{C_k - \lambda_k(t)\} \frac{s_k e^t}{\lambda_k(t)}, \\ I_A(t) &= \sum_{k \in A} \frac{s_k^2 e^{2t}}{s_k e^t + d_k}, \quad I_A(t) = \sum_{k \in A} \lambda_k(t) \quad \text{at } d_k = 0. \end{aligned} \quad (\text{D.13})$$

Therefore $1/I_A(t)$ is the scalar Cramér–Rao reference for a regular unbiased estimator of a *common log-intensity shift* with carriers, offsets, and all other path coordinates held fixed — an **oracle common-shift diagnostic**, *not* the nuisance-path quotient CRB. For bias $b(t) = \mathbb{E}_t \hat{t} - t$ the biased CR inequality reads

$$\text{MSE}_t(\hat{t}) \geq b(t)^2 + \frac{\{1 + b'(t)\}^2}{I_A(t)}.$$

MSE *below* $1/I_A$ is compatible with shrinkage bias, and MSE *above* $1/I_A$ does not establish unbiasedness. For a path model $\ell = U\theta$ at $d = 0$ with estimable contrast coordinates separated from the constant direction and $D_\lambda = \text{diag}(\lambda_1, \dots, \lambda_N)$, the information is $I_\theta = U^\top D_\lambda U$; eliminating a common nuisance scale m entering as $\ell + m\mathbf{1}$ gives the Schur complement

$$I_{\theta|m} = U^\top D_\lambda U - U^\top D_\lambda \mathbf{1} (\mathbf{1}^\top D_\lambda \mathbf{1})^{-1} \mathbf{1}^\top D_\lambda U,$$

the quotient Fisher object of this known-carrier parametric path model. It does not include unknown-object or unknown-carrier nuisance parameters; a contrast outside the range of the information matrix has no finite CR bound, and overlapping windows create cross-window covariance, so an average of $1/I_n$ is not an exact whole-record centered-loss CRB. Sharp minimax constants beyond these local calculations are not claimed.

j) Remark D.4.1 (instantiation in the Fig. 5 protocol).

There $d = 0$, and the carriers b_k are *oracle* values supplied from the simulation truth — the noiseless bucket carrier of the simulated object, times the photon-budget normalization; operationally a flat-field-calibrated benchmark, *not* programmed design intensities. With Λ_0 fixed by convention, the safe log-gain interval is exactly the theorem's intersection

$$\Theta = \left[\max_k \log \frac{\lambda_-}{\Lambda_0 b_k}, \min_k \log \frac{\lambda_+}{\Lambda_0 b_k} \right]$$

with the physical parameter range, and must be nonempty. The implemented estimator solves $M_n(\hat{\theta}_n) = y_n$ per window by monotone bisection using the homogeneous table: for $C \sim \text{Pois}(\Lambda_0 e^\theta b_k)$, $\mathbb{E} \psi_\alpha(C) = m_\alpha(\Lambda_0 e^\theta b_k)$. It uses the clamp of (D.8a) and truncated, renormalized distinct-frame boundary windows (no replicate padding; requested width $W = 64$ realizes centered odd effective width 65 in the interior). Calibration accuracy: linear interpolation of m_α against $\log \lambda$ has measured midpoint error $\varepsilon_{\text{cal}} = 1.6410165 \times 10^{-6}$ over all 2999 geometric midpoints (maximum near $\lambda \approx 0.92$), an **empirical midpoint maximum, not a certified continuous-interval bound**; (D.8c) gives the certificate route. Sixty bisections give $\varepsilon_{\text{bis}} \leq 1.5396755 \times 10^{-17}$ in log-parameter units. Per (D.8b), these enter RMSE as $\varepsilon_{\text{cal}}/\kappa_n + \varepsilon_{\text{bis}}$. The earlier (r3) arm inverted m_α at the window mean; that mean-Poisson proxy ignores within-window carrier variation and coincides with (D.8a) only for a constant carrier. Both arms are reported in Fig. 5, but only (D.8a) is the theorem estimator.

k) Remark D.4.2 (the $\beta \leq 1$ theorem and the precise obstruction above one). For $\beta \in (0, 1]$, the pointwise modulus $|\ell_k - \ell_n| \leq L_a |k - n|^\beta$ and the mean value theorem give

$$|q_k(\ell_k) - q_k(\ell_n)| \leq \bar{\kappa}_k L_a |k - n|^\beta,$$

so the triangle-inequality bias bound used in Theorem C is valid for arbitrary known carriers.

For $\beta \in (1, 2]$, standard smoothness has the local form

$$\ell_k - \ell_n = v_n u_k + r_{nk}, \quad u_k = \frac{k - n}{N}, \quad |r_{nk}| \leq L|u_k|^\beta,$$

rather than a pointwise modulus of order β on the function itself. Let $w_k \geq 0$ be the window weights and write $q_k(t) = m_\alpha(\Lambda_0 e^t b_k + d)$. Taylor's theorem gives

$$\begin{aligned} |B_n| \leq |v_n| \left| \sum_k w_k q'_k(\ell_n) u_k \right| + Q_1 L \sum_k w_k |u_k|^\beta \\ + Q_2 \left\{ v_n^2 \sum_k w_k u_k^2 + L^2 \sum_k w_k |u_k|^{2\beta} \right\}, \end{aligned} \quad (\text{D.14})$$

where $Q_1 = \sup_{k, t \in \Theta} q'_k(t)$ and $Q_2 = \sup_{k, t \in \Theta} |q''_k(t)|$.

The first term in (D.14) is the load-bearing obstruction: an $O(h_n^\beta)$ deterministic bias requires the *sensitivity-weighted first-moment balance*

$$|v_n| \left| \sum_k w_k q'_k(\ell_n) u_k \right| \lesssim \kappa_n (W/N)^\beta.$$

An *internal symmetric equal-weight window with a homogeneous carrier* satisfies it exactly (q'_k constant and $\sum_k w_k u_k = 0$ — the interior identity). A heterogeneous carrier generally does not: ordinary centering cancels $\sum_k w_k u_k$ but not the carrier-weighted moment, which is then $O(W/N)$, so an $O((W/N)^\beta)$ claim is false for $\beta > 1$ under the present assumptions. Crucially, a **truncated one-sided edge window fails the balance even with a homogeneous carrier**: for the implementation's first truncated window $k = 0, \dots, 32$ with target $n = 0$,

$$\frac{1}{33} \sum_{k=0}^{32} \frac{k}{N} = \frac{16}{N},$$

first order in the window radius, so the current local-constant estimator cannot support an all-frame uniform $\beta > 1$ rate — the $\beta > 1$ statement is restricted to *internal balanced windows*.

Under the balance condition the remaining nonlinear term is $O((W/N)^2)$ and is therefore no larger than $O((W/N)^\beta)$ for $1 < \beta \leq 2$, provided the local slope is bounded. Thus the calibrated inverse recovers the β -order deterministic bias on internal windows under homogeneous or sensitivity-balanced carriers, with a constant depending also on the slope bound and on Q_2 .

A second-order Taylor expansion of M_n^{-1} by itself does not remove an unbalanced first moment. For arbitrary carriers, or for edge frames, recovery of the $\beta > 1$ rate requires carrier-adapted moment conditions, carrier-adapted local-linear weights, or a separately proved local-polynomial edge estimator (for $\beta > 2$, moments through order $\lfloor \beta \rfloor$ must be cancelled). Theorem C therefore remains restricted to $\beta \leq 1$, while the internal-window homogeneous-carrier and sensitivity-balanced extensions above are valid weaker statements.

l) Remark D.4.3 (unknown random carrier; nonstationary finite-W covariance bound).: Let $\{B_k\}$ be an unobserved stationary random carrier with known law, independent of the parameter, and let

$$\bar{m}_{\alpha, k}(t) = \mathbb{E}_B m_\alpha(\Lambda_0 e^t B_k + d).$$

Because the gain path ℓ_k may vary, stationarity of $\{B_k\}$ does not in general make the soft-log sequence stationary. Define

$$z_k = \psi_\alpha(C_k) - \mathbb{E} \psi_\alpha(C_k), \quad g_k(b) = m_\alpha(\Lambda_0 e^{\ell_k} b + d).$$

Conditional independence of the Poisson randomization gives, for $i \neq j$,

$$\text{Cov}(z_i, z_j) = \text{Cov}(g_i(B_i), g_j(B_j)).$$

Consequently the exact finite-window identity is

$$\begin{aligned} \text{Var} \left(W^{-1} \sum_{i=1}^W z_i \right) &= W^{-2} \sum_{i=1}^W \text{Var}(z_i) \\ &\quad + 2W^{-2} \sum_{h=1}^{W-1} \sum_{i=1}^{W-h} \text{Cov}(z_i, z_{i+h}). \end{aligned} \quad (\text{D.15})$$

Let $p = 2 + \eta$, $r_\eta = \eta/(2 + \eta)$, and assume

$$v_* := \sup_i \text{Var}(z_i) < \infty,$$

$$G_p := \sup_i \|g_i(B_i) - \mathbb{E} g_i(B_i)\|_p < \infty.$$

If the carrier is strongly mixing with coefficient $\alpha_B(h)$, Davydov's inequality [32] gives

$$|\text{Cov}(z_i, z_{i+h})| \leq 8 \alpha_B(h)^{r_\eta} G_p^2.$$

Substitution into (D.15) yields the explicit finite-W bound

$$\begin{aligned} \text{Var} \left(W^{-1} \sum_{i=1}^W z_i \right) &\leq \frac{1}{W} \left[v_* + 16 G_p^2 \sum_{h=1}^{W-1} \left(1 - \frac{h}{W} \right) \alpha_B(h)^{r_\eta} \right] \\ &\leq \frac{v_* + 16 G_p^2 \sum_{h \geq 1} \alpha_B(h)^{r_\eta}}{W}. \end{aligned}$$

The factor 16 is the constant obtained from Davydov's constant 8 and the two covariance triangles. It is sharper to apply Davydov to the conditional means $g_i(B_i)$ than to z_i itself, because independent Poisson shot noise contributes only to the diagonal variance v_* .

If ℓ_k is constant and the resulting marked process is strictly stationary, write $\gamma_\psi(h) = \text{Cov}(z_0, z_h)$. Then

$$\begin{aligned} \text{Var} \left(W^{-1} \sum_{k=1}^W z_k \right) &= W^{-1} \sum_{|h| < W} \left(1 - \frac{|h|}{W} \right) \gamma_\psi(h) \\ &\leq \frac{\sum_h |\gamma_\psi(h)|}{W}, \end{aligned}$$

and when $\sum_h |\gamma_\psi(h)| < \infty$,

$$\begin{aligned} \text{Var} \left(W^{-1} \sum_{k=1}^W z_k \right) &= \frac{\sigma_{\psi, \text{LR}}^2}{W} + o(W^{-1}), \\ \sigma_{\psi, \text{LR}}^2 &= \sum_h \gamma_\psi(h). \end{aligned}$$

The multiplicative notation $\sigma_{\psi, \text{LR}}^2 W^{-1} \{1 + o(1)\}$ is valid only when $\sigma_{\psi, \text{LR}}^2 > 0$. For $z_n = \epsilon_n - \epsilon_{n-1}$ the signed long-run variance is zero while the finite-window variance is $2\sigma_\epsilon^2/W^2$.

m) Remark D.4.4 (clamp probability: a diagnostic, not an extra risk term).: By the global Lipschitz property of the clamped inverse, (D.8) already covers clamped, all-zero, and out-of-range windows; no term $D_\Theta^2 \mathbb{P}(\mathcal{E}_n^c)$ is added to the risk. The clamp probability remains useful as a *diagnostic* for whether the localized variance regime of the low-photon expansion applies (low total information drives clamp saturation, in which case the inverse-information reading of (D.10) is off scope). With $\mathcal{E}_n = \{y_n \in M_n(\Theta)\}$, use the notation B_n, V_n, κ_n of the finite-window proof and put

$$r_n = \min\{M_n(\ell_n) - M_n(\theta_{\min}), M_n(\theta_{\max}) - M_n(\ell_n)\}.$$

If ℓ_n has margin Δ_n from $\partial\Theta$, then $r_n \geq \kappa_n \Delta_n$. Since $y_n - M_n(\ell_n) = B_n + Z_n$, with $\mathbb{E}Z_n = 0$ and $\text{Var}(Z_n) = V_n$,

$$\mathcal{E}_n^c \subseteq \{|B_n + Z_n| \geq r_n\}.$$

Therefore

$$\mathbb{P}(\mathcal{E}_n^c) \leq \min \left\{ 1, \frac{B_n^2 + V_n}{r_n^2} \right\} \leq \min \left\{ 1, \frac{B_n^2 + V_n}{(\kappa_n \Delta_n)^2} \right\}.$$

If $|B_n| < r_n$, the sharper centered bound is

$$\mathbb{P}(\mathcal{E}_n^c) \leq \min \left\{ 1, \frac{V_n}{(r_n - |B_n|)^2} \right\}.$$

An exponential Bernstein bound requires a positive residual margin $r_n - |B_n|$; it is not available merely from a positive parameter margin when the deterministic smoothing bias can reach the boundary.

On $\mathcal{E}_n^c$, both the clamped estimate and the target lie in Θ , so the clamp-event slice of the MSE is trivially at most $D_\Theta^2 \mathbb{P}(\mathcal{E}_n^c)$ — but this bound is *never added* to (D.8), which already dominates it through the global Lipschitz argument; it is recorded only to size the diagnostic.

For a window containing W distinct, conditionally independent Poisson counts,

$$\begin{aligned} \mathbb{P}(C_k = 0 \text{ for all } k \in W_n) \\ = \exp \left(- \sum_{k \in W_n} \lambda_k(\ell_k) \right) \leq e^{-W\lambda_-}. \end{aligned}$$

If a padding convention repeats observations, the sum and the cardinality in this formula are over distinct frames after repeated weights have been aggregated; in that case the exponent need not be $W\lambda_-$ (the first width-65 replicate-padded window of the former convention had only 33 distinct frames, so its all-zero probability was $e^{-33\lambda}$, not $e^{-65\lambda}$ — at $\lambda = 0.25$, 2.6×10^{-4} versus 8.8×10^{-8}).

n) Remark D.4.5 (scope: oracle carriers, known carrier law, and blind recovery — three distinct estimators).: Theorem C is conditional on the realized per-frame carriers (b_k) being *known*: this is precisely the oracle-known-carrier arm implemented in the low-photon experiment (Fig. 5), and it does **not** establish blind low-photon recovery from an unknown carrier law. Theorem B, by contrast, is the *blind high-count* stationary-carrier result; the two must not be merged. If an unobserved

carrier B has a fully *known* law F_B , one may instead calibrate the different marginal curve

$$\bar{q}_{F_B}(t) = \mathbb{E}_{B \sim F_B} m_\alpha(\Lambda_0 e^t B + d),$$

a *third*, law-calibrated estimator that needs its own variance or long-run-variance theorem (Remark D.4.3 supplies the covariance tool); it is not the per-frame known-carrier estimator above. If F_B is *unknown*, $\bar{q}_{F_B}$ is an unknown nonlinear function which, unlike the high-count logarithmic model, cannot generally be absorbed into one additive gauge constant: with $s = \Lambda_0 e^t$,

$$m_\alpha(sB) = \log \alpha + c_\alpha sB + a_{2,\alpha} s^2 B^2 + O(s^3 B^3),$$

$$a_{2,\alpha} = \frac{1}{2} \log \frac{\alpha(\alpha+2)}{(\alpha+1)^2} \neq 0,$$

so the laws $B \equiv 1$ and $\Pr(B = \frac{1}{2}) = \Pr(B = \frac{3}{2}) = \frac{1}{2}$ — same mean, different second moments — produce *different* soft-log response curves. The known-carrier theorem plus one homogeneous calibration inverse therefore does not yield an unknown-law blind theorem (though this does not rule out every joint semiparametric method).

o) Centered-log quotient risk.: Let $e = \hat{\ell} - \ell$ and $P_N = I - N^{-1} \mathbf{1}\mathbf{1}^\top$. Since P_N is an orthogonal projection,

$$\frac{1}{N} \mathbb{E} \|P_N e\|_2^2 \leq \frac{1}{N} \sum_{n=1}^N \mathbb{E} e_n^2,$$

so the pointwise bounds transfer directly to the record-centered log-gain loss of Fig. 5; overlapping windows need not be independent for this expectation bound.

Shrinkage-bias caveat (load-bearing, per Sec. 6). For $\bar{\lambda} < 1$ the windowed soft-log statistic is *bias-fragile*: the mass at $C = 0$ pulls the weighted mean of ψ_α toward $\log \alpha$, and the amplification factor $(\kappa_{\max}/\kappa_n)^2$ in the specialized bias term of (D.8) blows up as the window sensitivity vanishes with $\bar{\lambda}_n \rightarrow 0$ (at $d = 0$; window-average form $M'_n(\ell_n) \approx c_\alpha \bar{\lambda}_n$). Empirically the MSE of the *uncalibrated* proxy (and of Anscombe) can then sit *below* the pointwise Fisher reference curve — that is bias, not super-efficiency. It does not contradict (D.11)–(D.12), whose minimax bounds apply to all estimators over a local parameter neighborhood: a biased estimator may beat $1/I_n$ at selected parameters only by paying risk elsewhere. The $1/(W\bar{\lambda})$ law is asserted only in the variance-limited regime above the $\bar{\lambda} \sim 1$ crossover (and under the local gain flatness of (D.10)); below it, honest accounting reports the bias floor, and all Fisher comparisons are made in the log-domain loss of this theorem (Fig. 5's `log_gain_mse` metric, against the *oracle common-shift diagnostic* $\text{mean}_n[1/I_n]$ with $I_n = \sum_{k \in W_n} \lambda_k$ at $d = 0$, per (D.13) — *not* a nuisance-path quotient CRB) — the gain-domain relMSE column retained for continuity is a different loss and is not Fisher-comparable. Recovering the rate below the crossover requires the offset-design, Anscombe [28] ($2\sqrt{C+3/8}$), or full Poisson-mixture MLE variants; for the mixture MLE the claim $\text{MSE} \approx 1/(WJ(\theta))$ with $J \sim \mathbb{E}\lambda$ at low photons is standard parametric asymptotics under regularity of the mixture likelihood and is *not* re-proven here.

p) What this does not claim.: The oracle experiment can identify absolute log gain because Λ_0 and the carriers are fixed and known; the implementation and reported loss deliberately center the record to align with the blind task, so no separate absolute-scale performance claim is reported. In the Fig. 5 protocol the carriers b_k are *oracle* values from the simulation truth (a flat-field-calibrated benchmark), so Fig. 5

upper-bounds what a real flat-field calibration could achieve rather than demonstrating blind carrier recovery — Theorem C is an oracle-known-carrier result, Theorem B is the blind high-count result, and the known-law estimator of Remark D.4.5 is a third object; none substitutes for the others. Miscalibration beyond the measured interpolation term ($\varepsilon_{\text{cal}}/\kappa_n + \varepsilon_{\text{bis}}$ of (D.8b), with ε_{cal} an empirical midpoint maximum pending the (D.8c) certificate) adds a bias not covered by (D.8). Constants in (D.8)–(D.9) are not tracked. Smoothness beyond $\beta = 1$ is claimed only in the internal-window homogeneous-carrier / sensitivity-balanced form of Remark D.4.2 (display (D.14)); for arbitrary carriers the carrier-weighted first moment obstructs every $\beta > 1$ rate, and the implementation’s truncated edge windows are not covered even under a homogeneous carrier. The unconditional $1/(W\bar{\lambda})$ law is withdrawn in favor of (D.10); the flat-gain specialization is the only $1/(W\bar{\lambda})$ statement retained. The minimax matching of (D.9) is at rate level only, *in the centered log-domain loss and at $d = 0$* , proved by the direct Poisson Le Cam/Assouad bounds (D.11)–(D.12) — not inherited from Theorem D, whose Gaussian model is not a submodel of the mixed-Poisson experiment; for $d > 0$ every information statement must replace $W\bar{\lambda}$ by $\sum_k \mathbb{E}_B s_k^2/(s_k + d)$. The plotted reference $\text{mean}_n[1/I_n]$ is the oracle common-shift diagnostic of (D.13); MSE above it does not establish unbiasedness, and MSE below it signals shrinkage bias, not super-efficiency. No optimality of any kind is claimed in the gain-domain (linear) metric, and no claim is made that the calibrated estimator beats bias-accepting transforms (e.g. Anscombe) in MSE below the $\bar{\lambda} \sim 1$ crossover — empirically it does not (Sec. 9). As $\Phi \rightarrow 0$ with no offset and no reference, Fisher information for the gain vanishes — an information limit no estimator evades (Sec. 10).

APPENDIX E

SRHT WHITENING: EXACT WALSH CONDITION AND PERMUTATION BOUNDS

This appendix proves the claims of Sec. 7: the exact covariance identity $\text{Cov}_D(Z_g, Z_h) = \hat{w}(g+h)$, the necessary-and-sufficient Walsh-flatness conditions for sign-only whitening (pairwise, spectral-window, and window-average forms), the Bernstein–Serfling permutation bound and its union-bound corollary, the permutation-alone carrier-variance identity with its flat-object counterexample, and the Hanson–Wright window-energy bound.

A. E.0 Setting and standing assumptions

- **(E1)** $K = 2^m$ and pixels are indexed by the Walsh group $G = \mathbb{F}_2^m$, with characters $\chi_g(j) = (-1)^{g \cdot j}$, unnormalized Hadamard matrix $H_{g,j} = \chi_g(j)$, and orthonormal carrier $U = K^{-1/2}H$. For every $g \neq 0$, χ_g is balanced: exactly $K/2$ entries equal $+1$.
- **(E2)** The object $T \in \mathbb{R}^K$ is deterministic and nonzero. Write $w_j = T_j^2$; S_1 , S_2 , and the spread functionals K_{eff} , K_∞ , K_4 are as in Appendix C. The Walsh transform of w is $\hat{w}(h) = \sum_j \chi_h(j) w_j$, so $\hat{w}(0) = S_2$.
- **(E3)** The SRHT coefficient vector is $x = UDPT$, where $D = \text{diag}(\eta_1, \dots, \eta_K)$ with i.i.d. Rademacher signs η_j ,

and P is a uniformly random pixel permutation independent of D . The normalized carrier is $Z_g = \sqrt{K} x_g = \sum_j \chi_g(j) \eta_j (PT)_j$. Physical nonnegative patterns are the offset form $B_g = \mu S_1 + \sigma Z_g$ with positivity margin $\mu S_1 > \sigma |Z_g|$; the bucket record is $R_n = a_n B_n$ with $\ell = \log a$ as in the main text. A *window* is a subset $A \subset G$ of $W = |A|$ rows.

Permutation leaves S_1 , S_2 , K_{eff} , K_4 , K_∞ invariant; it only reorders w , i.e. replaces $\hat{w}$ by $\hat{w}^P$ with $w_j^P = T_{Pj}^2$.

B. E.1 The exact covariance identity

Lemma E.1. *Under (E1)–(E3) with P fixed (in particular $P = \text{id}$), $\mathbb{E}_D Z_g = 0$ and for all $g, h \in G$*

$$\text{Cov}_D(Z_g, Z_h) = \sum_j \chi_{g+h}(j) (PT)_j^2 = \widehat{w^P}(g+h). \quad (\text{E.1})$$

Proof. $\mathbb{E} \eta_j = 0$ gives the mean. Since $\mathbb{E} \eta_i \eta_j = \delta_{ij}$,

$$\begin{aligned} \mathbb{E}_D Z_g Z_h &= \sum_{i,j} \chi_g(i) \chi_h(j) (PT)_i (PT)_j \mathbb{E} \eta_i \eta_j \\ &= \sum_j \chi_g(j) \chi_h(j) (PT)_j^2, \end{aligned}$$

and $\chi_g \chi_h = \chi_{g+h}$ by the character property of the Walsh group. ■

Consequently the covariance of any window A is the pattern matrix $\Gamma_A(g, h) = \widehat{w^P}(g+h)$, and, normalized by the common marginal variance S_2 , $\Gamma_A/S_2 = I_A + R_A$ with off-diagonal entries $R_A(g, h) = \widehat{w^P}(g+h)/S_2$, $g \neq h$. The entire gain–object coupling of sign-randomized acquisition is therefore carried by the non-DC Walsh spectrum of the squared (permuted) object — nothing else.

C. E.2 Sign-only whitening: necessary and sufficient conditions

Theorem E.2 (exact Walsh condition). *Under (E1)–(E3) with no permutation ($P = \text{id}$):*

- 1) (exact pairwise) $\text{Cov}_D(Z_g, Z_h) = S_2 \delta_{g,h}$ for all g, h **iff** $\hat{w}(q) = 0$ for every $q \neq 0$, equivalently **iff** $T_j^2 = S_2/K$ for all j (T^2 flat).
- 2) (pairwise ε -whitening) $|\text{Cov}_D(Z_g, Z_h)| \leq \varepsilon S_2$ for all $g \neq h$ **iff** $\max_{q \neq 0} |\hat{w}(q)| \leq \varepsilon S_2$.
- 3) (spectral window) For a window class $\mathcal{A}$, $\|\Gamma_A/S_2 - I_A\|_{\text{op}} \leq \varepsilon$ for all $A \in \mathcal{A}$ **iff** $\sup_{A \in \mathcal{A}} \|R_A\|_{\text{op}} \leq \varepsilon$.
- 4) (window average) With $\bar{Z}_A = W^{-1} \sum_{g \in A} Z_g$ and multiplicity $m_A(q) = \#\{(g, h) \in A \times A : g \neq h, g+h=q\}$, the variance $\text{Var}(\bar{Z}_A)$ lies in $[(1 \pm \varepsilon) S_2/W]$ for every $A \in \mathcal{A}$ **iff** $\sup_{A \in \mathcal{A}} |\sum_{q \neq 0} m_A(q) \hat{w}(q)| \leq \varepsilon W S_2$.

Proof. (1) If whitening holds, then for any $q \neq 0$ pick g, h with $g+h=q$ (always possible in G); (E.1) forces $\hat{w}(q) = 0$. Conversely, vanishing non-DC coefficients make (E.1) equal $S_2 \delta_{g,h}$. The flatness equivalence is Walsh inversion: the transform is invertible, so $\hat{w}$ supported on $\{0\}$ **iff** w is constant, i.e. $T_j^2 = S_2/K$. (2) Immediate from (E.1) after dividing by

S_2 . (3) The normalized window covariance is $I_A + R_A$; the statement is definitional. (4) Direct expansion:

$$\begin{aligned}\text{Var}(\bar{Z}_A) &= W^{-2} \sum_{g,h \in A} \hat{w}(g+h) \\ &= \frac{S_2}{W} + W^{-2} \sum_{q \neq 0} m_A(q) \hat{w}(q),\end{aligned}$$

and the two-sided requirement is exactly the stated inequality. ■

Two standard reductions relate the forms. If $\mathcal{A}$ contains all two-point windows, (3) implies (2), and (2) is the sharpest scalar condition. In the converse direction, since R_A has zero diagonal, Gershgorin's disc theorem gives $\|R_A\|_{\text{op}} \leq (W-1) \max_{q \neq 0} |\hat{w}(q)|/S_2$, so pairwise condition (2) at tolerance $\varepsilon/(W_{\max}-1)$ implies spectral condition (3) for all windows with $W \leq W_{\max}$. This Gershgorin step is a (generally loose) sufficient bound, not an equivalence.

The obstruction is sharp: if T^2 is aligned with the positive support of a single character χ_q , then $|\hat{w}(q)| = S_2$ and the row pair (g, h) with $g+h = q$ is *perfectly* correlated under sign randomization. Signs cannot remove this; it is a property of the object's pixel ordering.

a) *What this does not claim.*: Theorem E.2 characterizes second-moment decorrelation over the sign draw for a *fixed* object and ordering; it does not assert Gaussianity of Z_g (a Rademacher sum, whose normal approximation requires $K_\infty \rightarrow \infty$ with Berry–Esseen error $\lesssim K_\infty^{-1/2}$), independence of rows, or anything about noise, gain drift, or estimation error — those enter through Secs. 5–6 and Appendix F. It is also a statement about the design/object pair, not a high-probability event: no randomness beyond D is involved.

D. E.3 Random permutation as a probabilistic Walsh-flattener

By Lemma E.1, Theorem E.2 holds verbatim for a realized permutation P with w replaced by w^P : the *deterministic* necessary-and-sufficient condition remains Walsh-flatness, now of the permuted squared object. A random P makes that condition likely.

Theorem E.3 (Bernstein–Serfling permutation bound). *Under (E1)–(E3), there exist universal constants $c, C > 0$ such that for every fixed $q \neq 0$ and $\varepsilon > 0$,*

$$\Pr(|\widehat{w^P}(q)| \geq \varepsilon S_2) \leq 2 \exp[-c \min(\varepsilon^2 K_4, \varepsilon K_\infty)], \quad (\text{E.2})$$

and by the union bound over the $K-1$ nonzero frequencies,

$$\begin{aligned}\Pr\left(\max_{q \neq 0} |\widehat{w^P}(q)| \geq \varepsilon S_2\right) \\ \leq 2(K-1) \exp[-c \min(\varepsilon^2 K_4, \varepsilon K_\infty)].\end{aligned} \quad (\text{E.3})$$

In particular, pairwise ε -whitening (Theorem E.2(2) for w^P) holds with probability at least $1 - \delta$ whenever

$$\min(\varepsilon^2 K_4, \varepsilon K_\infty) \geq C \log(K/\delta). \quad (\text{E.4})$$

Proof. Fix $q \neq 0$ and let $A_q = \{j : \chi_q(j) = +1\}$, so $|A_q| = K/2$ by (E1). Then

$$\widehat{w^P}(q) = \sum_{j \in A_q} w_j^P - \sum_{j \notin A_q} w_j^P = 2 \sum_{j \in A_q} w_j^P - S_2,$$

a centered, doubled sum of $n = K/2$ draws *without replacement* from the finite population $\{w_1, \dots, w_K\}$ with mean $\mu_w = S_2/K$. Writing $S_q = \sum_{j \in A_q} w_j^P$, its exact variance is

$$\begin{aligned}\text{Var}(S_q) &= \frac{K^2}{4(K-1)} \sigma_w^2, \\ \text{Var}(2S_q - S_2) &= \frac{K^2}{K-1} \sigma_w^2 \\ &\leq 2 \sum_j w_j^2.\end{aligned}$$

where $\sigma_w^2 = K^{-1} \sum_j (w_j - \mu_w)^2$ and $K \geq 2$; also $0 \leq w_j \leq \|T\|_\infty^2 = S_2/K_\infty$. Thus the variance proxy is bounded by a universal constant times $\sum_j w_j^2 = S_2^2/K_4$. The Bernstein inequality for sampling without replacement (Serfling [31]; the Bernstein–Serfling form of Bardenet–Maillard [43]) then gives, for $t > 0$,

$$\Pr(|\widehat{w^P}(q)| \geq t) \leq 2 \exp\left[-c \min\left(\frac{t^2}{S_2^2/K_4}, \frac{t}{S_2/K_\infty}\right)\right].$$

Setting $t = \varepsilon S_2$ yields (E.2); (E.3) is the union bound; (E.4) follows by requiring the right side of (E.3) to be at most δ and absorbing the factor $2(K-1)$ into $C \log(K/\delta)$. ■

a) *What this does not claim.*: (E.2)–(E.4) are *sufficient* high-probability conditions with unspecified universal constants; they are not the deterministic criterion for a realized acquisition — that remains Walsh-flatness of w^P (Theorem E.2), which can and should be checked numerically for the ordering actually used. For spectral whitening of all windows up to size $W_{\max}$, the simple route substitutes $\varepsilon \rightarrow \varepsilon/(W_{\max}-1)$ in (E.4) via Gershgorin; sharper window-size dependence via matrix-Bernstein or Hanson–Wright arguments (Sec. E.5) is a concentration refinement, again sufficient rather than necessary. Both branches of the min collapse when energy concentrates on few pixels (K_4, K_∞ small): permutation cannot flatten a spike.

E. E.4 Permutation alone: exact carrier variance and the flat-object counterexample

Theorem E.4 (permutation-alone variance). *Under (E1)–(E2), with no signs and a uniformly random permutation P , the carrier $Z_g(P) = \sum_j \chi_g(j) T_{Pj}$ of any non-DC row $g \neq 0$ satisfies $\mathbb{E}_P Z_g = 0$ and*

$$\text{Var}_P Z_g = \frac{K S_2 - S_1^2}{K-1} = S_2 \frac{K - K_{\text{eff}}}{K-1}. \quad (\text{E.5})$$

If additionally $S_1 \neq 0$ and $K_{\text{eff}} \geq 1$ — in particular, for a nonnegative physical object — then for offset patterns $B_g = \mu S_1 + \sigma Z_g(P)$ with positivity margin,

$$\frac{\text{Var}_P B_g}{(\mathbb{E}_P B_g)^2} = \left(\frac{\sigma}{\mu}\right)^2 \frac{K - K_{\text{eff}}}{(K-1) K_{\text{eff}}} \leq \left(\frac{\sigma}{\mu}\right)^2 \frac{1}{K_{\text{eff}}}. \quad (\text{E.6})$$

No matching lower bound of order $1/K_{\text{eff}}$ holds uniformly over objects: for flat T (T_j constant), $K_{\text{eff}} = K$ while $Z_g(P) = 0$ for *every* permutation and every $g \neq 0$.

Proof. For $g \neq 0$, $\sum_i \chi_g(i) = 0$, so $\mathbb{E}_P Z_g = (\sum_i \chi_g(i)) \mathbb{E}_P T_{P1} = 0$. Exchangeability of $(T_{P1}, \dots, T_{PK})$

gives $\mathbb{E}_P T_{P_i}^2 = S_2/K$ and, for $i \neq j$, $\mathbb{E}_P T_{P_i} T_{P_j} = (S_1^2 - S_2)/(K(K-1))$. Hence

$$\mathbb{E}_P Z_g^2 = S_2 + \frac{S_1^2 - S_2}{K(K-1)} \sum_{i \neq j} \chi_g(i) \chi_g(j),$$

and $\sum_{i \neq j} \chi_g(i) \chi_g(j) = (\sum_i \chi_g(i))^2 - K = -K$, giving (E.5); the second form follows from $K_{\text{eff}} = S_1^2/S_2$. For (E.6), $\mathbb{E}_P B_g = \mu S_1$ and $\text{Var}_P B_g = \sigma^2 \text{Var}_P Z_g$; divide. The counterexample: if $T_j \equiv t$, every non-DC Walsh sum of a constant vector vanishes identically, while $S_1^2/S_2 = K$. Note (E.5) also exhibits the finite-population correction — the relative variance is $(\sigma/\mu)^2(1/K_{\text{eff}} - 1/K) \cdot K/(K-1)$. ■

a) *What this does not claim.*: Theorem E.4 does not say pixel permutation alone whitens or stationarizes time. In the $K_{\text{eff}} \geq 1$ physical-object regime, a shared random P_{col} gives every non-DC row the same one-dimensional mean and variance and an $O(1/K_{\text{eff}})$ upper variance scale; it supplies neither a stationary/mixing realized time series nor full joint row exchangeability. It is not equivalent to i.i.d. Rademacher or nonnegative random-pattern carrier noise, and it guarantees no $\Theta(1/K_{\text{eff}})$ excitation — a flat object has exactly zero. Two-sided coefficientwise control requires the sign flip and the spectral-spread condition (E.4) together. Full finite-population exchangeability instead belongs to a random acquisition order P_{row} (with adjacent complementary pairs treated as blocks). A common P_{col} fails joint row exchangeability by the triple-moment counterexample (E.9). Once any randomization draw is conditioned upon, the sequence is deterministic; concentration is with respect to the declared ensemble, not a claim that every realization is stationary or mixing.

Proposition E.4' (Walsh triple-moment counterexample to joint row exchangeability). *Let $K = 2^m \geq 8$, let χ_g be the Walsh characters, and $Z_g(P) = \sum_j \chi_g(j) T_{P_j}$ under one uniform pixel permutation P shared by all rows. Let $\bar{T} = K^{-1} \sum_j T_j$ and $S_{3,c} = \sum_j (T_j - \bar{T})^3$. For distinct nonzero rows g, h, r ,*

$$\mathbb{E}_P [Z_g Z_h Z_r] = \begin{cases} \frac{K^2}{(K-1)(K-2)} S_{3,c}, & g+h+r=0, \\ 0, & g, h, r \text{ lin. indep.} \end{cases} \quad (\text{E.9})$$

For a generic object with $S_{3,c} \neq 0$, relational triples ($g+h+r=0$) and independent triples therefore have different joint laws, disproving full row exchangeability under a common pixel permutation.

Proof. Replace T_j by $x_j = T_j - \bar{T}$ (non-DC Walsh coefficients unchanged). For the without-replacement sample $X_i = x_{P(i)}$, standard finite-population moments give $\mathbb{E} X_i^3 = S_{3,c}/K$, $\mathbb{E} X_i^2 X_j = -S_{3,c}/\{K(K-1)\}$ for $i \neq j$, and $\mathbb{E} X_i X_j X_k = 2S_{3,c}/\{K(K-1)(K-2)\}$ for distinct i, j, k . If $g+h+r=0$, character orthogonality gives sign sums K over the diagonal class, $-K$ over each exactly-two-equal pairing, and $2K$ over the all-distinct class; substitution gives the first line of (E.9). If $g+h+r \neq 0$, all the corresponding character sums vanish. ■

F. E.5 Hanson–Wright window-energy bound

Whitening of window *energy* — the quantity the window estimator of Sec. 6 actually averages — admits a direct quadratic-form bound. For a window A of W rows let P_A select those rows and set

$$Q_A = \|P_A U D P T\|_2^2 = \eta^\top M \eta, \\ M = \text{diag}(P T) U_A^\top U_A \text{diag}(P T),$$

with η the Rademacher sign vector and U_A the selected rows of U .

Proposition E.5 (window-energy concentration). *Condition on any realized P . Then $\mathbb{E}_D Q_A = (W/K) S_2$, and there is a universal $c > 0$ with*

$$\Pr_D \left(|Q_A - \frac{W}{K} S_2| \geq \varepsilon \frac{W}{K} S_2 \right) \\ \leq 2 \exp \left[-c \min \left(\varepsilon^2 \frac{W K_\infty}{K}, \varepsilon \frac{W K_\infty}{K} \right) \right]. \quad (\text{E.7})$$

Over a specified family of M windows, uniform relative energy whitening at tolerance $\varepsilon \leq 1$ holds with probability $\geq 1 - \delta$ once

$$\frac{W K_\infty}{K} \geq C \varepsilon^{-2} \log(M/\delta). \quad (\text{E.8})$$

Proof. Since U has orthonormal rows, $U_A U_A^\top = I_W$, so $U_A^\top U_A$ is an orthogonal projector and $(U_A^\top U_A)_{jj} = \sum_{g \in A} U_{gj}^2 = W/K$ (Hadamard entries are $\pm K^{-1/2}$). Hence $M \succeq 0$, and $\mathbb{E}_D Q_A = \text{tr } M = \sum_j (P T)_j^2 (W/K) = (W/K) S_2$. Because $\eta_j^2 = 1$, the diagonal of M contributes deterministically and $Q_A - \mathbb{E} Q_A$ is a pure off-diagonal Rademacher chaos, to which the Hanson–Wright inequality applies [Hanson–Wright [44]; sub-Gaussian form of Rudelson–Vershynin [45]]:

$$\Pr(|Q_A - \text{tr } M| \geq t) \leq 2 \exp \left[-c \min \left(\frac{t^2}{\|M\|_F^2}, \frac{t}{\|M\|_{\text{op}}} \right) \right].$$

The proxies: $\|M\|_{\text{op}} \leq \|\text{diag}(P T)\|_{\text{op}}^2 \|U_A^\top U_A\|_{\text{op}} \leq \|T\|_\infty^2 = S_2/K_\infty$, and for PSD M , $\|M\|_F^2 \leq \|M\|_{\text{op}} \text{tr } M \leq (W/K) S_2 \|T\|_\infty^2$. Substituting $t = \varepsilon (W/K) S_2$ gives $t^2/\|M\|_F^2 \geq \varepsilon^2 W K_\infty/K$ and $t/\|M\|_{\text{op}} \geq \varepsilon W K_\infty/K$, i.e. (E.7); for $\varepsilon \leq 1$ the quadratic branch is active. A union bound over M windows gives (E.8). ■

For dense objects of bounded dynamic range, $K_\infty \asymp K_4 \asymp K_{\text{eff}} \asymp K$ and (E.8) reduces to the mild $W \gtrsim \varepsilon^{-2} \log M$ — the window only needs logarithmic length. For an object supported on m comparable pixels, $K_\infty \asymp m$ and the condition hardens to $W m/K \gtrsim \varepsilon^{-2} \log M$: this is precisely where K_{eff} alone is the wrong functional and the three spread parameters separate (Sec. 6).

a) *What this does not claim.*: Proposition E.5 controls window *energy* over the sign draw for a pre-specified finite window family; it is a sufficient concentration tool, not the deterministic necessary-and-sufficient condition (which remains the spectral condition of Theorem E.2(3) applied to w^P), and it does not cover all $\binom{K}{W}$ windows simultaneously without paying for the larger union. The constants c, C are universal but not optimized, and no claim is made that (E.8) is tight in its W - or K_∞ -dependence — Hanson–Wright and matrix-Bernstein refinements can improve window-family dependence

for structured $\mathcal{A}$, but they do not change the deterministic criterion. Finally, all of Appendix E concerns the carrier statistics of the *design*; the translation of whitening into gain-estimation rates ($1/W$ per window) and into image error (leverage $B_L = 1$ for full SRHT inversion) is carried out in Appendix D and Appendix F respectively.

G. E.6 Two randomization semantics: pixel randomization (P_{col} , D) vs. acquisition-order randomization (P_{row})

This subsection separates two operations that earlier drafts conflated under the single word “permutation”. Write P_{col} for a uniformly random **pixel (column) permutation** of the object — the P of assumption (E3) — write D for the i.i.d. Rademacher **pixel sign** diagonal of (E3), and write P_{row} for a uniformly random reordering of the **acquisition (Hadamard row / time) order**. These control different claims:

- **Appendix E’s theorems govern cross-row covariance.** By Lemma E.1, the sign-draw covariance $\text{Cov}_D(Z_g, Z_h) = w^P(g+h)$ is a function of the permuted squared object w^P alone; it is therefore **invariant to the acquisition order** — P_{row} does not enter Lemma E.1 or Theorems E.2–E.3 at all. The whitening randomization of the theory is $x = UDP_{\text{col}}T$: pixel signs plus pixel permutation, in *natural* acquisition order.
- **The manuscript’s temporal-stationarity screen is controlled by acquisition-order structure.** The Brown–Forsythe/Levene screen [29], [30] of Sec. 7 probes temporal homogeneity of the acquired time series; it responds to whatever breaks the coarse-to-fine sequency-energy ordering of natural Hadamard rows — which either P_{col} or P_{row} accomplishes.

a) Factorized ablation (empirical attribution; results/perm_ablation_r1/).: All arms share the same ordered natural-Hadamard baseline ($K = 4096$, 64×64 , 8192 paired frames, noiseless carrier), the same 10-object panel, and the same Levene protocol (8 chunks, reject at $p < 10^{-3}$); 32 draws per randomized arm. Levene rejection rates:

arm	P_{col}	D	P_{row}	rate
ordered (baseline)	–	–	–	0.700
P_{col} only (theory’s P)	Y	–	–	0.056
P_{row} only	–	–	Y	0.122
D only	–	Y	–	0.434
$P_{\text{col}} + D$ (theory)	Y	Y	–	0.119
$P_{\text{row}} + D$ (code’s <code>srht_paired</code>)	–	Y	Y	0.059
$P_{\text{col}} + P_{\text{row}}$ (old Exp.-B arm)	Y	–	Y	0.059

The $P_{\text{col}} + P_{\text{row}}$ arm reproduces the historical headline ($19/320 = 5.9\%$) exactly, confirming the seed-compatible reconstruction of the old experiment; that historical 5.9% is attributable to *either* permutation alone (P_{col} : 5.6%; P_{row} : 12.2%).

b) Code caveat (stated plainly).: The implemented basis `make_srht_paired_basis` (`src/basis.py`, `permute_rows=True`) applies pixel signs D but permutes **Hadamard rows** (acquisition order), i.e. it implements $P_{\text{row}} + D$ — **not** the theorem’s $P_{\text{col}} + D$. Both combinations are empirically sufficient for the temporal-stationarity screen

(5.9% and 11.9%, versus 70% ordered), while **D alone is not** (43.4%): sign flips whiten carrier *amplitudes* (chunk-std CV collapses from 0.76 to 0.03) but leave the temporal sequency ordering intact, so the chunk-variance test still rejects. Text and captions must therefore attach each claim to its own arm: the covariance-whitening claims of E.1–E.3 attach to $P_{\text{col}} (+D)$; the temporal-screen results attach to the arm actually acquired ($P_{\text{row}} + D$ for the code’s SRHT).

c) What this does not claim.: The table is an empirical attribution on one object panel and one protocol, not a theorem; no claim is made that P_{row} enjoys the concentration guarantees (E.2)–(E.4), which are proved for P_{col} . The small increase from adding D on top of P_{col} (5.6% $\rightarrow$ 11.9%) is a property of the variance-homogeneity test on near-white sequences, not a whitening failure. Nothing here alters the theorems of E.1–E.5; it fixes which experimental arm may cite them.

APPENDIX F

THE MASTER FINITE-NOISE IDENTITY AND THE FLIP BOUNDARY

This appendix proves Theorem 1 (the master finite-noise relMSE identity) and Prop 3 (the finite- N flip boundary) of Sec. 8, together with the noise plug-ins and the three basis specializations quoted there. Throughout, $T \in \mathbb{R}^K$ is the object, with S_1, S_2, K_{eff} as in Appendix C; $A \in \mathbb{R}^{N \times K}$ is the design, $b = AT$ the ideal bucket vector; the raw bucket is $R_n = a_n B_n$ with gain $a_n = e^{\ell_n}$, $\ell \in S$ (the drift class of Sec. 4, $p = \dim S$). After the pipeline applies a gain estimate $\hat{a}_n = e^{\hat{\ell}_n}$, the corrected bucket is

$$z_n = R_n / \hat{a}_n = (1 + \delta_n) b_n + \xi_n, \quad \delta_n = \frac{a_n}{\hat{a}_n} - 1 = e^{\ell_n - \hat{\ell}_n} - 1, \quad (\text{F.1})$$

where δ_n is the *residual* multiplicative gain error and ξ_n is additive bucket-domain noise after correction, with $\mathbb{E} \delta = m_\delta$, $\text{Cov}(\delta) = V_\delta$, $\mathbb{E} \xi = 0$, $\text{Cov}(\xi) = \Sigma_\xi$, and $\delta \perp \xi$ (uncorrelated, $C_{\delta\xi} = \text{Cov}(\delta, \xi) = 0$ — a hypothesis, revisited in the cross-term note below). The reconstruction is $\hat{T} = Lz$ for any fixed linear operator $L \in \mathbb{R}^{K \times N}$ (exact inverse, pseudoinverse, DGI correlator, or a regularized estimator linearized around a fixed design).

A. F.1 The master identity

Theorem 1 (exact second-moment bridge). *Assume model (F.1) with A, T, L fixed (all expectations conditional on them), and assume — as part of the hypothesis, not as background convention — that:*

- (H1) $\mathbb{E} \xi = 0$;
- (H2) δ and ξ are uncorrelated, $C_{\delta\xi} := \text{Cov}(\delta, \xi) = 0$;
- (H3) the gain estimate $\hat{a}$ is **exogenous** with respect to the reconstruction record — computed independently of the noise realization ξ (injected/synthetic residuals, an independent calibration record, or a cross-fitted estimate from held-out frames) — so that (H1)–(H2) hold conditionally on the record being reconstructed;
- (H4) the second moments $m_\delta, V_\delta, \Sigma_\xi$ are finite.

Then, exactly and with no Gaussian or small-noise approximation,

$$\frac{\mathbb{E}\|\hat{T} - T\|_2^2}{S_2} = \underbrace{\frac{\|(LA - I)T + L \text{diag}(b) m_\delta\|_2^2}{S_2}}_{\text{bias}} + \underbrace{\frac{\text{tr}(L \text{diag}(b) V_\delta \text{diag}(b) L^\top)}{S_2}}_{\text{residual gain}} + \underbrace{\frac{\text{tr}(L \Sigma_\xi L^\top)}{S_2}}_{\text{additive noise}}. \quad (\text{F.2})$$

(a) If moreover $\mathbb{E} \delta_n = 0$ and the δ_n are uncorrelated with common variance v , the residual-gain term equals $v B_L(A, T)$ with the **leverage**

$$B_L(A, T) = \frac{1}{S_2} \sum_{n=1}^N b_n^2 \|Le_n\|_2^2. \quad (\text{F.3})$$

(b) If instead the residual gains are stationary with $\text{Cov}(\delta_n, \delta_m) = v r_\delta(n - m)$, the residual-gain term equals

$$R_{\text{gain}} = \frac{v}{S_2} \sum_{n,m} r_\delta(n - m) b_n b_m \langle Le_n, Le_m \rangle. \quad (\text{F.4})$$

Proof. Write $z = b + \text{diag}(b) \delta + \xi$, so $\hat{T} - T = (LA - I)T + L \text{diag}(b) \delta + L\xi$. Decompose $\delta = m_\delta + (\delta - m_\delta)$. The mean of $\hat{T} - T$ is $(LA - I)T + L \text{diag}(b) m_\delta$ (using (H1)); its squared norm is the bias term. By the bias–variance decomposition $\mathbb{E}\|X\|^2 = \|\mathbb{E}X\|^2 + \text{tr Cov}(X)$ and (H2), the covariance of $\hat{T}$ is $L \text{diag}(b) V_\delta \text{diag}(b) L^\top + L \Sigma_\xi L^\top$; taking traces and dividing by S_2 gives (F.2). For (a), $V_\delta = vI$ and $\text{tr}(L \text{diag}(b) \text{diag}(b) L^\top) = \sum_n b_n^2 \|Le_n\|_2^2$, giving (F.3). Case (b) is the coordinate expansion of the same trace with $(V_\delta)_{nm} = v r_\delta(n - m)$. ■

a) *Remark F.1.1 (same-record gain estimates violate the hypotheses; scope of the exact identity).*: Hypothesis (H3) is not decorative. When $\hat{a}$ is estimated from the *same* noisy record that carries ξ — the generic situation for a blind pipeline — the conditional mean $\mathbb{E}[\xi \mid \text{record}]$ is generically nonzero and δ is correlated with ξ , so (H1)–(H2) fail: the mean of $\hat{T} - T$ acquires an additional bias term $L m_\xi$ (with $m_\xi = \mathbb{E} \xi \neq 0$) inside the bias norm, and the covariance acquires the cross-term $(2/S_2) \text{tr}[L \text{diag}(b) C_{\delta\xi} L^\top]$. Neither term is included in (F.2). The exact identity therefore covers the **exogenous / injected-residual case** — which is precisely what the injected-residual protocol of Sec. 9 (experiment E4, Fig. 10) validates — and cross-fitted estimates that restore (H1)–(H2) by construction; for same-record blind pipelines, (F.2) is exact only up to the two extra terms above.

b) *What this does not claim.*: With (H1)–(H3) now stated in the theorem, the caveat is Remark F.1.1: (F.2) is not claimed for same-record blind gain estimation, where the $L m_\xi$ bias and the $C_{\delta\xi}$ cross-term appear. Beyond that, (F.2) is a second-moment accounting identity for a *linear* reconstructor with the design held fixed; it does not bound nonlinear or data-adaptive L , and for random-pattern pipelines the constants below are obtained by additionally averaging over the pattern draw. It

says nothing about how small v can be made — that is the estimation-rate question of Sec. 6. Finally, the scalar collapse $v B_L$ requires uncorrelated residuals: for smooth (coherent) residual gain — the generic output of a windowed blind estimator — only the matrix form (F.4) is correct. In particular, the simple summary “orthogonal inversion gives v ” holds only for independent coefficient-wise residuals under exact inversion.

B. F.2 Noise plug-ins

Proposition F.2 (read and physical nonnegative-Poisson noise, exact). (a) Gaussian read. Suppose detector-domain noise $\eta_n \sim \mathcal{N}(0, \sigma_{\text{read},n}^2)$ is independent across frames and the correction factor $\hat{a}_n$ is exogenous to η_n as required by (H3). Conditional on $(a, \hat{a})$, $\xi_n = \eta_n / \hat{a}_n$ and

$$\text{Var}(\xi_n \mid a, \hat{a}) = \frac{\sigma_{\text{read},n}^2}{\hat{a}_n^2} = \frac{\sigma_{\text{read},n}^2}{a_n^2} \left(\frac{a_n}{\hat{a}_n} \right)^2.$$

For fixed calibrated gains $\hat{a}_n = a_n$, this is $\tau_{G,n}^2 := \sigma_{\text{read},n}^2 / a_n^2$. (b) Poisson. Suppose $C_n \mid T, a \sim \text{Pois}(\Phi_n)$ with $\Phi_n = \gamma_n a_n b_n$, $z_n = C_n / (\gamma_n \hat{a}_n)$, and $\hat{a}_n$ is exogenous to C_n conditional on (T, a) . Then exactly

$$\mathbb{E}[z_n \mid T, a, \hat{a}] = \frac{a_n}{\hat{a}_n} b_n, \quad \text{Var}(z_n \mid T, a, \hat{a}) = \frac{b_n^2}{\Phi_n} \left(\frac{a_n}{\hat{a}_n} \right)^2. \quad (\text{F.5})$$

provided $b_n \geq 0$, $\gamma_n > 0$, and every used count row has $\Phi_n > 0$. Thus b_n is a physical nonnegative intensity, not an abstract signed transform coefficient.

Proof. (a) is scaling of a Gaussian. (b): a Poisson variable has mean = variance = Φ_n ; dividing by $\gamma_n \hat{a}_n$ scales the mean by $(\gamma_n \hat{a}_n)^{-1}$ and the variance by $(\gamma_n \hat{a}_n)^{-2}$, and $\Phi_n / (\gamma_n \hat{a}_n) = (a_n / \hat{a}_n) b_n$, $\Phi_n / (\gamma_n \hat{a}_n)^2 = (b_n^2 / \Phi_n) (a_n / \hat{a}_n)^2$. ■

Hence, for fixed calibrated gains $\hat{a}_n = a_n$, $\Sigma_\xi = \text{diag}(\tau_{G,n}^2 + b_n^2 / \Phi_n)$ plugs into (F.2), and the Poisson term is **exact at all photon counts**, including $\Phi_n \ll 1$; no log or Gaussian approximation enters the reconstruction layer. If exogenous $\hat{a}$ is random, the unconditional diagonal uses the expectation of the displayed conditional variances (and the residual-gain term of Theorem 1 accounts separately for $a/\hat{a} - 1$); a same-record $\hat{a}$ falls under Remark F.1.1 instead. Under a total budget $\Phi_n = \omega_n \Phi_{\text{tot}}$, $R_{\text{Pois}} = (S_2 \Phi_{\text{tot}})^{-1} \sum_n (b_n^2 / \omega_n) \|Le_n\|^2$.

a) *What this does not claim.*: The displayed plug-ins are exact conditionally on the gains and exogenous correction factors, and the Poisson formula applies only to physical nonnegative measurement rows. A signed coefficient $b_n = (UT)_n$ cannot be inserted as a Poisson mean. Signed transforms must be implemented by complementary masks or a positive offset, whose photon allocation and differencing covariance are separate objects. The optimal allocation $\omega_n \propto b_n \|Le_n\|$ requires knowing T and is not claimed to be achievable.

C. F.3 Specializations

a) *F.3.1 Orthogonal / full-SRHT inversion ($B_L = 1$).*: If $A = U$ is orthonormal ($N = K$, signed coefficients available) and $L = U^\top$, then $LA = I$ and $\|Le_n\| = 1$, so $B_L =$

$(1/S_2) \sum_n b_n^2 = 1$ by Parseval. With independent residuals and additive *signed-coefficient-domain* noise of covariance Σ_c ,

$$\text{relMSE}_{\text{orth}} = v + \frac{\text{tr}(\Sigma_c)}{S_2}. \quad (\text{F.6})$$

Exact orthogonal inversion transmits independent coefficient-wise residual gain at unit leverage. Equation (F.6) does not assign a direct Poisson variance to signed $b = UT$: a complementary-mask or positive-offset realization must first derive its physical count covariance and then supply the resulting Σ_c (F.3.2 gives the complementary-pair path). A full SRHT acquisition $A = P_{\text{row}} U D P_{\text{col}}$, $L = A^\top$ has the same algebraic propagation once its signed coefficients and Σ_c are defined; SRHT's role is *not* a better inverse but temporal whitening of the randomized ensemble so that the blind estimator can shrink v (Sec. 5/7).

b) *F.3.2 Pairwise Hadamard: exact perturbation law.*

Let $c_k = h_k^\top T$, $h_k \in \{\pm 1\}^K$, with physical masks $B_k^\pm = (S_1 \pm c_k)/2$ and reconstruction $T = K^{-1} H^\top c$, so coefficient errors e_k give $\text{relMSE} = (K S_2)^{-1} \sum_k \mathbb{E} e_k^2$. If the paired masks carry gains a_k^+ and $a_k^- = a_k^+ (1 + \Delta_k)$, the pairwise-normalized estimator satisfies the **exact** algebraic law

$$\hat{c}_k - c_k = -\frac{S_1 \Delta_k (1 - x_k^2)}{2 + \Delta_k (1 - x_k)}, \quad x_k = c_k / S_1, \quad (\text{F.7})$$

where the normalization is the true S_1 . For the production normalization $\hat{S}_1 = \gamma S_1$, provided the observed pair denominator is not clamped, the corresponding coefficient error obeys the second exact identity

$$\begin{aligned} e_k^{(\gamma)} &:= \hat{c}_k^{(\gamma)} - c_k = S_1 \frac{2x_k(\gamma - 1) - \Delta_k(1 - x_k)(\gamma + x_k)}{2 + \Delta_k(1 - x_k)} \\ &= \gamma e_k^{(1)} + (\gamma - 1)c_k. \end{aligned} \quad (\text{F.7a})$$

Consequently the recordwise raw risk decomposes exactly as

$$\frac{\sum_k (e_k^{(\gamma)})^2}{K S_2} = \gamma^2 \frac{\sum_k (e_k^{(1)})^2}{K S_2} + (\gamma - 1)^2 + \frac{2\gamma(\gamma - 1)}{K S_2} \sum_k e_k^{(1)} c_k. \quad (\text{F.7b})$$

The last term is the normalization-increment cross term; scale alignment can absorb much of the middle term, whereas raw MSE cannot. The coefficient identity, its $\gamma = 1$ reduction, and (F.7b) are regression-tested before use in the empirical-status paragraph. whence, exactly (Parseval),

$$R_{\text{pair,gain}} = \frac{K_{\text{eff}}}{K} \sum_{k=1}^K \mathbb{E} \left[\frac{\Delta_k^2 (1 - x_k^2)^2}{\{2 + \Delta_k (1 - x_k)\}^2} \right]. \quad (\text{F.8a})$$

Second-moment expansion (bounded increments). If $|\Delta_k| \leq \eta < 1$ almost surely, then for $g_x(t) = \{2 + t(1 - x)\}^{-2}$ the mean value theorem with $|x| \leq 1$, $|t| \leq \eta$ gives $|g_x(t) - g_x(0)| \leq |t|/\{2(1 - \eta)^3\}$, whence

$$\begin{aligned} &\left| R_{\text{pair,gain}} - \frac{K_{\text{eff}}}{4K} \sum_{k=1}^K (1 - x_k^2)^2 \mathbb{E} \Delta_k^2 \right| \\ &\leq \frac{K_{\text{eff}}}{2(1 - \eta)^3 K} \sum_{k=1}^K \mathbb{E} |\Delta_k|^3. \end{aligned} \quad (\text{F.8b})$$

If the pair increments share a common marginal distribution (no cross-pair independence required),

$$\begin{aligned} R_{\text{pair,gain}} &= \frac{K_{\text{eff}}}{4} D_H(T) \mathbb{E} \Delta^2 + O_\eta(K_{\text{eff}} \mathbb{E} |\Delta|^3), \\ D_H(T) &= \frac{1}{K} \sum_k (1 - x_k^2)^2. \end{aligned} \quad (\text{F.8})$$

The second moment is $\mathbb{E} \Delta^2$, not $\text{Var}(\Delta)$: the two may be interchanged only under the *extra* assumption $\mathbb{E} \Delta = 0$. A deterministic mismatch $\Delta \equiv \epsilon$ is the immediate counterexample — its $\text{Var}(\Delta) = 0$, yet (F.8a) gives a strictly positive risk of order $K_{\text{eff}} \epsilon^2$. For most non-DC Hadamard coefficients of a nonnegative object $|x_k| \ll 1$, so $D_H \approx 1$; D_H is nevertheless a *pipeline constant to be measured*, not assumed.

OU increment corollary (exact lognormal moments). For a stationary Gaussian OU/AR(1) log gain with $\text{Var}(\ell_n) = s^2$ and adjacent correlation $e^{-\rho}$, put $r(\rho) = 1 - e^{-\rho}$ and $u = s^2 r(\rho)$. The adjacent log increment $X = \ell_{n+1} - \ell_n$ is $\mathcal{N}(0, 2u)$ and $\Delta = e^X - 1$, so

$$\mathbb{E} \Delta = e^u - 1, \quad \mathbb{E} \Delta^2 = e^{4u} - 2e^u + 1 = 2u + 7u^2 + O(u^3), \quad (\text{F.8c})$$

with $\text{Var}(\Delta) = e^{4u} - e^{2u}$ and $\mathbb{E} \Delta^2 - \text{Var}(\Delta) = (e^u - 1)^2$. The bounded-increment proposition does not apply directly to this lognormal Δ ; instead, split (F.8a) on $\{|X| \leq c\}$: on the central event Taylor's theorem gives $\Delta^2(1 - x^2)^2 / \{2 + \Delta(1 - x)\}^2 = \frac{1}{4}(1 - x^2)^2 X^2 + O_c(|X|^3)$ uniformly in $x \in [-1, 1]$, while on the complement the exact fraction admits an exponential envelope $C_c(1 + e^{2|X|})$ whose Gaussian-tail expectation is $o(u^m)$ for every m . Since $\mathbb{E} X^2 = 2u$ and $\mathbb{E} |X|^3 = O(u^{3/2})$,

$$R_{\text{pair,gain}} = \frac{1}{2} K_{\text{eff}} D_H(T) s^2 r(\rho) + O(K_{\text{eff}} u^{3/2}) : \quad (\text{F.8d})$$

the leading pair law survives, but its proof must begin from $\mathbb{E} \Delta^2$. The K_{eff} amplification is structural: pairwise normalization converts a differential gain error into a coefficient error of order S_1 , and Parseval then divides by S_2 . For read noise, $2\sigma_{\text{read}}^2/S_2$ is exact only for a *fixed/noiseless normalization*, where the estimator is the linear difference of two independent equal-variance buckets. The unregularized Gaussian ratio has no finite second moment because its noisy denominator has positive density at zero. Under a common pair gain a and high SNR, the delta method instead gives

$$\begin{aligned} \text{Var}(e_k | c_k) &\approx \frac{2\sigma_{\text{read}}^2}{a^2} (1 + x_k^2), \\ R_{\text{pair,read}} &\approx \frac{2\sigma_{\text{read}}^2}{a^2 S_2} \left(1 + \frac{1}{K_{\text{eff}}} \right). \end{aligned}$$

A clamped ratio has finite but regularization-dependent risk. For Poisson counts with pair budget Φ_{pair} , using $B_k^+ + B_k^- = S_1$, the coefficient shot variance is $\text{Var}(e_k | c_k) \approx S_1^2 / \Phi_{\text{pair},k}$ — an approximation, exact only up to the c_k -dependent split between the two buckets — giving $R_{\text{pair,Pois}} = K_{\text{eff}} / \Phi_{\text{pair}}$ (equal-pair budget) or $(K_{\text{eff}} + 1) / (2\Phi_{\text{mask}})$ (equal-mask budget); these differ by allocation convention.

c) *What F.3.2 does not claim.*: (F.7)–(F.8) presuppose a stable DC/S_1 normalization; at low photons the denominator bucket sum can be noisy or zero, and a robust implementation needs offset/clipping or a likelihood-ratio estimator. Equations

(F.7) and (F.7a) are algebra on the unclamped branch; only the moment step (F.8) invokes small Δ and a common marginal distribution across coefficient indices. No cross-pair independence is required for the expectation of the summed squared error.

d) *F.3.3 Random patterns / DGI: floor, leverage, and population-moment constants.*: **Three distinct constants must carry three distinct names; no generic “pipeline C_0 ” label is permitted.** For a fixed design A , linear reconstruction L , object T , and $b = AT$, define the *raw fixed-design sampling floor* and the *residual leverage*

$$C_0^{\text{floor}}(A, L, T) := \frac{N}{S_2} \|(LA - I)T\|_2^2, \quad (\text{F.9a})$$

$$\Lambda_{\text{lev}}(A, L, T) := N B_L(A, T) = \frac{N}{S_2} \sum_{n=1}^N b_n^2 \|Le_n\|_2^2, \quad (\text{F.9b})$$

$$C_0^{\text{lev}}(A, L, T) := \Lambda_{\text{lev}} - 1. \quad (\text{F.9c})$$

The first is the raw-MSE sampling floor of the fixed operator; the second propagates exogenous multiplicative residuals and equal-count Poisson noise (C_0^{lev} is merely the shifted notation $\Lambda_{\text{lev}} - 1$). Third, for N iid pattern draws $I_n = \mu \mathbf{1} + \sigma x_n$ with iid coordinates ($\mathbb{E}x = 0$, $\mathbb{E}x^2 = 1$, $\mathbb{E}x^3 = \gamma_3$, $\mathbb{E}x^4 = \beta_4$), the ideal known- (μ, σ) correlator $\hat{T}_{\text{DGI}} = (N\sigma)^{-1} \sum_n x_n z_n$ and one-sample vector $Z = \sigma^{-1}x(\mu S_1 + \sigma x^\top T)$ define the *population raw-moment constant*

$$\frac{\mathbb{E}\|\hat{T}_{\text{DGI}} - T\|_2^2}{S_2} = \frac{C_0^{\text{raw, moment}}}{N}, \quad (\text{F.9})$$

$$C_0^{\text{raw, moment}} := \frac{\mathbb{E}\|Z - T\|_2^2}{S_2},$$

whose evaluation is a moment calculation: expanding $\mathbb{E}\|Z\|_2^2$ with $Z = (\mu S_1/\sigma)x + x x^\top T$ and subtracting $\mathbb{E}\|Z\|_2^2 = S_2$,

$$C_0^{\text{raw, moment}} = K + \beta_4 - 2 + K_{\text{eff}}[K(\mu/\sigma)^2 + 2\gamma_3(\mu/\sigma)]$$

$$\xrightarrow{\mu=0, \gamma_3=0} K + \beta_4 - 2. \quad (\text{F.10})$$

The $K_{\text{eff}}K(\mu/\sigma)^2$ term is the familiar nonnegative-background penalty. For the ideal correlator, independence and unbiasedness give $\mathbb{E}_A C_0^{\text{floor}} = C_0^{\text{raw, moment}}$ and $\mathbb{E}_A \Lambda_{\text{lev}} = 1 + C_0^{\text{raw, moment}}$; **for a mean-subtracted or scale-aligned implementation these are not definition-level identities** — each constant must be measured for the pipeline that is actually run. The clean-commit R15 contract output reproduces the historical grid’s mean-subtracted *raw* fixed-design floor near 10^3 : objectwise range 980.110–1114.529, with the ten-object *mean* agreeing with $K + \beta_4 - 2 = 1023.800$ to 0.8%. The historical *scale-aligned* floor is near $6.2\text{--}7.2 \times 10^2$ and the nonnegative-background drift leverage near 10^6 ; conflating any two is a factor-1.5 to factor- 10^3 error.

Proposition (exact pipeline specialization). *Suppose the residuals are exogenous to detector noise and design, with*

common mean m_δ , common second moment $q_\delta = \mathbb{E}\delta_n^2$, and $\text{Cov}(\delta) = (q_\delta - m_\delta^2)I$. Under the hypotheses of Theorem 1,

$$\frac{\mathbb{E}\|\hat{T} - T\|_2^2}{S_2} = \frac{\|(LA - I)T + m_\delta LAT\|_2^2}{S_2} + \frac{\Lambda_{\text{lev}}}{N}(q_\delta - m_\delta^2) + R_\xi, \quad (\text{F.10a})$$

with $R_\xi = \text{tr}(L\Sigma_\xi L^\top)/S_2$. If $m_\delta = 0$,

$$\frac{\mathbb{E}\|\hat{T} - T\|_2^2}{S_2} = \frac{C_0^{\text{floor}}}{N} + \frac{1 + C_0^{\text{lev}}}{N} q_\delta + R_\xi; \quad (\text{F.10b})$$

and if $\Sigma_\xi = \tau_G^2 I + \Phi_{\text{frame}}^{-1} \text{diag}(b^2)$,

$$R_\xi = \frac{\tau_G^2 \text{tr}(LL^\top)}{S_2} + \frac{1 + C_0^{\text{lev}}}{N \Phi_{\text{frame}}}. \quad (\text{F.10c})$$

Proof. The residual mean vector is $m_\delta \mathbf{1}$ and $L \text{diag}(b) \mathbf{1} = Lb = LAT$; substitute this mean and the stated covariance into the exact identity (F.2). The Poisson specialization uses $(S_2 \Phi_{\text{frame}})^{-1} \sum_n b_n^2 \|Le_n\|_2^2 = \Lambda_{\text{lev}}/(N \Phi_{\text{frame}})$. ■

For the ideal iid correlator with centered residuals the pattern-averaged bookkeeping is the familiar display

$$\text{relMSE}_{\text{DGI}} = \frac{C_0^{\text{floor}}}{N} + \frac{(1 + C_0^{\text{lev}})v}{N} + \frac{K\tau_G^2}{N\sigma^2 S_2} + \frac{1 + C_0^{\text{lev}}}{N \Phi_{\text{frame}}}, \quad (\text{F.11})$$

where for the ideal correlator $\mathbb{E}_A \Lambda_{\text{lev}} = 1 + C_0^{\text{raw, moment}}$, so in residual-injection experiments the measurable slope is $NB_L \rightarrow \Lambda_{\text{lev}}$. **Status (labeled).** This pattern-averaging step treats the residuals δ as independent of the pattern draw — exact for injected residuals (the Fig. 10 protocol), a heuristic decoupling for blind pipelines, where $\hat{a}$ is computed from the same record the patterns generated. There is no universal long-run-variance multiplier for correlated residuals. If $vB_L > 0$, one may *define* a pipeline-specific $\chi_{\text{eff}} := R_{\text{gain}}/(vB_L)$ using the exact matrix expression (F.4), but it depends on (A, L, T) and the full covariance; it is not generally $1 + 2 \sum_h \text{corr}(\delta_0, \delta_h)$.

e) *Correction (drift-leverage constant of the gain term; measured, no fitted parameters).*: As previously written, the gain term of (F.11) evaluated with the *scale-aligned floor* constant in place of the leverage is **wrong for mean-subtracted DGI pipelines** under time-structured (drift-like) residual gain: on the published grid it underpredicts the measured random-arm drift excess by a median factor $\approx 1.4 \times 10^3$ (1364 / 1527 / 1521 across the none / AGC / SCGI-proxy corrections; white-drift diagnostic, results/prop3_nofreeparam_r1/). The correct leverage is the **nonnegative-background constant** $\Lambda_{\text{lev}} = 1 + C_0^{\text{lev, raw background}}$, obtained from (F.10) evaluated at the measured raw pattern moments (at the grid’s operating point $C_0^{\text{lev, raw background}} \approx 0.7\text{--}1.2 \times 10^6$, versus the scale-aligned floor $\approx 6.2\text{--}7.2 \times 10^2$ and the *raw* fixed-design floor $\approx 1.0\text{--}1.1 \times 10^3$); with it the measured excess is reproduced to 0.3–4.7% (median ratios 1.003 / 1.047 / 1.039), still with zero fitted parameters. Mechanism: mean subtraction removes the $K_{\text{eff}}K(\mu/\sigma)^2$ background penalty from the **sampling floor** — hence the small floor constant in the C_0^{floor}/N term, confirmed

exactly by the flat oracle arm — but **not** from the **drift leverage**: a time-varying gain modulates the large DC background coherently, and the correlator pays the full background leverage. For mean-subtracted pipelines the gain term of (F.11) must therefore carry $\Lambda_{\text{lev}} = 1 + C_0^{\text{lev,raw background}}$, while the sampling-floor term keeps the (raw or aligned, *consistently chosen*) floor constant. This vindicates — and extends to the gain term — the “no universal C_0 ” warning below, which earlier versions applied to the floor term only.

f) *What F.3.3 does not claim.*: There is **no universal C_0** : (F.10) is exact only for the ideal known- (μ, σ) correlator with iid coordinates. Mean subtraction, reference normalization, and photon allocation all change the constants, so each must be measured for the pipeline that is run. The sampling floor and drift leverage differ by about 10^3 here, and raw versus scale-aligned floors by about 1.54. Table I now reports the exact fixed-operator column-norm calculation, averaged over the ten bridge objects, $NB_L \in [7.527, 7.544] \times 10^5$ across N ; it is neither a Monte Carlo residual probe nor the Prop. 3 floor. The flux-limited convention $\Phi_n = \gamma b_n$ applies only to nonnegative physical b_n and replaces the last term of (F.11) by $(N\sigma^2 S_2)^{-1} \mathbb{E}[\|x\|^2 b_n / \gamma]$.

D. F.4 Prop 3: the finite- N flip boundary

a) *Drift model and assumptions.*: Let s^2 be the variance scale of the log-gain process ℓ and let ρ_{pair} be the **adjacent-pair decorrelation parameter**: for an Ornstein–Uhlenbeck log-gain the paired-mask gain increment obeys $\mathbb{E}[\Delta_k^2] = 2s^2 r(\rho_{\text{pair}}) + O(s^4 r^2)$ with $r(\rho) = 1 - e^{-\rho}$ (exact moments: (F.8c); $r(\rho) = \rho + O(\rho^2)$ at small ρ). Assume the small-mismatch regime of (F.8) (so $R_{\text{pair,gain}}(\rho_{\text{pair}}) \approx \frac{1}{2} K_{\text{eff}} D_H(T) s^2 r(\rho_{\text{pair}})$, per (F.8d)) and the DGI decomposition (F.11) with blind residual variance $v_{\text{blind}}(\rho_{\text{pair}}, N)$, treated here as an **externally measured function of (ρ_{pair}, N)** (the runner measures it on the grid). **No deterministic-Lipschitz $\rho^{2/3}$ rate is attributed to the OU drift**: OU paths are not Lipschitz, and the correct interior OU window analysis (lemma below, (F.14)–(F.16)) gives stochastic drift error $s^2 \rho W / 6$ to first order and window-optimized risk $\propto \rho^{1/2}$, not $\rho^{2/3}$.

Prop 3 (raw-MSE flip boundary; fixed-point form). *All risks in this proposition are raw relative MSEs of Theorem 1 (the metric caution below explains why a scale-aligned analogue is a different number). Define $\rho^* = \inf\{\rho_{\text{pair}} \geq 0 : R_{\text{pair}}(\rho_{\text{pair}}) \geq R_{\text{rand}}(\rho_{\text{pair}}, N)\}$, where $R_{\text{pair}} = R_{\text{pair,gain}} + R_{\text{pair,read}} + R_{\text{pair,Pois}}$ and, for the random arm,*

$$R_{\text{rand}}(\rho, N) = \frac{C_0^{\text{floor}}}{N} + \Gamma_{\text{blind}}(\rho, N) + R_{\text{rand,noise}}. \quad (\text{F.12a})$$

For an oracle-known-gain arm, $\Gamma_{\text{blind}} = 0$ (the validated skeleton below). For an exogenous, centered, frame-uncorrelated residual with $q_{\text{blind}}(\rho, N) = \mathbb{E}\delta_n^2$,

$$\Gamma_{\text{blind}}(\rho, N) = \frac{1 + C_0^{\text{lev}}}{N} q_{\text{blind}}(\rho, N), \quad (\text{F.12b})$$

with C_0^{lev} the pipeline’s drift-leverage constant of (F.9c) (equal to $C_0^{\text{lev,raw background}} \approx 10^6$ for the mean-subtracted pipeline

— the F.3.3 correction). **No scalar representation is asserted for a same-record blind estimator**: there the full covariance, nonzero-mean, and residual-noise cross terms of Theorem 1 are required, and any scalar display must be labeled a conditional proxy. At first order define

$$A_T = \frac{1}{2} K_{\text{eff}} D_H(T) s^2, \quad Q(\rho) = \frac{\Delta_R(\rho)}{A_T},$$

$$\Delta_R(\rho) = \frac{C_0^{\text{floor}}}{N} + \Gamma_{\text{blind}}(\rho, N) + R_{\text{rand,noise}} - R_{\text{pair,noise}}.$$

Assume $A_T > 0$ (equivalently $s > 0$ and $D_H(T) > 0$) and that the risks, hence Q , are continuous in ρ on the comparison interval. Then the leading boundary is the **fixed-point equation**

$$1 - e^{-\rho^*} = Q(\rho^*), \quad (\text{F.12})$$

up to the remainder in (F.8d). If $\Delta_R(0) \leq 0$ then $\rho^* = 0$ (the flip is immediate, not absent). If $\Delta_R(0) > 0$, a finite crossing on $[0, \rho_{\text{max}}]$ exists when $1 - e^{-\rho} \geq Q(\rho)$ somewhere on that interval, and is unique only under an additional single-crossing condition (e.g. strict increase of $1 - e^{-\rho} - Q(\rho)$); if the difference stays negative, no crossing occurs in range. Only if $Q(\rho) \equiv Q_0$ is constant may one write the explicit solution: $Q_0 \leq 0 \Rightarrow \rho^* = 0$; $0 < Q_0 < 1 \Rightarrow \rho^* = -\log(1 - Q_0)$; $Q_0 \geq 1 \Rightarrow$ no finite crossing in the leading OU model. When q_{blind} depends on ρ , the logarithmic display is a fixed point, **not an explicit solution**.

Proof. At the boundary, $R_{\text{pair}}(\rho^*) = R_{\text{rand}}(\rho^*, N)$. Substituting $R_{\text{pair,gain}} = \frac{1}{2} K_{\text{eff}} D_H s^2 r(\rho^*)$ (from (F.8d), whose proof starts from $\mathbb{E}\Delta^2$, not $\text{Var } \Delta$) and (F.12a)–(F.12b), and solving for $r(\rho^*) = 1 - e^{-\rho^*}$, gives (F.12). Existence under the stated sign/continuity conditions is the intermediate value theorem applied to $\rho \mapsto 1 - e^{-\rho} - Q(\rho)$; the infimum is a crossing by continuity; uniqueness is exactly the single-crossing assumption and is not otherwise proven. ■

Empirical status of Prop 3 (two estimators and a separate blind proxy). The equal-frame grid ($N = 2048$, $K = 1024$, OU drift, 10 objects, five seeds, all constants measured, zero fitted parameters) is re-evaluated in raw MSE at `results/prop3_estimator_contract_r15/`. Its clean-commit manifest classifies it as the submission-authoritative R15 contract output; the prior raw-metric directory remains historical evidence rather than submission-authoritative output.

(a) *Same-estimator theorem validation (true S_1).* The theorem-faithful arm passes the true S_1 to the pair estimator and compares it with the same raw oracle random arm. For the 40 cells with $\sigma_a \geq 0.1$, all 40 raw crossings are observed; the prediction/observation agreement factor has median 1.02041 and maximum 1.03686. The pooled empirical slope is -2.12044 ($R^2 = 0.97669$), versus the no-free-parameter prediction -2.11371 ($R^2 = 0.97807$). This is the validation of (F.7)–(F.8).

(b) *Production-estimator robustness (median pair sum).* Keeping the production $\hat{S}_1 = \text{median}_k(R_k^+ + R_k^-)$ on the identical traces gives 40/40 crossings, median factor 1.04394, and maximum factor 1.19985. Its pooled slope is -2.00656 ($R^2 = 0.97271$), against the same -2.11371 leading prediction. Equation (F.7a), not (F.7) alone, describes this arm: the full-grid maximum coefficient-identity discrepancy is 6.1×10^{-7} after normalization by S_1 , no denominator clamp fires, and (F.7b) closes up to float32 accumulation. The high- σ_a gap therefore includes the random normalization gauge and its cross term; it is not assigned solely to the cubic remainder. The earlier $-2.11 / -2.07$ sentence came from a metric/estimator mismatch and is replaced by these two explicit slope comparisons.

(c) *Same-record blind arm (conditional noiseless proxy, not a scalar theorem validation).* The empirical ordered-pair arm remains better than blind random+DGI in all 100 previously scanned cells. In the R15 diagnostic, raw $q_\delta = \mathbb{E}(a/\hat{a} - 1)^2$ exceeds the scale-aligned gain error by a median factor 1.019, and the residual lag-one correlation has median 0.882. The reconstruction-space decomposition using the projected mean and covariance closes with maximum absolute error 1.74×10^{-5} ; the scalar leverage proxy is close in this particular random-design protocol (median 1.016, maximum 1.041 relative to exact projected gain energy) but remains a proxy, not (F.12b). No detector noise is present, so $C_{\delta\xi} = 0$ structurally rather than being empirically estimated. The corrected scalar no-flip curve is retained only as a conditional engineering check consistent with the 100/100 observation.

b) *Leading-order heuristic (labeled as such).*: If $r(\rho) = \rho$, $D_H = 1$, noise differences are negligible, v_{blind} is negligible, Q is treated as constant, and ρ means adjacent-pair decorrelation, (F.12) collapses to the engineering rule

$$\rho^* \approx \frac{2C_0^{\text{floor}}}{N K_{\text{eff}} s^2}, \quad (\text{F.13})$$

with first corrections $\rho^* \approx \frac{2(C_0^{\text{floor}} + (1 + C_0^{\text{lev}})v_{\text{blind}})/[N K_{\text{eff}} D_H s^2]}{2(R_{\text{DGI,noise}} - R_{\text{pair,noise}})/(K_{\text{eff}} D_H s^2)} +$ and the nonlinear replacement $\rho \mapsto r^{-1}(\cdot)$ when adjacent increments are not infinitesimal. Note that for mean-subtracted pipelines “negligible v_{blind} ” means $(1 + C_0^{\text{lev,raw background}})v_{\text{blind}} \ll C_0^{\text{floor}}$ — a far stronger requirement than $v_{\text{blind}} \ll C_0^{\text{floor}}$, and the reason the blind-arm flip vanished on the published grid (empirical-status point (c) above). (F.13) is a *leading-order heuristic*, not a theorem; the theorem is (F.12).

c) *Lemma (OU window risk; not a Lipschitz-path theorem).*: Let ℓ_n be stationary Gaussian with $\text{Cov}(\ell_n, \ell_{n+h}) = s^2 e^{-\rho|h|}$, and let ε_n be an independent centered stationary carrier-noise sequence with autocovariance $\gamma_\varepsilon(h)$. For an interior centered window $W = 2m + 1$ and $\hat{\ell}_n = W^{-1} \sum_{j=-m}^m (\ell_{n+j} + \varepsilon_{n+j})$,

$$\begin{aligned} \mathbb{E}(\hat{\ell}_n - \ell_n)^2 &= s^2 \left[1 - \frac{2}{W} \sum_{j=-m}^m e^{-\rho|j|} \right. \\ &\quad \left. + \frac{1}{W^2} \sum_{i,j=-m}^m e^{-\rho|i-j|} \right] \\ &\quad + \frac{1}{W^2} \sum_{i,j=-m}^m \gamma_\varepsilon(i-j). \end{aligned} \quad (\text{F.14})$$

If $\rho W \rightarrow 0$, the OU contribution is

$$s^2 \rho \frac{W^2 - 1}{6W} + O(s^2 \rho^2 W^2), \quad (\text{F.15})$$

and if $\sum_h |\gamma_\varepsilon(h)| \leq \sigma_{\text{abs}}^2$,

$$\mathbb{E}(\hat{\ell}_n - \ell_n)^2 \leq \frac{\sigma_{\text{abs}}^2}{W} + C s^2 \rho W. \quad (\text{F.16})$$

When $1 \ll W^* \ll \min(N, 1/\rho)$, an interior optimizer has $W^* \asymp \sigma_{\text{abs}}/(s\sqrt{\rho})$ and $\text{MSE}^* \asymp \sigma_{\text{abs}} s\sqrt{\rho}$ (for iid carrier noise, $W^* \sim \sqrt{6} \sigma_\varepsilon/(s\sqrt{\rho})$ and $\text{MSE}^* \sim \sqrt{2/3} \sigma_\varepsilon s\sqrt{\rho}$); outside this feasible band the appropriate boundary window applies. **The deterministic Lipschitz $\rho^{2/3}$ rate is therefore not an OU output.** Prop 3 may treat $q_{\text{blind}}(\rho, N)$ as an

externally measured function — as the empirical analysis does — but any *OU estimator theorem* must use this stochastic $\rho^{1/2}$ analysis instead.

d) *Metric and implementation cautions (raw versus scale-aligned).*: The Prop.-3 runner’s scale-aligned gain error $\min_c \|\hat{c}a - a\|^2/\|a\|^2$ is not $q_\delta = N^{-1} \sum_n (a_n/\hat{a}_n - 1)^2$ and is not $\text{Var}(\delta)$; they agree only to first order after a fixed small-error gauge. Likewise, a scale-aligned reconstruction floor is not the raw C_0^{floor} required by Prop. 3. If $\hat{T} = T + e$, put $a = \langle e, T \rangle / S_2$ and $b = \|e\|^2 / S_2$; the implementation’s scale-aligned error, minimizing over $c \in \mathbb{R}$, is

$$R_{\text{align}} = \frac{b - a^2}{1 + 2a + b}, \quad (\text{F.17})$$

which is not generally proportional to the raw risk b (even for $e \perp T$ it is $b/(1+b)$, not b). Therefore a raw-MSE theorem and an aligned-MSE empirical boundary must not be called an exact no-free-parameter validation of one another. The R15 true- S_1 arm compares raw with raw using the same estimator on both sides in the clean-commit contract directory.

e) *ρ -convention warning.*: Throughout, ρ_{pair} (adjacent-pair decorrelation) is the variable of (F.12)–(F.13); it is **not** the normalized bandwidth ρ_{bw} of the tall-design thresholds ($p \approx \rho_{\text{bw}} N$, Sec. 4). If a bandwidth parameter with adjacent-increment variance $\sim 2s^2 \rho_{\text{bw}} / N$ is substituted, the explicit $1/N$ in (F.13) cancels and the boundary reads $\rho_{\text{bw}}^* \approx 2C_0/(K_{\text{eff}} s^2)$ — same physics, different symbol. Every phase-diagram axis must declare its convention.

f) *What Prop 3 does not claim.*: The boundary compares two *specific* pipelines (pairwise-normalized Hadamard vs. blind random+DGI with the ideal correlator); it does not rank SRHT-paired pipelines, externally calibrated systems, or prior-regularized reconstructions, each of which shifts Δ_R . It inherits every assumption of its ingredients: small mismatch (F.8b)–(F.8d), measured constants C_0^{floor} , C_0^{lev} , and D_H (with the floor/leverage distinction of the F.3.3 correction and the raw/aligned metric distinction of (F.17)), a stationary drift with a declared $r(\rho)$, and a q_{blind} that is an externally measured function, not an OU theorem (the $\rho^{2/3}$ attribution is withdrawn; the OU lemma gives $\rho^{1/2}$). Every prediction and every observation must be stated in the *same* raw or aligned metric. The strict no-free-parameter claim attaches only to the R15 true- S_1 raw-vs-raw comparison reported in the clean-commit R15 contract directory; a purely aligned comparison is a different empirical analogue. It is a mean-square crossing, not a guarantee about any single realization; the single-crossing property itself is assumed, not derived; and at ρ_{pair} beyond the small-drift expansion only the defining infimum — not (F.12) — is meaningful. Finally, the validated regime is the $v = 0$ skeleton (empirical-status point (a)); no scalar flip theorem is currently claimed for the blind arm — the corrected conditional proxy and the data both give no flip on the published grid.

REFERENCES

- [1] T. B. Pittman, Y. H. Shih, D. V. Strekalov, and A. V. Sergienko, “Optical imaging by means of two-photon quantum entanglement,” *Phys. Rev. A* **52**, R3429–R3432 (1995).

- [2] R. S. Bennink, S. J. Bentley, and R. W. Boyd, “‘Two-Photon’ Coincidence Imaging with a Classical Source,” *Phys. Rev. Lett.* **89**, 113601 (2002).
- [3] J. H. Shapiro, “Computational ghost imaging,” *Phys. Rev. A* **78**, 061802(R) (2008).
- [4] Y. Bromberg, O. Katz, and Y. Silberberg, “Ghost imaging with a single detector,” *Phys. Rev. A* **79**, 053840 (2009).
- [5] M. Bina *et al.*, “Backscattering Differential Ghost Imaging in Turbid Media,” *Phys. Rev. Lett.* **110**, 083901 (2013).
- [6] J. Cheng, “Ghost imaging through turbulent atmosphere,” *Opt. Express* **17**(10), 7916–7921 (2009).
- [7] Y. Peng and W. Chen, “Deep learning-enhanced ghost imaging through dynamic and complex scattering media with supervised corrections of dynamic scaling factors,” *Appl. Phys. Lett.* **124**, 181104 (2024).
- [8] Y. Peng, Y. Xiao, and W. Chen, “High-fidelity optical wireless transmission in complex environments around a corner using the design of a single-layer neural network for data encoding,” *Opt. Express* **33**(14), 30123–30135 (2025).
- [9] M. F. Duarte *et al.*, “Single-Pixel Imaging via Compressive Sampling,” *IEEE Signal Process. Mag.* **25**(2), 83–91 (2008).
- [10] M. P. Edgar, G. M. Gibson, and M. J. Padgett, “Principles and prospects for single-pixel imaging,” *Nat. Photonics* **13**, 13–20 (2019).
- [11] Z. Zhang, X. Wang, G. Zheng, and J. Zhong, “Hadamard single-pixel imaging versus Fourier single-pixel imaging,” *Opt. Express* **25**(16), 19619–19639 (2017).
- [12] W.-K. Yu, “Super Sub-Nyquist Single-Pixel Imaging by Means of Cake-Cutting Hadamard Basis Sort,” *Sensors* **19**(19), 4122 (2019).
- [13] M. Kech and F. Kraher, “Optimal Injectivity Conditions for Bilinear Inverse Problems with Applications to Identifiability of Deconvolution Problems,” *SIAM J. Appl. Algebra Geom.* **1**(1), 20–37 (2017).
- [14] Y. Li, K. Lee, and Y. Bresler, “Identifiability in Bilinear Inverse Problems With Applications to Subspace or Sparsity-Constrained Blind Gain and Phase Calibration,” *IEEE Trans. Inf. Theory* **63**(2), 822–842 (2017).
- [15] A. Ahmed, B. Recht, and J. Romberg, “Blind Deconvolution Using Convex Programming,” *IEEE Trans. Inf. Theory* **60**(3), 1711–1732 (2014).
- [16] S. Choudhary and U. Mitra, “Identifiability Scaling Laws in Bilinear Inverse Problems,” arXiv:1402.2637 (2014).
- [17] R. Kueng, H. Rauhut, and U. Terstiege, “Low rank matrix recovery from rank one measurements,” *Appl. Comput. Harmon. Anal.* **42**(1), 88–116 (2017).
- [18] F. Ferri, D. Magatti, L. A. Lugiato, and A. Gatti, “Differential Ghost Imaging,” *Phys. Rev. Lett.* **104**, 253603 (2010).
- [19] B. Sun *et al.*, “Normalized ghost imaging,” *Opt. Express* **20**(15), 16892–16901 (2012).
- [20] A. J. Weiss and B. Friedlander, “Eigenstructure methods for direction finding with sensor gain and phase uncertainties,” *Circuits Syst. Signal Process.* **9**(3), 271–300 (1990).
- [21] L. Balzano and R. Nowak, “Blind calibration of sensor networks,” in *Proc. IPSN*, 2007, pp. 79–88.
- [22] S. Ling and T. Strohmer, “Self-calibration and biconvex compressive sensing,” *Inverse Problems* **31**(11), 115002 (2015).
- [23] X. Nie *et al.*, “Noise-robust computational ghost imaging with pink noise speckle patterns,” *Phys. Rev. A* **104**, 013513 (2021).
- [24] Y. Han *et al.*, “Generalized sparse Hadamard single-pixel imaging,” *Opt. Express* **33**(20), 43324–43341 (2025).
- [25] A. B. Tsybakov, *Introduction to Nonparametric Estimation*, Springer Series in Statistics. New York: Springer, 2009.
- [26] R. D. Gill and B. Y. Levit, “Applications of the van Trees inequality: a Bayesian Cramér–Rao bound,” *Bernoulli* **1**(1–2), 59–79 (1995).
- [27] M. S. Pinsker, “Optimal filtering of square-integrable signals in Gaussian noise,” *Probl. Inf. Transm.* **16**(2), 120–133 (1980).
- [28] F. J. Anscombe, “The transformation of Poisson, binomial and negative-binomial data,” *Biometrika* **35**(3–4), 246–254 (1948).
- [29] M. B. Brown and A. B. Forsythe, “Robust tests for the equality of variances,” *J. Amer. Statist. Assoc.* **69**(346), 364–367 (1974).
- [30] H. Levene, “Robust tests for equality of variances,” in *Contributions to Probability and Statistics: Essays in Honor of Harold Hotelling*, I. Olkin *et al.*, Eds. Stanford, CA: Stanford Univ. Press, 1960, pp. 278–292.
- [31] R. J. Serfling, “Probability inequalities for the sum in sampling without replacement,” *Ann. Statist.* **2**(1), 39–48 (1974).
- [32] Yu. A. Davydov, “Convergence of distributions generated by stationary stochastic processes,” *Theory Probab. Appl.* **13**(4), 691–696 (1968).
- [33] F. Merlevède, M. Peligrad, and E. Rio, “A Bernstein type inequality and moderate deviations for weakly dependent sequences,” *Probab. Theory Relat. Fields* **151**(3–4), 435–474 (2011).
- [34] H. C. P. Berbee, *Random Walks with Stationary Increments and Renewal Theory*, Mathematical Centre Tracts, vol. 112. Amsterdam: Mathematisch Centrum, 1979.
- [35] E. Rio, *Asymptotic Theory of Weakly Dependent Random Processes*, Probability Theory and Stochastic Modelling, vol. 80. Berlin: Springer, 2017.
- [36] R. Vershynin, *High-Dimensional Probability: An Introduction with Applications in Data Science*, Cambridge Series in Statistical and Probabilistic Mathematics, vol. 47. Cambridge: Cambridge Univ. Press, 2018.
- [37] L. D. Brown and M. G. Low, “Asymptotic equivalence of nonparametric regression and white noise,” *Ann. Statist.* **24**(6), 2384–2398 (1996).
- [38] C. R. Rao and S. K. Mitra, *Generalized Inverse of Matrices and Its Applications*. New York: Wiley, 1971.
- [39] P. Stoica and T. L. Marzetta, “Parameter estimation problems with singular information matrices,” *IEEE Trans. Signal Process.* **49**(1), 87–90 (2001).
- [40] E. L. Lehmann and G. Casella, *Theory of Point Estimation*, 2nd ed., Springer Texts in Statistics. New York: Springer, 1998.
- [41] C. A. J. Klaassen, “On an inequality of Chernoff,” *Ann. Probab.* **13**(3), 966–974 (1985).
- [42] S. G. Bobkov and M. Ledoux, “On modified logarithmic Sobolev inequalities for Bernoulli and Poisson measures,” *J. Funct. Anal.* **156**(2), 347–365 (1998).
- [43] R. Bardenet and O.-A. Maillard, “Concentration inequalities for sampling without replacement,” *Bernoulli* **21**(3), 1361–1385 (2015).
- [44] D. L. Hanson and F. T. Wright, “A bound on tail probabilities for quadratic forms in independent random variables,” *Ann. Math. Statist.* **42**(3), 1079–1083 (1971).
- [45] M. Rudelson and R. Vershynin, “Hanson–Wright inequality and sub-Gaussian concentration,” *Electron. Commun. Probab.* **18**, paper no. 82, 1–9 (2013).
- [46] P. Billingsley, *Probability and Measure*, 3rd ed. New York: Wiley, 1995.
- [47] V. V. Petrov, *Sums of Independent Random Variables*. Berlin: Springer-Verlag, 1975.
- [48] A. W. van der Vaart, *Asymptotic Statistics*. Cambridge, U.K.: Cambridge Univ. Press, 1998.